REAL ANALYSIS II · SEMESTER 5 · MODULE 3

Prepare: The Riemann Integral

Everything you need before you open the question paper, from Plus Two calculus and Real Analysis I up to Bartle & Sherbert §7.1 (7.1.1–7.1.7), §7.2 (7.2.1–7.2.13) and §7.3 (7.3.1–7.3.18): tagged partitions, the Cauchy and Squeeze criteria, step, continuous and monotone functions, the Fundamental Theorem of Calculus, Substitution, Lebesgue's criterion, Integration by Parts and Taylor's theorem.

File 1 of 3	**Prepare** (this file): learn the ideas and proofs
File 2	Model paper: all 74 past questions + new questions
File 3	Answer key: full formal solutions

Each answer in File 3 starts with topic codes such as **T15** **P08** **M08** . If a code is new to you, read that topic here first.

How to read this file

> Do, don't re-read: cover the proof, write the skeleton from memory, then check every "Why" line.

DEFINITION / THEOREM · LEARN WORD FOR WORD

Exact wording from Bartle & Sherbert 4e.

In plain words: what the symbols actually say when you strip away the notation.

SKELETON · WRITE THIS BEFORE YOU HAVE ANY IDEA

Let $\varepsilon > 0$ be given. Choose $\delta_\varepsilon =$? . Let $\dot{P}$ be any tagged partition of [a,b] with $\|P\| < \delta_\varepsilon$. Then $|S(f;\dot{P}) - L| \leq$? $< \varepsilon$. Hence $f \in \mathcal{R}[a,b]$ with $\int_a^b f = L$. ∎

Know The hypotheses unpacked into inequalities you can use.

Want The target statement unpacked into what you must bound.

SCRATCH · HOW WE FIND Δ OR N (NEVER COPIED INTO THE EXAM)

Work backwards from $|S(f;\dot{P}) - L| < \varepsilon \Leftarrow C\|P\| < \varepsilon \Leftarrow \|P\| < \varepsilon/C$.

PROOF · WHAT YOU WRITE IN THE EXAM

1. Forward logical step starting from $\varepsilon > 0$. ∎
   Every step names the definition, theorem or inequality that justifies it.

HYPOTHESES · WHAT BREAKS IF YOU DROP ONE

☐ each condition the theorem needs
what goes wrong without it, with its counterexample tag C7

YOUR INTUITION WILL LIE TO YOU HERE

A naive mental picture that clashes with the formal definition, and how to fix it.

CHECK YOURSELF

A quick prompt to test if you can use the idea cold.

The complete short answer so you can check immediately.

One idea for the whole module

YOU ARE HANDED E > 0; YOU MUST HAND BACK A MESH BOUND $\delta_\varepsilon > 0$

Cut the interval [a,b] into pieces $[x_{i-1}, x_i]$ and pick a sample point (**tag**) t_i in each piece. Build a rectangle of width $x_i - x_{i-1}$ and signed height $f(t_i)$. The sum of these rectangle areas is the **Riemann sum** $S(f;\dot{P}) = \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1})$.

A function f is **Riemann integrable** with integral L if, **whatever** $\varepsilon > 0$ an opponent hands you, you can hand back a cut-size limit $\delta_\varepsilon > 0$ so that **every** partition whose widest piece has length $\|P\| < \delta_\varepsilon$—*and* **every** legal choice of tags inside those pieces—produces a sum $S(f;\dot{P})$ within ε of L.

CONCEPT	OPPONENT PICKS	YOU PICK	ALLOWED TO DEPEND ON	MUST NOT DEPEND ON
$f \in \mathcal{R}[a,b] (Def 7.1.1)$	$\varepsilon > 0$	$\delta_\varepsilon > 0$	$\varepsilon, f, [a,b]$	**partition P, tags** t_i
Cauchy Criterion (7.2.1)	$\varepsilon > 0$	$\eta_\varepsilon > 0$	$\varepsilon, f, [a,b]$	$\dot{P}, \dot{Q}$ **with norm <** η_ε
Squeeze Theorem (7.2.3)	$\varepsilon > 0$	$\alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b]$	$\varepsilon, f, [a,b]$	**individual points x**

THE CHAIN THAT CARRIES THE MODULE

Proving integrability straight from Definition 7.1.1 needs you to guess L first. To avoid guessing L, Bartle builds one chain:

Definition 7.1.1 $(T03) \Rightarrow$ **Cauchy Criterion** $(T10) \Rightarrow$ **Squeeze Theorem** (T12) + **Step functions** $\in \mathcal{R}$ $(T14) \Rightarrow$ **Continuous** $\in \mathcal{R}$ (T15) & **Monotone** $\in \mathcal{R}$ (T16).

Then **Mean Value Theorem** (P12) connects derivatives to Riemann sums to give **FTC First Form** (T20), while **Additivity** (T17) + continuity at c gives **FTC Second Form** (T23), unlocking **Substitution** (T25), **Integration by Parts** (T31) and **Taylor's Remainder** (T32).

The chain in one table

> Read down: each result is proved from the ones above it.

STEP	WHAT IT SAYS	BUILT FROM
T01 – T03 Definition	$\forall \varepsilon > 0\ \exists \delta_\varepsilon > 0 :\ \|P\| < \delta_\varepsilon \Rightarrow \|S(f;\dot{P}) - L\| < \varepsilon$	partitions, tags (M02, M03)
T04 – T07 Basic laws	unique L; finite changes don't matter; linear and order-preserving	T03, triangle ineq. (P05)
T08 Boundedness	$f \in \mathcal{R}[a,b] \Rightarrow \|f\| \leq M$ (unbounded $\Rightarrow \notin \mathcal{R}$)	T03, reverse triangle (P05)
T09 Thomae	popcorn function $h \in \mathcal{R}[0,1]$ with $\int_0^1 h = 0$	T03, tag splitting (M10)
T10 – T11 Cauchy	$\|S(f;\dot{P}) - S(f;\dot{Q})\| < \varepsilon$; Dirichlet $\notin \mathcal{R}[0,1]$	T03, Cauchy seq. (P06)
T12 Squeeze	$\alpha_\varepsilon \leq f \leq \omega_\varepsilon$ with $\int(\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon$	T10, T07
T13 – T14 Step fns	$\int_a^b \varphi_J =$ d-c; $\int_a^b \sum k_j \varphi_{J_j} = \sum k_j(d_j - c_j)$	T03, T06, T07
T15 Continuous $\Rightarrow \mathcal{R}$	continuous on $[a,b] \Rightarrow$ integrable	T12, T14, UCT (P08), Max–Min (P07)
T16 Monotone $\Rightarrow \mathcal{R}$	monotone on $[a,b] \Rightarrow$ integrable (staircase telescopes)	T12, T14, M05

T17 – T19 Additivity	$\int_a^b f = \int_a^c f + \int_c^b f$; restrictions; oriented $\int_\alpha^\beta$	T10, T08, T03
T20 – T21 FTC 1	F cont., F'=f off finite E, $f \in \mathcal{R} \Rightarrow \int_a^b f = F(b) - F(a)$	T03, MVT (P12), M05
T22 – T24 FTC 2	$F(z) = \int_a^z f$ is Lipschitz; f cont. at $c \Rightarrow F'(c) = f(c)$	T17, T07
T25 Substitution	$\int_\alpha^\beta f(\varphi(t))\varphi'(t)\,dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx$	T20, T23, Chain Rule (P11)
T26 – T30 Lebesgue	$f \in \mathcal{R} \iff$ bounded & cont. a.e.; $\varphi \circ f$, $\lvert f\rvert$, fg $\in \mathcal{R}$	null sets (M07), T07, T08
T31 – T32 Parts & Taylor	$\int_a^b fG = [FG]_a^b - \int_a^b Fg$; Taylor integral remainder R_n	T20, T30, Product Rule (P11)

Codes and map

P01–P14	Plus Two, Real Analysis I, and Modules 1–2 bridge
M01–M10	Maths tools built from zero
T01–T32	Module 3 topics (Bartle §7.1–§7.3)
C7	a counterexample from the library
Q14	question 14 of the model paper

SYLLABUS UNIT	BARTLE	TOPICS
12. Riemann Integral	7.1.1–7.1.7	T01–T09
13. Riemann Integrable Functions	7.2.1–7.2.13	T10–T19
14. Fundamental Theorem, Substitution, Lebesgue, Taylor	7.3.1–7.3.18	T20–T32

T01 Partitions, subintervals and norm (7.1 eq. (1))
T02 Tags, tagged partitions and Riemann sums (7.1 eq. (2))
T03 Definition of the Riemann integral; $\mathcal{R}[a,b]$(7.1.1)
T04 Uniqueness of the integral (7.1.2)
T05 Integrals from the definition: k, step g, h(x)=x (7.1.4a–c)
T06 Changing a function at finitely many points (7.1.3, 7.1.4d)
T07 Linearity and monotonicity of the integral (7.1.5)
T08 Boundedness Theorem: $f \in \mathcal{R}[a,b] \Rightarrow$ bounded (7.1.6)
T09 Thomae's function is integrable with integral 0 (7.1.7)

T10 Cauchy Criterion for integrability (7.2.1)

T11 Dirichlet's function (and relatives) not integrable (7.2.2b)

T12 Squeeze Theorem for integrals (7.2.3, 7.2.6b)

T13 Elementary step functions φ_J(7.2.4)

T14 Step functions are integrable (7.2.5, 7.2.6a)

T15 Continuous functions on [a,b] are integrable (7.2.7)

T16 Monotone functions on [a,b] are integrable (7.2.8)

T17 Additivity Theorem on [a,c] and [c,b] (7.2.9)

T18 Restriction and subdivision corollaries (7.2.10, 7.2.11)

T19 Oriented integrals $\int_\alpha^\beta f$ for any α, β, γ(7.2.12, 7.2.13)

T20 Fundamental Theorem of Calculus, First Form (7.3.1)

T21 Examples and limits of FTC 1: when it works and fails (7.3.2)

T22 Indefinite integral $F(z) = \int_a^z f$ is Lipschitz (7.3.3, 7.3.4)

T23 Fundamental Theorem of Calculus, Second Form (7.3.5, 7.3.6)

T24 When $F(z) = \int_a^z f$ is not an antiderivative (7.3.7)

T25 Substitution Theorem and change of limits (7.3.8, 7.3.9)

T26 Null sets and "almost everywhere"; $\mathbb{Q} \cap [0,1]$ is null (7.3.10, 7.3.11)

T27 Lebesgue's Integrability Criterion and examples (7.3.12, 7.3.13)

T28 Composition Theorem: $\varphi \circ f \in \mathcal{R}[a,b]$(7.3.14)

T29 Absolute value: $|f| \in \mathcal{R}[a,b]$ and $|\int f| \leq \int |f|$(7.3.15)

T30 Product Theorem: f,g $\in \mathcal{R}[a,b] \Rightarrow$ fg $\in \mathcal{R}[a,b]$(7.3.16)

T31 Integration by Parts (7.3.17)

T32 Taylor's Theorem with the Integral Remainder (7.3.18)

The whole module in one story

THE STORY IN FIVE SENTENCES

1. A Riemann sum $S(f;\dot{P}) = \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1})$ approximates signed area using rectangles on a partition P with sample tags t_i, and $f \in \mathcal{R}[a,b]$ means all sums with mesh $\|P\| < \delta_\varepsilon$ cluster within ε of one number $L = \int_a^b f$ ($T01$–$T04$).

2. Every integrable function must be bounded (T08), may be altered at finitely many points without changing its integral (T06), and obeys linearity and order (T07), though direct ε–δ proofs work cleanly only for simple functions like constants, step functions, x, and Thomae's function (T05, T09).

3. The Cauchy Criterion (T10) and Squeeze Theorem (T12) eliminate the need to know L in advance: by trapping f between step functions (T13–T14), we prove that every continuous function (T15) and every monotone function (T16) on [a,b] is integrable, while Dirichlet's function is not (T11), and integrals split additively across subintervals (T17–T19).

4. The two forms of the Fundamental Theorem of Calculus connect $\mathcal{R}[a,b]$ to derivatives: FTC 1 evaluates $\int_a^b f = F(b) - F(a)$ via the Mean Value Theorem when F'=f off a finite set and $f \in \mathcal{R}[a,b]$ ($T20$–$T21$), while FTC 2 differentiates the Lipschitz indefinite integral $F(z) = \int_a^z f$ at any point where f is continuous (T22–T24), yielding the Substitution Theorem (T25).

5. Lebesgue's Criterion crowns the theory—a bounded function on [a,b] is Riemann integrable if and only if its discontinuities form a null set (T26–T27)—which immediately proves the Composition, Absolute Value, and Product Theorems (T28–T30), and hence Integration by Parts (T31) and Taylor's Theorem with Integral Remainder (T32).

The hypotheses to check every time

MASTER CHECKLIST ACROSS MODULE 3

☐ for $f \in \mathcal{R}[a,b]$: f must be **bounded** on [a,b] (Theorem 7.1.6)

1/x on (0,1] with f(0)=0 is unbounded, so $\notin \mathcal{R}[0,1]$ C16

☐ for FTC 1 (Theorem 7.3.1): F **continuous on [a,b]**, F'=f off a **finite** set E, and $f \in \mathcal{R}[a,b]$

$K(x) = x^2\cos(1/x^2)$ is differentiable everywhere, but K' is unbounded so $K' \notin \mathcal{R}[0,1]$ C19

☐ for FTC 2 (Theorem 7.3.5) : $f \in \mathcal{R}[a,b]$ and f is **continuous at c**

$f = \operatorname{sgn}$ jumps at 0, so $F(x) = |x| - 1$ has a corner at 0 C20

☐ for Substitution (Theorem 7.3.8) : φ' is **continuous on** $[\alpha,\beta]$ and f is **continuous on** $\varphi([\alpha,\beta])$

$\varphi(t) = \sqrt{t}$ on [0,4] has $\varphi'(t) = 1/(2\sqrt{t})$ blowing up at 0

☐ for Lebesgue's Criterion (Theorem 7.3.12): f is **bounded** and its discontinuity set is a **null set**

Dirichlet is bounded, but discontinuous on [0,1] (not null) C7

Index

Tap a line to jump there. Every topic heading has an **index** ↑ link back here, and every topic chip such as T5 in the text is a link too.

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PREREQUISITES · P01–P14

What you bring from earlier courses

P01 Plus Two integrals: area vs antiderivative

index ↑

> Two ideas you treated as the same in school; this module proves why they agree.

WHAT YOU ALREADY KNOW

In Plus Two you used $\int_a^b f(x)\,dx$ in two ways: (1) the signed area between y=f(x) and the x-axis on [a,b], and (2) F(b)-F(a) where F'(x)=f(x) (such as $\int x^n\,dx = \dfrac{x^{n+1}}{n+1}$ and $\int \sin x\,dx = -\cos x$).

In Real Analysis, **Definition 7.1.1** defines $\int_a^b f$ purely from rectangle sums (view 1), and the **Fundamental Theorem of Calculus** (Theorem 7.3.1) is the bridge proving that F(b)-F(a) gives the exact same number.

P02 Sigma (Σ) notation and linearity of finite sums

index ↑

> Every Riemann sum is a finite sum, so standard algebra applies without limits.

RULES FOR $\sum_{i=1}^{n}$

For any real numbers a_i, b_i and constant $c \in \mathbb{R}$:

$\sum_{i=1}^{n} c\,a_i = c\sum_{i=1}^{n} a_i, \qquad \sum_{i=1}^{n}(a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$

.

If $a_i \le b_i$ for each $i = 1, \ldots, n$, then $\sum_{i=1}^{n} a_i \le \sum_{i=1}^{n} b_i$. And summing a constant 1 over n terms gives $\sum_{i=1}^{n} 1 = n$.

P03 Standard finite and geometric sums

index ↑

> Three closed-form sums used in squeeze and null-set proofs.

FORMULAS TO KNOW COLD

Integers: $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$, so $\sum_{k=1}^{n-1} k = \frac{(n-1)n}{2}$.

Squares: $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

Halving geometric series: $\sum_{k=1}^{\infty} \frac{1}{2^k} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = \frac{1/2}{1 - 1/2} = 1$, so $\sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon$.

P04 Supremum, infimum and completeness of $\mathbb{R}$

index ↑

> Every bounded non-empty set of reals has a least upper bound $\sup$ and greatest lower bound $\inf$.

REAL ANALYSIS I REFRESHER

If $A \subseteq \mathbb{R}$ is non-empty and bounded above, $s = \sup A$ means: $(i) a \leq s$ for all $a \in A$, and (ii) whatever $\varepsilon > 0$ you are handed, there is some $a \in A$ with $a > s - \varepsilon$.

P05 Triangle inequality and the "arbitrary ε" rule

> The workhorse inequalities of every ε–δ proof.

THREE RULES USED IN ALMOST EVERY PROOF

1. Unpacking absolute value: $|a| < \varepsilon \iff -\varepsilon < a < \varepsilon$, and $|S - L| < \varepsilon \iff L - \varepsilon < S < L + \varepsilon$.

2. Triangle and reverse triangle inequalities: $|a + b| \leq |a| + |b|$ and $|a + b| \geq ||a| - |b|| \geq |a| |b|$.

3. Arbitrary ε rule: If $x \leq y + \varepsilon$ for **every** $\varepsilon > 0$, then $x \leq y$. In particular, if $|A - B| < \varepsilon$ for every $\varepsilon > 0$, then A = B. (Why? If $|A - B| > 0$, taking $\varepsilon = \frac{1}{2}|AB| > 0$ gives a contradiction.)

P06 Cauchy sequences in $\mathbb{R}$ converge (Bartle 3.5.5)

index ↑

How we prove a limit A exists without knowing A beforehand.

THEOREM 3.5.5 · CAUCHY CONVERGENCE CRITERION

A sequence (a_n) of real numbers converges to some real number $A \in \mathbb{R}$ if and only if for every $\varepsilon > 0$ there is $K \in \mathbb{N}$ such that $|a_m - a_n| < \varepsilon$ for all $m, n \geq K$.

In Theorem 7.2.1 we build a sequence of Riemann sums $a_m = S(f; \dot{P}_m)$ satisfying $|a_m a_n| < 1/n$ for all m > n; that immediately forces (a_m) to be Cauchy and therefore convergent to a real number A.

P07 Continuous on [a,b]: Boundedness, Max–Min, IVT

index ↑

Module 1 theorems on a closed bounded interval [a,b].

THEOREMS 5.3.2, 5.3.4, 5.3.7

If f : $[a, b] \to \mathbb{R}$ is continuous on the closed bounded interval [a,b], then:

- **Boundedness (5.3.2):** there is M > 0 with $|f(x)| \leq M$ for all $x \in [a, b]$.
- **Maximum–Minimum (5.3.4):** f attains its minimum and maximum on [a,b] (and on every closed subinterval $I_i = [x_{i-1}, x_i]$): there exist $u_i, v_i \in I_i$ with $f(u_i) \leq f(x) \leq f(v_i)$ for all $x \in I_i$.
- **Bolzano's IVT (5.3.7):** f takes every value between $\min_{[a,b]} f$ and $\max_{[a,b]} f$.

P08 Uniform Continuity Theorem (Bartle 5.4.3)

index ↑

One δ works across the whole closed bounded interval [a,b].

THEOREM 5.4.3 · UNIFORM CONTINUITY THEOREM

If f : $[a,b] \to \mathbb{R}$ is continuous on the closed bounded interval [a,b], then f is **uniformly continuous** on [a,b]: whatever $\varepsilon > 0$ you are handed, there is a single $\delta > 0$ such that for **all** $u, v \in [a,b]$ with $|uv| < \delta$, we have $|f(u) - f(v)| < \varepsilon$.

In Theorem 7.2.7 we apply this with $\varepsilon/(b-a)$ in place of ε, so that on every subinterval of length $< \delta$ the gap $\max f - \min f$ is Simultaneously less than $\varepsilon/(b-a)$.

P09 Step functions and Lipschitz functions (Bartle 5.4.9, 5.4.4)

index ↑

> Two special classes of functions from Module 1.

DEFINITIONS 5.4.9 AND 5.4.4

Step function (5.4.9): $\varphi : [a,b] \to \mathbb{R}$ is a step function if [a,b] is the union of finitely many non-overlapping intervals $J_1, \ldots, J_m$ (open, closed, or half-open) on each of which φ is constant.

Lipschitz function (5.4.4): F : $[a,b] \to \mathbb{R}$ is Lipschitz if there is M > 0 such that $|F(z) - F(w)| \leq M|z - w|$ for all $z, w \in [a,b]$. Every Lipschitz function is uniformly continuous (and hence continuous): take $\delta = \varepsilon/M$.

P10 Monotone functions and their jumps (Bartle 5.6.4)

index ↑

> Increasing or decreasing functions can jump, but only countably many times.

MONOTONE FUNCTIONS

f : $[a, b] \to \mathbb{R}$ is **increasing** if $x_1 \le x_2 \Rightarrow f(x_1) \le f(x_2)$, and **decreasing** if $x_1 \le x_2 \Rightarrow f(x_1) \ge f(x_2)$. On any subinterval $[x_{k-1}, x_k]$, an increasing function automatically has its minimum at the left endpoint x_{k-1} and its maximum at the right endpoint x_k. By Theorem 5.6.4 (cited in 7.3.13(b)), the set of discontinuities of a monotone function is at most countable.

P11 Derivative rules: continuity, product, chain (Bartle §6.1)

index ↑

Three differentiation facts from Module 2.

THEOREMS 6.1.2, 6.1.3(C), 6.1.6

- **Differentiable ⇒ continuous (6.1.2):** If F is differentiable at c, then F is continuous at c.
- **Product Rule (6.1.3c):** (FG)' = F'G + FG'. (Reversed in Theorem 7.3.17 to give Integration by Parts.)
- **Chain Rule (6.1.6):** If $H(t) = F(\varphi(t))$, then $H'(t) = F'(\varphi(t))\,\varphi'(t)$. (Used in Theorem 7.3.8 for Substitution.)

P12 Mean Value Theorem (Bartle 6.2.4)

index ↑

The single point where Module 2 enters the proof of the Fundamental Theorem of Calculus.

THEOREM 6.2.4 · MEAN VALUE THEOREM

If F is continuous on the closed interval $[x_{i-1}, x_i]$ and differentiable on the open interval (x_{i-1}, x_i), then there exists a point $u_i \in (x_{i-1}, x_i)$ such that

$$F(x_i) - F(x_{i-1}) = F'(u_i)(x_i - x_{i-1}).$$

Notice that $F'(u_i)(x_i - x_{i-1})$ has the exact shape of one term of a Riemann sum with tag u_i!

P13 Countable sets and the Archimedean property

index ↑

Listing a set in a sequence $r_1, r_2, r_3, \ldots$ and choosing $n > 1/\varepsilon$.

COUNTABILITY AND ARCHIMEDES

Archimedean Property: Given any $\varepsilon > 0$, there is $n \in \mathbb{N}$ with $n > 1/\varepsilon$, i.e. 1/n < ε.

Countable set: A set whose elements can be listed in a sequence $r_1, r_2, r_3, \ldots$ (finite or countably infinite). In particular, the rational numbers $\mathbb{Q}$ and $\mathbb{Q} \cap [0,1]$ are countable.

P14 Dirichlet and Thomae functions (Bartle 5.1.6(g), (h))

index ↑

> Two benchmark functions from Module 1 that test where Riemann integration succeeds and fails.

EXAMPLES 5.1.6(G) AND 5.1.6(H)

Dirichlet's function C7 **:** f(x) = 1 if $x \in \mathbb{Q}$, $f(x) = 0$ if $x \notin \mathbb{Q}$. Discontinuous at **every** point of [0,1] because both rationals and irrationals are dense in $\mathbb{R}$.

Thomae's function C8 **:** On [0,1], h(x) = 0 if x is irrational, h(0) = 1, and h(m/n) = 1/n if $m, n \in \mathbb{N}$ have no common integer factor except 1. Continuous at every irrational in [0,1] and discontinuous at every rational in [0,1].

MATHS TOOLS · M01–M10

The ten tools built from zero

M01 Telescoping sums: why middle terms vanish

index ↑

> When you add consecutive differences $(a_i - a_{i-1})$, only the first and last survive.

FROM ZERO · WRITE OUT FOUR TERMS

Suppose you have numbers $a_0, a_1, a_2, \ldots, a_n$ and you add the differences $(a_i - a_{i-1})$:

$\sum_{i=1}^{n}(a_i - a_{i-1}) = (\cancel{a_1} - a_0) + (\cancel{a_2} - \cancel{a_1}) + (\cancel{a_3} - \cancel{a_2}) + \cdots + (a_n - \cancel{a_{n-1}}) = a_n - a_0$.

Three places we use this in Module 3:

- **Lengths of subintervals:** If $a = x_0 < x_1 < \cdots < x_n = b$, then $\sum_{i=1}^{n}(x_i - x_{i-1}) = x_n x_0 = b - a$. The pieces of a partition always add up to the total length of [a,b].
- **Squares (Example 7.1.4(c)):** $\sum_{i=1}^{n} \frac{1}{2}(x_i^2 - x_{i-1}^2) = \frac{1}{2}(x_n^2 - x_0^2) = \frac{1}{2}(b^2 - a^2)$.
- **Cubes (Exercise 7.1.14):** $\sum_{i=1}^{n} \frac{1}{3}(x_i^3 - x_{i-1}^3) = \frac{1}{3}(b^3 - a^3)$.

M02 Partitions and mesh: why $\|P\| \to 0$ is stronger than $n \to \infty$

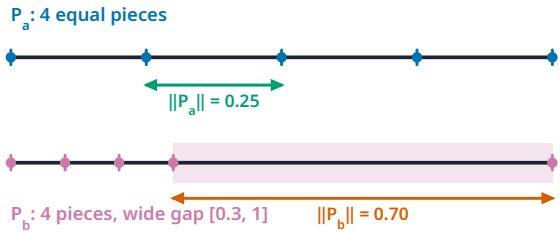
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> The norm $\|P\|$ is the length of the single widest subinterval, not the number of pieces.

HOW TO FIND THE NORM OF A PARTITION

Given $P = (x_0, x_1, \ldots, x_n)$ on [a,b], list the n consecutive gaps $\Delta x_i = x_i - x_{i-1}$ and pick the **largest** one: $\|P\| = \max\{\Delta x_1, \ldots, \Delta x_n\}$.

For an **equal partition** into n pieces of equal width, every gap is (b-a)/n, so $\|P\| = (b-a)/n$. Here $n \to \infty$ makes $\|P\| \to 0$. For an **unequal partition**, having many pieces does NOT force $\|P\|$ to be small!



What to notice: Both $P_a = (0, 0.25, 0.5, 0.75, 1)$ and $P_b = (0, 0.1, 0.2, 0.3, 1)$ cut [0,1] into n=4 pieces, but $\|P_a\| = 0.25$ while $\|P_b\| = 0.7$ because of the wide right-hand piece [0.3,1]. **What it does not prove:** Sending $n \to \infty$ alone never proves integrability unless the widest gap $\|P\| \to 0$.

M03 Tags and endpoints: one point can tag two intervals

index ↑

A tag t_i can be any point in $[x_{i-1}, x_i]$, including the left or right endpoint.

WHY "AT MOST 2N SUBINTERVALS" APPEARS IN PROOFS

In each subinterval $I_i = [x_{i-1}, x_i]$ we pick one tag $t_i \in [x_{i-1}, x_i]$. What if a special point c happens to be the partition point x_k between $I_k = [x_{k-1}, x_k]$ and $I_{k+1} = [x_k, x_{k+1}]$? Then an opponent could legally pick $t_k = c$ as the right-endpoint tag of I_k **and** $t_{k+1} = c$ as the left-endpoint tag of I_{k+1}.

So a single point c can be the tag of **at most 2** subintervals, and a finite set E of n points can be the tag of **at most 2n** subintervals.

M04 Bounding a sum by the largest term times total length

How to turn $|\sum_{i=1}^{n}(f(t_i) - g(t_i))\Delta x_i|$ into a single simple product.

TWO STANDARD SUM ESTIMATES

Since subinterval lengths $\Delta x_i = x_i - x_{i-1}$ are always positive, the triangle inequality gives

$|\sum_{i=1}^{n} a_i\,\Delta x_i| \le \sum_{i=1}^{n} |a_i|\,\Delta x_i \le (\max_{1\le i\le n}|a_i|)\sum_{i=1}^{n}\Delta x_i = (\max_{1\le i\le n}|a_i|)(b-a)$.

Case 1 (small height everywhere): If $|a_i| < \varepsilon/(b-a)$ on every piece, the whole sum is $< \dfrac{\varepsilon}{b-a}(b-a) = \varepsilon$.

Case 2 (one point c changed, Theorem 7.1.3): If f(x) = g(x) for all $x \neq c$, then $f(t_i) - g(t_i) = 0$ except on the (at most 2) subintervals whose tag is c. Each has width $\Delta x_i \le \|P\|$ and height $|f(c) - g(c)| \le |f(c)| + |g(c)|$, so $|S(f;\dot{P}) - S(g;\dot{P})| \le 2(|f(c)| + |g(c)|)\|P\|$.

M05 Telescoping with monotonicity and with FTC

index ↑

Two places where telescoping turns n rectangles into f(b)-f(a) or F(b)-F(a).

STAIRCASE CANCELLATION AND MVT CANCELLATION

1. Monotone staircase (Theorem 7.2.8): On n equal pieces of width (b-a)/n, an increasing function f has $\max_{I_k} f = f(x_k)$ and $\min_{I_k} f = f(x_{k-1})$. The difference between upper and lower step integrals is

$\dfrac{b-a}{n}\sum_{k=1}^{n}(f(x_k) - f(x_{k-1})) = \dfrac{b-a}{n}(f(b) - f(a))$.

2. FTC First Form (Theorem 7.3.1): Applying the MVT to F on each $[x_{i-1}, x_i]$ gives $F(x_i) - F(x_{i-1}) = f(u_i)\Delta x_i$. Summing over $i = 1, \ldots, n$ telescopes the left side to F(b) - F(a) and turns the right side into $S(f;\dot{P}_u)$.

M06 Elementary step functions φ_J as building blocks

index ↑

Write any step function as $\sum_{j=1}^{m} k_j \varphi_{J_j}$ and read off its integral immediately.

INDICATOR RECIPE

For any subinterval $J \subseteq [a,b]$ with endpoints c < d, the **elementary step function** $\varphi_J(x)$ equals 1 on J and 0 outside J. Its integral is simply the length of J : $\int_a^b \varphi_J = d - c$ (Lemma 7.2.4).

If φ equals k_1 on J_1, k_2 on $J_2, \ldots, k_m$ on J_m (where $J_1, \ldots, J_m$ partition [a,b]), then $\varphi(x) = \sum_{j=1}^m k_j \varphi_{J_j}(x)$, so $\int_a^b \varphi = \sum_{j=1}^m k_j(d_j - c_j)$: **sum of (height $\times$ width)**.

M07 The $\varepsilon/2^k$ covering trick for countable sets

index ↑

How an infinite sequence of points can be covered by intervals of total length $\leq \varepsilon$.

WHY COUNTABLE SETS HAVE ZERO TOTAL LENGTH

Suppose $Z = \{r_1, r_2, r_3, \ldots\}$ is countable. If we gave every point r_k an interval of the same length δ, the total length $\delta + \delta + \cdots$ would be ∞. Instead, we shrink the interval width by half at each step!

Put r_k inside the open interval $J_k = (r_k \frac{\varepsilon}{2^{k+1}},\ r_k + \frac{\varepsilon}{2^{k+1}})$. The length of J_k is $2 \cdot \frac{\varepsilon}{2^{k+1}} = \frac{\varepsilon}{2^k}$. By P03, the total length of all infinitely many intervals is $\sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon$.

M08 The $\varepsilon/2 (or \varepsilon/3)$ argument and $\delta = \min\{\delta_1, \delta_2\}$

index ↑

How to combine two or three separate error estimates at the same time.

COMBINING TOLERANCES

When a proof splits an error into two parts, $|A - B| \leq \text{Error}_1 + \text{Error}_2$, ask for each part to be $< \varepsilon/2$ (*or* $< \varepsilon/3$ if there are three parts, as in the Additivity Theorem 7.2.9). If $\text{Error}_1 < \varepsilon/2$ needs $\|P\| < \delta_1$ and $\text{Error}_2 < \varepsilon/2$ needs $\|P\| < \delta_2$, choose

$$\delta = \min\{\delta_1, \delta_2\} > 0.$$

Then any partition with $\|P\| < \delta$ satisfies **both** $\|P\| < \delta_1$ and $\|P\| < \delta_2$ simultaneously, so $\text{Error}_1 + \text{Error}_2 < \varepsilon/2 + \varepsilon/2 = \varepsilon$.

M09 Choosing δ_ε explicitly from $C\delta < \varepsilon$

index ↑

Do the algebra in scratch work first so your proof starts with a clean choice of δ_ε.

BACKWARDS IN SCRATCH, FORWARDS IN PROOF

If your scratch work shows that $\|P\| < \delta$ implies $|S(f;\dot{P}) - L| \leq C\delta$ (*for* instance $|S(g;\dot{P}) - 8| \leq 5\delta$ in Example 7.1.4(b)), then to force $C\delta < \varepsilon$ you simply pick any $\delta_\varepsilon < \varepsilon/C$ (such as $\delta_\varepsilon = \varepsilon/(2C)$, e.g. $\varepsilon/10$). If your estimate isalready strict ($|S - L| < \delta$), then $\delta_\varepsilon = \varepsilon$ works directly.

M10 Splitting a Riemann sum by where the tags land

index ↑

How Bartle handles jumps and popcorn functions (Examples 7.1.4(b),(d), 7.1.7; Exercises 7.1.4–5).

TWO WAYS TO SPLIT $\dot{P} = \dot{P}_1 \cup \dot{P}_2$

1. Tag in a subinterval $[c_1, c_2]$ (Exercises 7.1.4–5): Suppose $\|P\| < \delta$ and $\dot{P}_1$ consists of the subintervals I_i whose tag t_i lands in $[c_1, c_2]$. Because each I_i has width $< \delta$, no such interval can stick out past $c_1 - \delta$ or $c_2 + \delta$, and every point of $[c_1 + \delta, c_2 - \delta]$ must lie in one of them. Thus the union U_1 of intervals in $\dot{P}_1$ is trapped between:

$$[c_1 + \delta,\ c_2 - \delta] \subseteq U_1 \subseteq [c_1 - \delta,\ c_2 + \delta],$$

so the total width of $\dot{P}_1$ lies between $(c_2 - c_1) - 2\delta$ and $(c_2 - c_1) + 2\delta$ (or between $1 - \delta$ and $1 + \delta$ when $c_1 = a = 0$ is the endpoint of [0,3]).

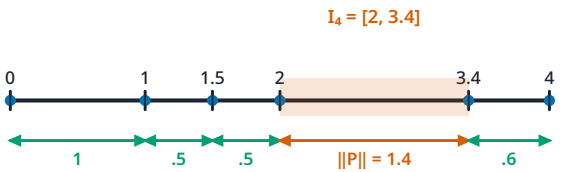
2. Tag in a finite "bad" set $E = \{x : f(x) \geq \varepsilon/2\}$ (Example 7.1.7): Put $\dot{P}_1 =$ subintervals with tag $t_i \in E$, and $\dot{P}_2 =$ subintervals with tag $t_i \notin E$. If E has n points, $\dot{P}_1$ has at most 2n subintervals (by M03) of total width $\leq 2n\|P\|$, while on $\dot{P}_2$ the height is $< \varepsilon/2$ and total width is $\leq$ b-a.

The Riemann Integral

T01 Partitions, subintervals and the norm

index ↑

> Cutting [a,b] into pieces and measuring the widest piece (PYQ MTS-26 Q4, MAT-22 Q3, MAT-21 Q5, MAT-22 MCQ 16).



What to notice: For P=(0,1,1.5,2,3.4,4) on [0,4], the five subinterval widths are 1, 0.5, 0.5, 1.4, 0.6; the norm $\|P\| = 1.4$ is the single largest gap $I_4 = [2, 3.4]$. **What it does not prove:** Knowing $\|P\|$ does not tell you how many subintervals P has or where the other cut points sit.

DEFINITION · PARTITION AND NORM (BARTLE §7.1, P. 199–200, EQ. (1))

If I := [a,b] is a closed bounded interval in $\mathbb{R}$, then a **partition** of I is a finite, ordered set $P := (x_0, x_1, \ldots, x_{n-1}, x_n)$ of points in I such that $a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$. The points of P are used to divide I = [a,b] into non-overlapping subintervals $I_1 := [x_0, x_1], \quad I_2 := [x_1, x_2], \quad \ldots, \quad I_n := [x_{n-1}, x_n]$. We define the **norm** (or **mesh**) of P to be the number $\|P\| := \max\{x_1 - x_0,\ x_2 - x_1,\ \ldots,\ x_n - x_{n-1}\}$.

In plain words: a partition is a strictly increasing list of cut points starting at a and ending at b; its norm $\|P\|$ is the length of the widest piece.

SKELETON · FIND THE NORM OF A GIVEN PARTITION

List the subinterval lengths $\Delta x_i = x_i - x_{i-1}$ for $i = 1, \ldots, n$: ? . Therefore $\|P\| = \max\{$? $\} =$? .

WORKED EXAMPLE · BARTLE EXERCISE 7.1.1 & PAST PAPERS

1. On I = [0, 4] (Bartle Ex 7.1.1; PYQ MTS-26 Q4, MAT-22 Q3):

- *P*1 = (0, 1, 2, 4): gaps are 1 − 0 = 1, 2 − 1 = 1, 4 − 2 = 2, so $\|P_1\| = \max\{1, 1, 2\} = 2$.
- *P*2 = (0, 2, 3, 4): gaps are 2, 1, 1, so $\|P_2\| = \max\{2, 1, 1\} = 2$.
- *P*3 = (0, 1, 1. 5, 2, 3. 4, 4): gaps are 1, 0. 5, 0. 5, 1. 4, 0. 6, so $\|P_3\| = 1.4$.
- *P*4 = (0, 0. 5, 2. 5, 3. 5, 4): gaps are 0. 5, 2, 1, 0. 5, so $\|P_4\| = 2$.

2. Non-example: (0, 2, 1, 4) is **not** a partition of [0, 4] (not strictly increasing: $2 \not< 1$), and (0, 1, 2, 3) is **not** a partition of [0, 4] because *x*3 = 3 ≠ 4 = *b*.

CHECK YOURSELF · PYQ MAT-22 MCQ 16

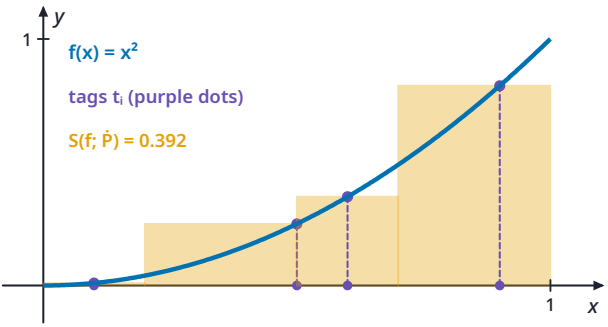
On I = [2, 9], which partition has the largest norm: *P*1 = (2, 4, 6, 8, 9), *P*2 = (2, 6, 9), or *P*3 = (2, 5, 7, 9)?

$\|P_1\| = \max 2, 2, 2, 1 = 2; \|P_2\| = \max\{4, 3\} = 4; \|P_3\| = \max\{3, 2, 2\} = 3$. So P_2 has the largest norm (4).

In the model paper: **Q1** (PYQ MTS-26 Q4, MAT-22 Q3, MAT-21 Q5, MAT-22 MCQ 16).

T02 Tags, tagged partitions and Riemann sums

> Picking a sample point t_i in each subinterval and adding the rectangle areas (PYQ MTS-22 Q4).



What to notice: For $f(x) = x^2$ on [0,1] with P=(0,0.2,0.5,0.7,1) and tags t=(0.1,0.5,0.6,0.9), each rectangle's top touches the curve at $(t_i, f(t_i))$, giving $S(f;\dot{P}) = 0.002 + 0.075 + 0.072 + 0.243 = 0.392$. **What it does not prove:** A single Riemann sum 0.392 is only an approximation, not the exact integral 1/3.

DEFINITION · TAGGED PARTITION AND RIEMANN SUM (BARTLE §7.1, P. 200, EQ. (2))

If a point t_i has been selected from each subinterval $I_i = [x_{i-1}, x_i]$, for $i = 1, 2, \ldots, n$, then the points are called **tags** of the subintervals I_i.A set of ordered pairs $\dot{P} := \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}$ of subintervals and corresponding tags is called a **tagged partition** of I; the dot over the P indicates that a tag has been chosen for each subinterval. The **norm** of a tagged partition is defined to be the same as for the corresponding ordinary partition and does not depend on the choice of tags. If f : $[a, b] \to \mathbb{R}$ is a function defined on [a,b], then the **Riemann sum** of f corresponding to the tagged partition $\dot{P}$ is the number $S(f;\dot{P}) := \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1})$.

In plain words: in each piece $[x_{i-1}, x_i]$, evaluate f at its chosen tag t_i, multiply by the piece's width $(x_i - x_{i-1})$, and add up all n products.

SKELETON · COMPUTE A RIEMANN SUM $S(f;\dot{P})$

List the subintervals $I_i = [x_{i-1}, x_i]$, widths $\Delta x_i = x_i - x_{i-1}$, and tags $t_i \in I_i$. Compute $f(t_i)$ for each i. Then $S(f;\dot{P}) = \sum_{i=1}^{n} f(t_i)\Delta x_i =$? .

WORKED EXAMPLE · BARTLE EXERCISE 7.1.2

Let $f(x) = x^2$ on [0,4]. We compute $S(f; \dot{P})$ for $P_1 = (0, 1, 2, 4)$ (widths 1, 1, 2) and $P_2 = (0, 2, 3, 4)$ (widths 2, 1, 1):

- $(a) P_1$ **with left endpoints** $t = (0, 1, 2) : S = f(0) \cdot 1 + f(1) \cdot 1 + f(2) \cdot 2 = 0(1) + 1(1) + 4(2) = 9.$
- $(b) P_1$ **with right endpoints** $t = (1, 2, 4) : S = f(1) \cdot 1 + f(2) \cdot 1 + f(4) \cdot 2 = 1(1) + 4(1) + 16(2) = 37.$
- $(c) P_2$ **with left endpoints** t = (0,2,3): S = 0(2) + 4(1) + 9(1) = 13.
- $(d) P_2$ **with right endpoints** t = (2,3,4): S = 4(2) + 9(1) + 16(1) = 33.

Non-example: Taking $t_1 = 1.5$ for $I_1 = [0, 1]$ is **not** a tagged partition because the tag t_1 must belong to its own subinterval [0,1].

YOUR INTUITION WILL LIE TO YOU HERE

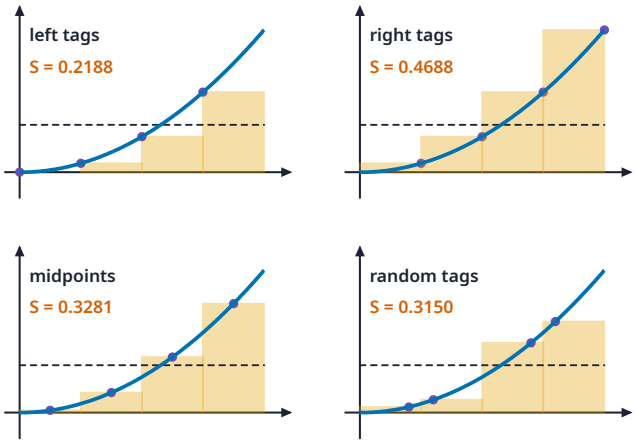
Trap 1: Writing $f(x_i)$ in the definition of a Riemann sum. The definition requires $f(t_i)$ for an **arbitrary** tag $t_i \in [x_{i-1}, x_i]$, not just the right endpoint x_i.

Trap 2: Thinking $S(f; \dot{P})$ is always positive. Where $f(t_i) <$ 0, the term $f(t_i)(x_i - x_{i-1})$ is negative (signed area).

In the model paper: **Q2** (PYQ MTS-22 Q4).

T03 Definition of the Riemann integral ($\mathcal{R}[a, b]$)

> The ε–δ_ε definition that governs the entire module (PYQ MAT-21 MCQ 9).



What to notice: On the same partition P=(0,0.25,0.5,0.75,1) of [0,1] for $f(x) = x^2$, left tags give 0.21875, right tags give 0.46875, midpoints give 0.328125, and random tags (0.2,0.3,0.7,0.8) give 0.315—all clustering near L=1/3. **What it does not prove:** Checking four tag choices on one partition does not prove $f \in \mathcal{R}[0,1]$; every tag choice on every partition with $\|P\| < \delta_\varepsilon$ must lie in $(L-\varepsilon, L+\varepsilon)$.

DEFINITION 7.1.1 · RIEMANN INTEGRABLE (BARTLE P. 201) · SAY IT WORD FOR WORD

A function f : $[a,b] \to \mathbb{R}$ is said to be **Riemann integrable** on [a,b] if there exists a number $L \in \mathbb{R}$ such that for every $\varepsilon > 0$ there exists $\delta_\varepsilon > 0$ such that if $\dot{P}$ is any tagged partition of [a,b] with $\|P\| < \delta_\varepsilon$, then $|S(f;\dot{P}) - L| < \varepsilon$. The set of all Riemann integrable functions on [a,b] is denoted by $\mathcal{R}[a,b]$. We write $L = \int_a^b f$ or $\int_a^b f(x)\,dx$.

In plain words: there is one target number L such that, whatever error tolerance $\varepsilon > 0$ you are handed, you can find a mesh bound $\delta_\varepsilon > 0$ forcing every Riemann sum with widest piece $< \delta_\varepsilon$ (*for* any choice of cut points and any choice of tags) to land within ε of L.

SKELETON · PROVE $f \in \mathcal{R}[a,b]$ WITH $\int_a^b f = L$ DIRECTLY FROM DEFINITION 7.1.1

Let $\varepsilon > 0$ be given. Choose $\delta_\varepsilon =$? > 0. Let $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be any tagged partition of [a,b] with $\|P\| < \delta_\varepsilon$. Then $|S(f;\dot{P}) - L| \le$? $< \varepsilon$. Therefore $f \in \mathcal{R}[a,b]$ and $\int_a^b f = L$. ∎

REMARK 7.1.1 & EXERCISE 7.1.3 · WHY THIS IS NOT AN ORDINARY LIMIT

Why $\lim_{\|P\|\to 0} S(f;\dot{P})$ is not an ordinary function limit: For a single number $\delta = \|P\|$, there are infinitely many partitions P having that exact norm and infinitely many tag choices t_i for each P. Thus $S(f;\dot{P})$ is **not a single-valued function of** $\|P\|$.

Strict < versus $\leq$ (Bartle Ex 7.1.3): Replacing $\|P\| < \delta_\varepsilon$ and $|S(f;\dot{P}) - L| < \varepsilon$ by $\leq$ gives an equivalent definition (given ε, apply the $\leq -version$ with $\varepsilon/2$ and take $\delta'_\varepsilon = \frac{1}{2}\delta_{\varepsilon/2}$).

Worked example & Non-example: f(x)=5 on [0,2] is in $\mathcal{R}[0,2]$ with L=10 since $S(f;\dot{P}) = 10$ for every $\dot{P}(any\delta_\varepsilon > 0$ works). By contrast, Dirichlet's function C7 on [0,1] is **not** in $\mathcal{R}[0,1]$ because on every partition, rational tags give S=1 and irrational tags give S=0, so no single number L can be within $\varepsilon = 1/2$ of both.

In the model paper: Q3 (PYQ MAT-21 MCQ 9).

T04 Uniqueness of the integral

A function in $\mathcal{R}[a,b]$ cannot have two different integrals (Bartle Theorem 7.1.2).

THEOREM 7.1.2 · UNIQUENESS OF THE INTEGRAL (BARTLE P. 201)

If $f \in \mathcal{R}[a,b]$, then the value of the integral is uniquely determined.

In plain words: if L' and L'' both satisfy Definition 7.1.1 for the same f on [a,b], then L' = L''.

HYPOTHESES · THEOREM 7.1.2

☐ $f \in \mathcal{R}[a,b](so\delta_\varepsilon$ works for **all** tagged partitions of norm $< \delta_\varepsilon)$

if you only require convergence along one chosen sequence of tagged partitions $(\dot{P}_n)$, different tag sequences can converge to different limits: for Dirichlet C7, rational tags give 1 while irrational tags give 0

Know L' and L'' both satisfy Definition 7.1.1 for f on [a,b].
Want L' = L'', by showing $|L' - L''| < \varepsilon$ for every $\varepsilon > 0$.

SCRATCH · HOW WE LINK L' AND L''

Add and subtract the same Riemann sum $S(f;\dot{P})$: $|L' - L''| \le |L' - S(f;\dot{P})| + |S(f;\dot{P}) L''|$. Make each piece $< \varepsilon/2$ by taking $\delta_\varepsilon = \min\{\delta'_{\varepsilon/2}, \delta''_{\varepsilon/2}\}$.

PROOF OF THEOREM 7.1.2 (BARTLE P. 201)

1. Assume that L' and L'' both satisfy Definition 7.1.1 and let $\varepsilon > 0$.
   To prove L' = L'', we will show $|L' - L''| < \varepsilon$ for arbitrary $\varepsilon > 0 (P05)$.

2. There exists $\delta'_{\varepsilon/2} > 0$ such that if $\dot{P}_1$ is any tagged partition with $\|P_1\| < \delta'_{\varepsilon/2}$, then $|S(f;\dot{P}_1) - L'| < \varepsilon/2$.
   Definition 7.1.1 applied to L' with tolerance $\varepsilon/2 > 0$.

3. Also there exists $\delta''_{\varepsilon/2} > 0$ such that if $\dot{P}_2$ is any tagged partition with $\|P_2\| < \delta''_{\varepsilon/2}$, then $|S(f;\dot{P}_2) - L''| < \varepsilon/2$.
   Definition 7.1.1 applied to L'' with tolerance $\varepsilon/2 > 0$.

4. Now let $\delta_\varepsilon := \min\{\delta'_{\varepsilon/2}, \delta''_{\varepsilon/2}\} > 0$ and let $\dot{P}$ be a tagged partition with $\|P\| < \delta_\varepsilon$.
   Minimum of two positive numbers is positive (M08); an equal partition with $n > (b-a)/\delta_\varepsilon$ guarantees such $a\dot{P}$ exists.

5. Since both $\|P\| < \delta'_{\varepsilon/2}$ and $\|P\| < \delta''_{\varepsilon/2}$, the Triangle Inequality gives $|L' - L''| = |L' - S(f;\dot{P}) + S(f;\dot{P}) - L''| \le |L' - S(f;\dot{P})| + |S(f;\dot{P}) - L''| < \varepsilon/2 + \varepsilon/2 = \varepsilon$.
   Steps 2 and 3 both apply to the same $\dot{P}$; Triangle Inequality (P05).

6. Since $\varepsilon > 0$ is arbitrary, it follows that L' = L''. ∎
   Arbitrary ε rule (P05): a non-negative number $|L' - L''| < \varepsilon$ for all $\varepsilon > 0$ must be 0.

WHERE THE HYPOTHESIS WAS USED & WORKED APPLICATION (EX 7.1.9)

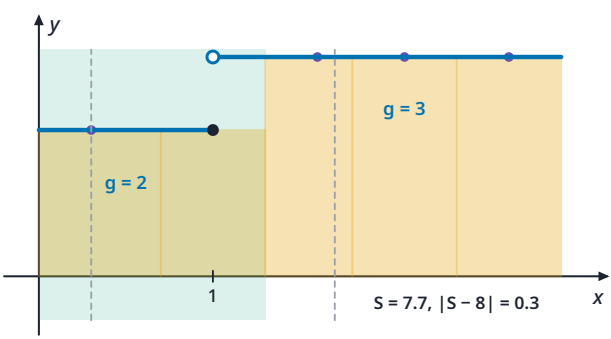
Where used: $f \in \mathcal{R}[a,b]$ guarantees that **one** partition $\dot{P}$ with $\|P\| < \delta_\varepsilon$ satisfies the closeness bound for L' and L'' simultaneously.

Worked example (Bartle Ex 7.1.9): If $f \in \mathcal{R}[a,b]$ and $(\dot{P}_n)$ is any sequence of tagged partitions with $\|P_n\| \to 0$, then $\lim_{n\to\infty} S(f;\dot{P}_n) = \int_a^b f$. Indeed, given $\varepsilon > 0$, take δ_ε from Definition 7.1.1; since $\|P_n\| \to 0$, there is $K \in \mathbb{N}$ with $\|P_n\| < \delta_\varepsilon$ for all $n \geq K$, so $|S(f;\dot{P}_n) - \int_a^b f| < \varepsilon$ for all $n \geq K$. By uniqueness of limits, $\lim S(f;\dot{P}_n)$ must equal the unique integral $\int_a^b f$.

In the model paper: **Q4** .

T05 Integrals straight from the definition: k, step g, and h(x)=x

> Three foundational calculations directly from Definition 7.1.1 (Bartle 7.1.4(a)–(c); PYQ MTS-25 Q4).



What to notice: For g(x)=2 on [0,1] and 3 on (1,3] with $\|P\| < \delta$, the subintervals with tag ≤ 1 cover $[0, 1-\delta]$ and lie inside $[0, 1+\delta]$, so the sum can only err in the $\delta - strip$ around the jump x=1 by at most 5δ. **What it does not prove:** The picture shows one δ; the proof verifies $|S(g;\dot{P}) - 8| \leq 5\delta < \varepsilon$ for every $\varepsilon > 0$ via $\delta_\varepsilon = \varepsilon/10$.

EXAMPLES 7.1.4(A), (B), (C) (BARTLE PP. 202–204)

(a) Constant function: Every constant function f(x) := k on [a,b] is in $\mathcal{R}[a,b]$ and $\int_a^b k = k(b-a)$.

(b) Two-piece step function: g : $[0,3] \to \mathbb{R}$ defined by g(x) := 2 for $0 \le x \le 1$, and g(x) := 3 for 1 < $x \le 3$ is in $\mathcal{R}[0,3]$ and $\int_0^3 g = 8$.

(c) Identity function: h(x) := x on [0,1] is in $\mathcal{R}[0,1]$ and $\int_0^1 x\,dx = \frac{1}{2}$.

In plain words: for a constant, every Riemann sum is exact; for a step function, the error comes only from the subinterval(s) straddling the jump; for x, midpoint tags telescope to 1/2 and any other tags differ by less than δ.

PROOF OF 7.1.4(A) · CONSTANT FUNCTION (PYQ MTS-25 Q4)

1. Let f(x) := k for all $x \in [a,b]$. For any tagged partition $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ of $[a,b]$, $S(f;\dot{P}) = \sum_{i=1}^n f(t_i)(x_i - x_{i-1}) = \sum_{i=1}^n k(x_i - x_{i-1}) = k\sum_{i=1}^n (x_i - x_{i-1}) = k(b-a)$.
   Factor out k (P02) and telescope $\sum(x_i - x_{i-1}) = x_n - x_0 = b - a (M01)$.

2. Hence for any $\varepsilon > 0$, we can choose $\delta_\varepsilon := 1$ so that if $\|P\|$ < δ_ε, then $|S(f;\dot{P}) k(b-a)| = 0$ < ε. Since $\varepsilon > 0$ is arbitrary, $f \in \mathcal{R}[a,b]$ and $\int_a^b k = k(b-a)$. ∎
   Definition 7.1.1 satisfied with L = k(b-a).

PROOF OF 7.1.4(B) · STEP FUNCTION G(X)=2 ON [0,1], 3 ON (1,3]

1. Let $\dot{P}$ be a tagged partition of [0,3] with $\|P\| < \delta$. Split $\dot{P}$ into $\dot{P}_1$ (tags $t_i \in [0,1]$, where g=2) and $\dot{P}_2$ (tags $t_i \in (1,3]$, where g=3), so $S(g;\dot{P}) = S(g;\dot{P}_1) + S(g;\dot{P}_2)$.
   Tag-splitting tool (M10).

2. Let U_1 be the union of all subintervals in $\dot{P}_1$. Since $\|P\| < \delta$, we have $[0, 1-\delta] \subseteq U_1 \subseteq [0, 1+\delta]$.
   If $u \in [0,1-\delta]$ lies in I_k, then $x_k \le u + \delta \le 1$, so $t_k \le 1$ and $I_k \in \dot{P}_1$; conversely if $t_k \le 1$, then $x_k \le t_k + \delta \le 1 + \delta (M10)$.

3. Since $g(t_i) = 2$ on $\dot{P}_1$, the sum $S(g;\dot{P}_1)$ equals 2 times the total length of U_1, so $2(1-\delta) \le S(g;\dot{P}_1) \le 2(1+\delta)$.
   Length of $[0,1-\delta]$ is $1-\delta$ and length of $[0,1+\delta]$ is $1+\delta$.

4. Similarly, the union U_2 of subintervals in $\dot{P}_2$ satisfies $[1+\delta, 3] \subseteq U_2 \subseteq [1-\delta, 3]$ (lengths $2-\delta$ and $2+\delta$), so $3(2-\delta) \le S(g;\dot{P}_2) \le 3(2+\delta)$.
   Same tag-inclusion argument on (1,3] with $g(t_i) = 3$.

5. Adding the inequalities in Steps 3 and 4 gives $8 - 5\delta \le S(g;\dot{P}) \le 8 + 5\delta$, or $|S(g;\dot{P}) - 8| \le 5\delta$.
   $2(1-\delta) + 3(2-\delta) = 8 - 5\delta$ and $2(1+\delta) + 3(2+\delta) = 8 + 5\delta$.

6. Given $\varepsilon > 0$, choose $\delta_\varepsilon := \varepsilon/10 > 0 (any \delta_\varepsilon < \varepsilon/5$ works). If $\|P\| < \delta_\varepsilon$, then $|S(g;\dot{P}) - 8| \le 5(\varepsilon/10) = \varepsilon/2 < \varepsilon$. Thus $g \in \mathcal{R}[0,3]$ and $\int_0^3 g = 8$. ∎
   Choosing δ_ε from $5\delta < \varepsilon (M09)$.

SCRATCH · EXAMPLE 7.1.4(C): HOW TO INTEGRATE H(X)=X ON [0,1] FROM DEFINITION 7.1.1

On subinterval $I_i = [x_{i-1}, x_i]$, choose the **midpoint tag** $q_i = \frac{1}{2}(x_{i-1} + x_i)$. Then $h(q_i)(x_i x_{i-1}) = \frac{1}{2}(x_i + x_{i-1})(x_i - x_{i-1}) = \frac{1}{2}(x_i^2 - x_{i-1}^2)$, which telescopes! What if the opponent picks arbitrary tags $t_i \in I_i$? Both t_i and q_i lie in I_i, so $|t_i - q_i| \le x_i - x_{i-1} \le \|P\| < \delta$.

PROOF OF $7.1.4(c)\cdot \int_0^1 x\,dx = 1/2$ **(BARTLE PP. 203–204)**

1. Let $\{I_i\}_{i=1}^n$ be any partition of [0,1], choose the midpoint $q_i := \frac{1}{2}(x_{i-1} + x_i)$ of each $I_i = [x_{i-1}, x_i]$, and let $\dot{Q} := \{(I_i, q_i)\}_{i=1}^n$. Then $h(q_i)(x_i - x_{i-1}) = \frac{1}{2}(x_i + x_{i-1})(x_i - x_{i-1}) = \frac{1}{2}(x_i^2 - x_{i-1}^2)$.
   Difference of two squares $(x_i + x_{i-1})(x_i - x_{i-1}) = x_i^2 x_{i-1}^2$.

2. Summing over $i = 1, \ldots, n$ telescopes: $S(h; \dot{Q}) = \sum_{i=1}^n \frac{1}{2}(x_i^2 - x_{i-1}^2) = \frac{1}{2}(x_n^2 - x_0^2) = \frac{1}{2}(1^2 0^2) = \frac{1}{2}$.
   Telescoping sum (M01) with $x_0 = 0$ and $x_n = 1$.

3. Now let $\dot{P} = \{(I_i, t_i)\}_{i=1}^n$ be an arbitrary tagged partition of [0,1] with $\|P\| < \delta$, and let $\dot{Q}$ have the same partition points with midpoint tags q_i. Since $t_i, q_i \in I_i$, we have $|t_i - q_i| < \delta$.
   Two points in the same subinterval of length $\leq \|P\| < \delta$ are within δ of each other.

4. By the Triangle Inequality, $|S(h; \dot{P}) - S(h; \dot{Q})| \leq \sum_{i=1}^n |t_i - q_i|(x_i - x_{i-1}) < \delta \sum_{i=1}^n (x_i - x_{i-1}) = \delta(x_n - x_0) = \delta$.
   Sum bound (M04) and telescoping $\sum(x_i - x_{i-1}) = 1 - 0 = 1 (M01)$.

5. Since $S(h; \dot{Q}) = \frac{1}{2}$, we have $|S(h; \dot{P})\frac{1}{2}| < \delta$ whenever $\|P\| < \delta$. Given $\varepsilon > 0$, take $\delta_\varepsilon := \varepsilon$; then $\|P\| < \delta_\varepsilon \Rightarrow |S(h; \dot{P}) - \frac{1}{2}| < \varepsilon$. Thus $h \in \mathcal{R}[0,1]$ and $\int_0^1 x\,dx = \frac{1}{2}$. ∎
   Definition 7.1.1 with $\delta_\varepsilon = \varepsilon$.

WORKED EXTENSION $\cdot\, Q(x) = x^2$ **ON** $[a, b], 0 \leq a <$ **B (BARTLE EXERCISE 7.1.14)**

Choose special tags $q_i := \sqrt{\frac{1}{3}(x_i^2 + x_i x_{i-1} + x_{i-1}^2)}$. Since $x_{i-1}^2 \leq \frac{1}{3}(x_i^2 + x_i x_{i-1} + x_{i-1}^2) \leq x_i^2$, we have $q_i \in [x_{i-1}, x_i]$. Then $Q(q_i)(x_i - x_{i-1}) = \frac{1}{3}(x_i^2 + x_i x_{i-1} + x_{i-1}^2)(x_i - x_{i-1}) = \frac{1}{3}(x_i^3 - x_{i-1}^3)$, so $S(Q; \dot{Q}) = \frac{1}{3}(b^3 - a^3)$. For any tags $t_i \in I_i, |t_i^2 - q_i^2| = |t_i + q_i||t_i q_i| < 2b\delta$, so $|S(Q; \dot{P}) - \frac{1}{3}(b^3 - a^3)| < 2b(b-a)\delta$. Taking $\delta_\varepsilon = \varepsilon/[2b(b-a)]$ proves $\int_a^b x^2\,dx = \frac{1}{3}(b^3 - a^3)$.

In the model paper: **Q5** (PYQ MTS-25 Q4).

T06 Changing a function at finitely many points

Altering $g \in \mathcal{R}[a,b]$ at finitely many points affects neither integrability nor the integral (Bartle 7.1.3, 7.1.4(d); PYQ MAT-22 MCQ 18, MAT-21 MCQ 13, MTS-22 MCQ 7).

THEOREM 7.1.3 (BARTLE P. 201) · SAY IT WORD FOR WORD

If g is Riemann integrable on [a,b] and if f(x) = g(x) except for a finite number of points in [a,b], then f is Riemann integrable and $\int_a^b f = \int_a^b g$.

In plain words: a single point (or finitely many points) has width zero, so changing the height of f at finitely many points changes any Riemann sum by at most a multiple of $\|P\|$, which vanishes as $\|P\| \to 0$.

HYPOTHESES · THEOREM 7.1.3

- ☐ $g \in \mathcal{R}[a,b]$
  if g is Dirichlet C7 and f changes g at 5 points, f is still not integrable
- ☐ the exceptional set where $f(x) \neq g(x)$ is **finite**
  on a countably infinite set, integrability can fail: g(x)=0 is integrable on [0,1], but changing g on $\{1/k : k \in \mathbb{N}\}$ to H(1/k)=k gives an unbounded function $H \notin \mathcal{R}[0,1]$ (*or* changing g=0 on $\mathbb{Q} \cap [0,1]$ to 1 gives Dirichlet C7)

SKELETON · APPLY THEOREM 7.1.3

Observe that f(x) = g(x) for all $x \in [a,b] \setminus \{c_1, \ldots, c_m\}$, a finite set, and $g \in \mathcal{R}[a,b]$ with $\int_a^b g =$? . By Theorem 7.1.3, $f \in R[a,b]$ and $\int_a^b f = \int_a^b g =$? . ∎

PROOF OF THEOREM 7.1.3 (BARTLE PP. 201–202)

1. It suffices to prove the case of one exceptional point $c \in [a, b]$; the case of m points follows by mathematical induction, changing one point at a time.
   Standard induction on the number of b exceptional points.

2. Let c be the exceptional point, let $L := \int g$, a and assume $f(x) = g(x)$ for all $x \neq c$.
   Unpack the hypothesis.

3. For any tagged partition $P^{\cdot}$, the sums $S(f; P^{\cdot})$ and $S(g; P^{\cdot})$ have identical terms except at most two (when c is a partition point tag of two adjacent subintervals). Hence $|S(f;\dot{P}) - S(g;\dot{P})| = |\sum(f(t_i) - g(t_i))(x_i x_{i-1})| \leq 2(|g(c)| + |f(c)|)\|P\|$.
   At most 2 subintervals have tag c (M03), each of width $\leq \|P\|$, and $|f(c) - g(c)| \leq |f(c)| + |g(c)|$ (M04). (Note: Bartle prints $f(xi) - g(xi)$ inside the sum; the summand is evaluated at the tags ti.)

4. Given $\varepsilon > 0$, choose $\delta 1 > 0$ with $\delta_1 < \varepsilon/(4(|f(c)| + |g(c)| + 1))$ ($so 2(|g(c)| + |f(c)|)\delta_1 < \varepsilon/2$), and choose $\delta 2 > 0$ such that $\|P\| < \delta_2 \Rightarrow |S(g;\dot{P}) - L| < \varepsilon/2$.
   Definition 7.1.1 for g with tolerance $\varepsilon/2$, and M09 for the exceptional-point error.

5. Let $\delta := \min\{\delta 1, \delta 2\} > 0$. If $\|P\| < \delta$, then $|S(f;\dot{P}) - L| \leq |S(f;\dot{P}) - S(g;\dot{P})| + |S(g;\dot{P}) - L| < \varepsilon/2 + \varepsilon/2 = \varepsilon$. Therefore $f \in R[a, b]$ with $b\ b \int f = L = \int g$. ∎
   $\varepsilon/2$ argument and $a\ a$ Triangle Inequality (M08, P05).

WORKED EXAMPLES · BARTLE EX 7.1.6 & EXAMPLE 7.1.4(D)

1. Bartle Ex 7.1.6: Let f(x) = 2 on [0,1) and 1 on [1,2]; let h(x) = 2 on [0,1), h(1) = 3, and h(x) = 1 on (1,2]. Both differ from the step function g(x)=2 on [0,1], 1 on (1,2] at most at the single point c = 1, so by Theorem 7.1.3, $\int_0^2 h = \int_0^2 f = 2(1) + 1(1) = 3$.

2. Countably many points where heights shrink to 0 (Bartle Example 7.1.4(d)): Let G(1/n) = 1/n for $n \in \mathbb{N}$ and G(x) = 0 elsewhere on [0,1]. Theorem 7.1.3 does NOT apply directly (infinitely many points!), but given $\varepsilon > 0$, the set $E = \{x : G(x) \geq \varepsilon\} = \{1, \frac{1}{2}, \ldots, \frac{1}{n}\}$ (where $n \leq 1/\varepsilon$) is **finite** with n points. Taking $\delta = \varepsilon/(2n)$, any $\dot{P}$ with $\|P\| < \delta$ splits into $\dot{P}_0$ (tags outside E, where G < ε and total width ≤ 1) and $\dot{P}_1$ (tags in E, at most 2n intervals of width $< \delta$ with $G \leq 1$), giving $0 \leq S(G; \dot{P}) < \varepsilon(1) + (2n)\delta(1) = 2\varepsilon$. Hence $G \in \mathcal{R}[0,1]$ and $\int_0^1 G = 0$.

EXAM TRAP · PYQ MTS-22 MCQ 7 VS MAT-22 MCQ 18

If f(x) = 0 except at **finitely many** points $c_1, \ldots, c_n \in [a,b]$, then by Theorem 7.1.3 (with $g \equiv 0$) **both** $f \in \mathcal{R}[a,b]$ **and** $\int_a^b f = 0$ hold (MAT-22 MCQ 18). If f(x) = g(x) off the **infinite** set $\{1/n : n \in \mathbb{N}\}$ ($MTS-22$ MCQ 7), knowing $f \in \mathcal{R}[0,1]$ does NOT force $g \in \mathcal{R}[0,1]$ unless g is also bounded (take f=0 and g(1/n)=n).

In the model paper: **Q6** (PYQ MAT-22 MCQ 18, MAT-21 MCQ 13, MTS-22 MCQ 7).

T07 Linearity and monotonicity of the integral

Integrals respect scalar multiples, sums, and $\leq$ inequalities (Bartle Theorem 7.1.5; PYQ MTS-25 Q5, MAT-21 Q29, MTS-22 Q5, MAT-22 MCQ 14).

THEOREM 7.1.5 · PROPERTIES OF THE INTEGRAL (BARTLE P. 204)

Suppose f and g are in $\mathcal{R}[a,b]$. Then:

(a) If $k \in \mathbb{R}$, the function kf is in $\mathcal{R}[a,b]$ and $\int_a^b \text{kf} = k\int_a^b f$.

(b) The function f + g is in $\mathcal{R}[a,b]$ and $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.

(c) If $f(x) \leq g(x)$ for all $x \in [a,b]$, then $\int_a^b f \leq \int_a^b g$.

In plain words: because every Riemann sum is a finite sum that is linear and order-preserving, passing to the integral preserves constants, sums, and $\leq$.

HYPOTHESES · THEOREM 7.1.5

☐ both $f \in \mathcal{R}[a,b]$ and $g \in \mathcal{R}[a,b]$

$f =$ Dirichlet C7 and g = 1 f have $f + g = 1 \in \mathcal{R}[0,1]$, yet neither $\int_0^1 f$ nor $\int_0^1 g$ exists

☐ in (c), $f(x) \leq g(x)$ on [a,b] (or all but finitely many points by 7.1.3)

if $f \leq g$ fails on an interval of positive length where $f > g$, $\int_a^b f \leq \int_a^b g$ fails

PROOF OF THEOREM 7.1.5 (BARTLE PP. 204–205)

1. For any tagged partition $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^{n}$ of [a,b], finite sum rules (P02) give $S(kf; \dot{P}) = kS(f; \dot{P}), \qquad S(f+g; \dot{P}) = S(f; \dot{P}) + S(g; \dot{P})$, and since $x_i - x_{i-1} > 0$, $f(x) \le g(x)$ on [a,b] implies $S(f; \dot{P}) \le S(g; \dot{P})$.
   Linearity and monotonicity of finite sums (P02).

2. **Part (a):** If k = 0, kf $= 0 \in \mathcal{R}[a,b]$ with integral $0 = 0 \int_a^b f$. If $k \neq 0$, given $\varepsilon > 0$ let $\delta_\varepsilon > 0$ be such that $\|P\| < \delta_\varepsilon \Rightarrow |S(f; \dot{P}) - \int_a^b f| < \varepsilon/|k|$; then $|S(kf; \dot{P}) - k\int_a^b f| = |k|\,|S(f; \dot{P}) - \int_a^b f| < \varepsilon$.
   Definition 7.1.1 applied with tolerance $\varepsilon/|k|$.

3. **Part (b):** Given $\varepsilon > 0$, choose $\delta_1, \delta_2 > 0$ for f, g with tolerance $\varepsilon/2$, and put $\delta_\varepsilon := \min\{\delta_1, \delta_2\} > 0$. If $\|P\| < \delta_\varepsilon$, then both $|S(f; \dot{P}) \int_a^b f| < \varepsilon/2$ and $|S(g; \dot{P}) - \int_a^b g| < \varepsilon/2$ hold.
   Common δ_ε argument (M08) as in Theorem 7.1.2.

4. By Step 1 and the Triangle Inequality, $|S(f+g; \dot{P}) - (\int_a^b f + \int_a^b g)| \le |S(f; \dot{P}) \int_a^b f| + |S(g; \dot{P}) - \int_a^b g| < \dfrac{\varepsilon}{2} + \dfrac{\varepsilon}{2} = \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $f + g \in \mathcal{R}[a,b]$ with integral $\int_a^b f + \int_a^b g$.
   Triangle Inequality (P05) and Definition 7.1.1.

5. **Part (c):** For $\|P\| < \delta_\varepsilon$ as in Step 3, unpacking the absolute values gives $\int_a^b f - \dfrac{\varepsilon}{2} < S(f; \dot{P}) \le S(g; \dot{P}) < \int_a^b g + \dfrac{\varepsilon}{2}$. Adding $\varepsilon/2$ yields $\int_a^b f < \int_a^b g + \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $\int_a^b f \le \int_a^b g$. ∎
   Unpacking $|S - L| < \varepsilon/2$ and the arbitrary ε rule (P05).

WORKED EXAMPLE · BOUNDING $|\int_a^b f|$ (BARTLE EXERCISE 7.1.8) & MCQ TRAP

Bartle Ex 7.1.8: If $f \in \mathcal{R}[a,b]$ and $|f(x)| \leq M$ for all $x \in [a,b]$, then -M $\leq f(x) \leq M$ on [a,b]. By Theorem 7.1.5(c) and Example 7.1.4(a), $-M(b-a) = \int_a^b (-M) \leq \int_a^b f \leq \int_a^b M = M(b-a)$, which means $|\int_a^b f| \leq M(b-a)$.

MCQ trap (MAT-22 MCQ 14): $f(x) \leq g(x)$ implies $\int_a^b f \leq \int_a^b g$ (with $\leq$), **never** strict < in general (for instance, take f = g, or take f=0 and g(0)=1, g(x)=0 for $x \in (0,1]$, where $f \leq g$ and f(0)

In the model paper: **Q21** (PYQ MAT-21 Q29, MTS-22 Q5, MTS-25 Q5, MAT-22 MCQ 14).

T08 The Boundedness Theorem

Every Riemann integrable function is bounded; unbounded $\Rightarrow \notin \mathcal{R}[a,b]$ (Bartle Theorem 7.1.6; PYQ MAT-22 MCQ 15, MAT-21 MCQ 11, 17, MTS-22 MCQ 8, 9).

THEOREM 7.1.6 · BOUNDEDNESS THEOREM (BARTLE P. 205) · SAY IT WORD FOR WORD

If $f \in \mathcal{R}[a,b]$, then f is bounded on [a,b].

In plain words: if f blows up to $\pm\infty$ anywhere on [a,b], you can pick a tag on that subinterval to make a single rectangle arbitrarily tall, destroying $|S(f;\dot{P}) - L| < 1$.

DIRECTION & CONVERSE · THEOREM 7.1.6

☐ **Necessary condition:** $f \in \mathcal{R}[a,b] \Rightarrow f$ is bounded on [a,b] (contrapositive: f unbounded on $[a,b] \Rightarrow f \notin \mathcal{R}[a,b]$)

Converse is FALSE: bounded does NOT imply integrable! Dirichlet's function C7 satisfies $0 \leq f(x) \leq 1$ on [0,1] (bounded), yet $f \notin \mathcal{R}[0,1]$. Also $f \in \mathcal{R}[a,b]$ does NOT imply f is continuous or monotone (C8 Thomae).

Know $f \in \mathcal{R}[a,b]$ with integral L. Assume for contradiction that f is unbounded on [a,b].

Want Construct a tagged partition $\dot{Q}$ with $\|Q\| < \delta$ but $|S(f;\dot{Q})| > |L| + 1$, contradicting Definition 7.1.1 for $\varepsilon = 1$.

PROOF OF THEOREM 7.1.6 (BARTLE PP. 205–206)

1. Assume that f is an unbounded function in $\mathcal{R}[a,b]$ with integral L. For $\varepsilon = 1$, there exists $\delta > 0$ such that if $\dot{P}$ is any tagged partition of [a,b] with $\|P\| < \delta$, then $|S(f;\dot{P}) - L| < 1$, which by the Triangle Inequality implies $|S(f;\dot{P})| \leq |S(f;\dot{P}) - L| + |L| < |L| + 1$.
   Definition 7.1.1 with $\varepsilon = 1$ and Triangle Inequality (P05).

2. Let $Q = \{[x_{i-1}, x_i]\}_{i=1}^{n}$ be a fixed partition of [a,b] with $\|Q\| < \delta$. Since f is unbounded on [a,b], f must be unbounded on at least one subinterval $[x_{k-1}, x_k]$.
   If f were bounded on each $[x_{i-1}, x_i]$ by M_i, it would be bounded on all of [a,b] by $\max\{M_1, \ldots, M_n\}$.

3. We now tag Q as follows: for $i \neq k$, fix $t_i := x_i$. Since f is unbounded on $[x_{k-1}, x_k]$ and $x_k - x_{k-1} > 0$, there exists a tag $t_k \in [x_{k-1}, x_k]$ such that $|f(t_k)(x_k - x_{k-1})| > |L| + 1 + |\sum_{i \neq k} f(t_i)(x_i x_{i-1})|$.
   For $i \neq k$ the sum is now a fixed number; $|f(t_k)|$ can be chosen arbitrarily large on $[x_{k-1}, x_k]$.

4. By the Reverse Triangle Inequality $|A + B| \geq |A| - |B|$, we obtain $|S(f;\dot{Q})| \geq |f(t_k)(x_k - x_{k-1})| - |\sum_{i \neq k} f(t_i)(x_i x_{i-1})| > |L| + 1$, which contradicts Step 1. Therefore f must be bounded on [a,b]. ∎
   Reverse Triangle Inequality (P05) yields the contradiction.

WORKED EXAMPLES · H(1/K)=K (PYQ MTS-22 MCQ 9, MAT-21 MCQ 17) & BARTLE EX 7.1.10

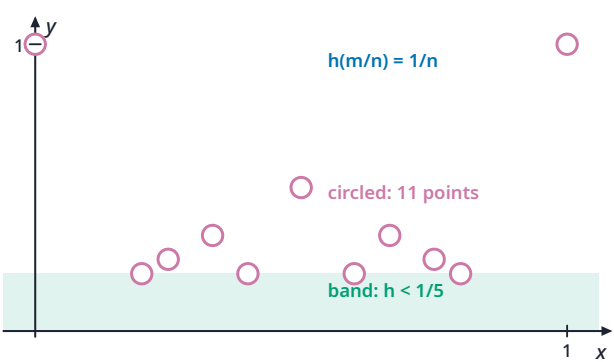
$1. H(1/k) = k (k \in \mathbb{N}), H(x) = 0$ **elsewhere on [0,1]:** Since $H(1/k) = k \to \infty$ as $k \to \infty$, H is **unbounded** on [0,1]. By Theorem 7.1.6, $H \notin \mathcal{R}[0,1]$ immediately! (Notice the contrast with Example 7.1.4(d) where G(1/n) = 1/n stays bounded and has integral 0.)

2. Why one convergent sequence of Riemann sums does not prove integrability (Bartle Ex 7.1.10): Let g(x) = 0 for rational $x \in [0,1]$ and g(x) = 1/x for irrational $x \in [0,1]$. Since g is unbounded near $0, g \notin \mathcal{R}[0,1]$ by Theorem 7.1.6. Yet if we pick the equal partition P_n with rational tags $t_i = i/n$, then $S(g; \dot{P}_n) = 0 \to 0$! You must control **all** tags, not just your favourite ones.

In the model paper: **Q7** (PYQ MAT-22 MCQ 15, MAT-21 MCQ 11, 17, MTS-22 MCQ 8, 9).

T09 Thomae's function is integrable (with integral 0)

> Discontinuous at every rational, continuous at every irrational, not monotone—yet in $\mathcal{R}[0,1]$ (Bartle Example 7.1.7).



What to notice: For Thomae's function h(m/n)=1/n, only 11 points in [0,1] reach or exceed height $\varepsilon/2 = 1/5$ (circled set E); every other point sits inside the shaded band h(x) < 1/5. **What it does not prove:** Drawing up to denominator n=5 shows the pattern, whereas the proof works for arbitrary $\varepsilon > 0$ because $1/n \geq \varepsilon/2 \iff n \leq 2/\varepsilon$ always leaves finitely many fractions.

EXAMPLE 7.1.7 · THOMAE'S FUNCTION C8 **(BARTLE P. 206)**

Define h : $[0,1] \to \mathbb{R}$ by h(x) := 0 if $x \in [0,1]$ is irrational, h(0) := 1, and h(x) := 1/n if $x \in [0,1]$ is the rational number x = m/n where $m, n \in \mathbb{N}$ have no common integer factor except 1. Then h is continuous at every irrational point and discontinuous at every rational point of [0,1], is not monotone on any subinterval, and $h \in \mathcal{R}[0,1]$ with $\int_0^1 h = 0$.

In plain words: for any threshold $\varepsilon/2 > 0$, only finitely many rationals m/n have denominator $n \le 2/\varepsilon$. We trap those finitely many tall spikes in tiny subintervals of total width $\le \varepsilon/2$, and everywhere else the height is $< \varepsilon/2$.

Know $0 \le h(x) \le 1$ on [0,1]; for any $\varepsilon > 0$, $E = \{x \in [0,1] : h(x) \ge \varepsilon/2\}$ is a finite set of n points.

Want Find $\delta > 0$ so that $\|P\| < \delta \Rightarrow 0 \le S(h; \dot{P}) < \varepsilon$.

SCRATCH · HOW WE CHOOSE δ

Split $\dot{P} = \dot{P}_1 \cup \dot{P}_2$ where $\dot{P}_1$ has tags in E (n points $\Rightarrow$ at most 2n subintervals). On $\dot{P}_1$: height ≤ 1, total width $\le 2n\delta$; want $2n\delta \le \varepsilon/2 \Leftarrow \delta = \varepsilon/(4n)$. On $\dot{P}_2$: height < $\varepsilon/2$, total width ≤ 1, so sum < $\varepsilon/2$. Total < ε.

PROOF OF EXAMPLE 7.1.7 (BARTLE P. 206)

1. Let $\varepsilon > 0$ be given. Since $h(m/n) = 1/n \geq \varepsilon/2 \iff n \leq 2/\varepsilon$, there are only finitely many denominators n and finitely many numerators $m \leq n$, so the set $E := \{x \in [0,1] : h(x) \geq \varepsilon/2\}$ is finite.
   Archimedean property (P13): only finitely many $n \in \mathbb{N}$ satisfy $n \leq 2/\varepsilon$.

2. Let n be the number of points in E and set $\delta := \varepsilon/(4n) > 0$. Let $\dot{P}$ be any tagged partition of [0,1] with $\|P\| < \delta$, and split $\dot{P}$ into $\dot{P}_1$ (tags $t_i \in E$) and $\dot{P}_2$ (tags $t_i \notin E$).
   Tag-splitting tool (M10).

3. Allowing for a point of E being an endpoint tag of two adjacent subintervals, $\dot{P}_1$ has at most 2n subintervals, of total length at most $2n\delta = 2n \cdot \dfrac{\varepsilon}{4n} = \dfrac{\varepsilon}{2}$. Since $0 < h(t_i) \leq 1$ on $\dot{P}_1$, we have $S(h; \dot{P}_1) \leq 1 \cdot 2n\delta = \varepsilon/2$.
   Endpoint tag count (M03) and sum bound (M04).

4. For tags t_i in $\dot{P}_2$, we have $h(t_i) < \varepsilon/2$, and the total length of the subintervals in $\dot{P}_2$ is at most 1 (the length of [0,1]), so $S(h; \dot{P}_2) < (\varepsilon/2) \cdot 1 = \varepsilon/2$.
   Definition of E: outside E, h(x) < $\varepsilon/2$.

5. Combining Steps 3 and 4 gives $0 \leq S(h; \dot{P}) = S(h; \dot{P}_1) + S(h; \dot{P}_2) < \dfrac{\varepsilon}{2} + \dfrac{\varepsilon}{2} = \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $h \in \mathcal{R}[0,1]$ and $\int_0^1 h = 0$. ∎
   Definition 7.1.1 with L = 0.

CHECK YOURSELF · BARTLE P. 206 NUMERICAL CHECK

List all points in $E = \{x \in [0,1] : h(x) \geq 1/5\}$ (when $\varepsilon/2 = 1/5$, i.e. $\varepsilon = 2/5$), count them, and give Bartle's δ.

We need denominators 1,2,3,4,5 in lowest terms (plus 0, where $h(0) = 1$) : $E = \{0, 1, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}\}$, which has n = 11 points. Thus $\delta = \varepsilon/(4n) = (2/5)/44 = 1/110$.

In the model paper: **Q22** .

Riemann Integrable Functions

T10 The Cauchy Criterion for integrability

index ↑

Testing integrability without knowing the value L in advance (Bartle Theorem 7.2.1; PYQ MAT-21 Q8).

THEOREM 7.2.1 · CAUCHY CRITERION (BARTLE P. 208) · SAY IT WORD FOR WORD

A function f : $[a,b] \to \mathbb{R}$ belongs to $\mathcal{R}[a,b]$ if and only if for every $\varepsilon > 0$ there exists $\eta_\varepsilon > 0$ such that if $\dot{P}$ and $\dot{Q}$ are any tagged partitions of [a,b] with $\|P\| < \eta_\varepsilon$ and $\|Q\| < \eta_\varepsilon$, then $|S(f;\dot{P}) - S(f;\dot{Q})| < \varepsilon$.

In plain words: f is integrable if and only if, whatever $\varepsilon > 0$ you are handed, you can find a mesh bound $\eta_\varepsilon > 0$ so that any two Riemann sums with mesh $< \eta_\varepsilon$ are within ε of each other—without ever naming L.

HYPOTHESES · THEOREM 7.2.1

☐ $|S(f;\dot{P}) - S(f;\dot{Q})| < \varepsilon$ holds for **all** pairs of tagged partitions $\dot{P}, \dot{Q}$ with $\|P\|, \|Q\| < \eta_\varepsilon$

for Dirichlet's function C7, two partitions with the same tiny mesh still give $|S(f;\dot{P}) - S(f;\dot{Q})| = |1-0| = 1 \not< 1/2$

☐ completeness of $\mathbb{R}$ (Cauchy Convergence Criterion 3.5.5 for sequences in $\mathbb{R}$)

used in the ($\Leftarrow$) direction to guarantee that the Cauchy sequence of numbers $(S(f;\dot{P}_m))$ converges to a real limit $A \in \mathbb{R}$

SKELETON · USE THE CAUCHY CRITERION TO PROVE $f \in \mathcal{R}[a,b]$

Let $\varepsilon > 0$ be given. Choose $\eta_\varepsilon =$? > 0. Let $\dot{P}, \dot{Q}$ be any tagged partitions of [a,b] with $\|P\| < \eta_\varepsilon$ and $\|Q\| < \eta_\varepsilon$. Then $|S(f;\dot{P}) - S(f;\dot{Q})| \leq$? $< \varepsilon$. By the Cauchy Criterion (Theorem 7.2.1), $f \in \mathcal{R}[a,b]$. ∎

Know ($\Rightarrow$) $f \in \mathcal{R}[a,b]$ with integral L. ($\Leftarrow$) For each $\varepsilon > 0$ there is $\eta_\varepsilon > 0$ making $|S(f;\dot{P}) - S(f;\dot{Q})| < \varepsilon$.

Want ($\Rightarrow$) Produce η_ε. ($\Leftarrow$) Construct the limit $A \in \mathbb{R}$ using $\varepsilon = 1/n$ and prove $\int_a^b f = A$.

SCRATCH · HOW (⇐) BUILDS THE MISSING NUMBER A

Take $\varepsilon = 1/n$ to get mesh bounds $\delta_n > 0$, forced decreasing ($\delta_n \geq \delta_{n+1}$). Pick one tagged partition $\dot{P}_n$ with $\|P_n\| < \delta_n$. For m > n, both $\dot{P}_m$ and $\dot{P}_n$ have norm < δ_n, so $|S(f;\dot{P}_m) - S(f;\dot{P}_n)| <$ 1/n: the real sequence $a_m = S(f;\dot{P}_m)$ is Cauchy! Call its limit A. Then any $\dot{Q}$ with $\|Q\| < \delta_K$ is within 1/K of $S(f;\dot{P}_K)$, which is within 1/K of A; choose $K > 2/\varepsilon$.

PROOF OF THEOREM 7.2.1 (BARTLE P. 208)

($\Rightarrow$)

1. If $f \in \mathcal{R}[a,b]$ with integral L, let $\eta_\varepsilon := \delta_{\varepsilon/2} > 0$ be the number from Definition 7.1.1 corresponding to $\varepsilon/2$. If $\dot{P}, \dot{Q}$ have $\|P\| < \eta_\varepsilon$ and $\|Q\| < \eta_\varepsilon$, then $|S(f;\dot{P}) - L| < \varepsilon/2$ and $|S(f;\dot{Q}) - L| < \varepsilon/2$.
   Definition 7.1.1 applied twice with tolerance $\varepsilon/2$.

2. By the Triangle Inequality, $|S(f;\dot{P}) - S(f;\dot{Q})| = |S(f;\dot{P}) - L + L - S(f;\dot{Q})| \leq |S(f;\dot{P}) - L| + |L - S(f;\dot{Q})| < \varepsilon/2 + \varepsilon/2 = \varepsilon$.
   Add and subtract L, then apply the Triangle Inequality (P05).

($\Leftarrow$)

1. For each $n \in \mathbb{N}$, let $\delta_n > 0$ be the number $\eta_{1/n}$ such that whenever $\dot{P}$ and $\dot{Q}$ have norms $< \delta_n$, we have $|S(f;\dot{P}) - S(f;\dot{Q})|$ < 1/n. We may assume $\delta_n \geq \delta_{n+1}$ for all n (otherwise replace δ_n by $\delta'_n := \min\{\delta_1, \ldots, \delta_n\}$) .
   Hypothesis applied with $\varepsilon = 1/n$; taking running minimums keeps δ_n decreasing so finer partitions satisfy all earlier bounds.

2. For each $n \in \mathbb{N}$, choose a tagged partition $\dot{P}_n$ with $\|P_n\| < \delta_n$. If m > n, then $\|P_m\| < \delta_m \leq \delta_n$ and $\|P_n\| < \delta_n$, so $|S(f;\dot{P}_m) - S(f;\dot{P}_n)|$ < 1/n.
   Both $\dot{P}_m$ and $\dot{P}_n$ have norm $< \delta_n$, so Step 1 applies to the pair $(\dot{P}_m, \dot{P}_n)$.

3. Consequently, the sequence $(S(f;\dot{P}_m))_{m=1}^{\infty}$ is a Cauchy sequence in $\mathbb{R}$. By the Cauchy Convergence Criterion for sequences (Theorem 3.5.5, P06), this sequence converges in $\mathbb{R}$; let $A := \lim_m S(f;\dot{P}_m)$.
   Completeness of $\mathbb{R}(P06)$.

4. Passing to the limit in Step 2 as $m \to \infty$ gives $|S(f;\dot{P}_n) - A| \leq 1/n$ for all $n \in \mathbb{N}$.
   Order property of sequence limits (strict < 1/n becomes $\leq 1/n$ in the limit).

5 To see that A is the Riemann integral of f, given $\varepsilon > 0$, choose $K \in \mathbb{N}$ with $K > 2/\varepsilon (so 1/K < \varepsilon/2)$. If $\dot{Q}$ is any tagged partition with $\|Q\| < \delta_K$, then $|S(f;\dot{Q}) - A| \leq |S(f;\dot{Q}) - S(f;\dot{P}_K)| + |S(f;\dot{P}_K) - A| < \dfrac{1}{K} + \dfrac{1}{K} = \dfrac{2}{K} < \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $f \in \mathcal{R}[a,b]$ with $\int_a^b f = A$. ∎

Archimedean property (P13), Step 1 for $(\dot{Q}, \dot{P}_K)$, Step 4 for n=K, and Triangle Inequality.

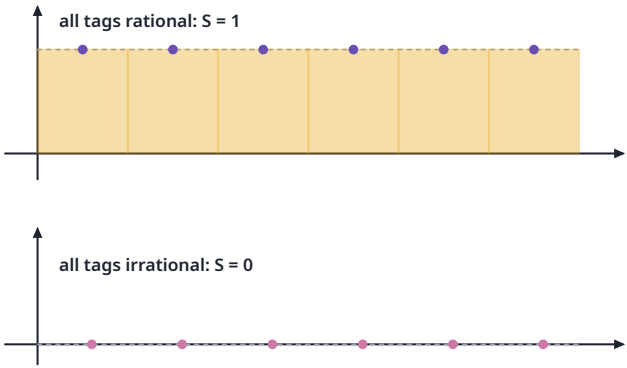
WORKED EXAMPLE · BARTLE EXAMPLE 7.2.2(A)

Let g(x) = 2 on [0,1] and 3 on (1,3] as in Example 7.1.4(b). For any $\dot{P}, \dot{Q}$ with $\|P\| < \delta$ and $\|Q\| < \delta$, we showed $8 - 5\delta \leq S(g;\dot{P}) \leq 8 + 5\delta$ and $8 - 5\delta \leq S(g;\dot{Q}) \leq 8 + 5\delta$. Subtracting gives $|S(g;\dot{P}) - S(g;\dot{Q})| \leq 10\delta$. Given $\varepsilon > 0$, choose $\eta_\varepsilon := \varepsilon/20$; then $\|P\|, \|Q\| < \eta_\varepsilon \Rightarrow |S(g;\dot{P}) - S(g;\dot{Q})| \leq 10(\varepsilon/20) = \varepsilon/2 < \varepsilon$. By Theorem 7.2.1, $g \in \mathcal{R}[0,3]$.

In the model paper: **Q23** (PYQ MAT-21 Q8).

T11 Dirichlet's function (and relatives) are not integrable

Negating the Cauchy Criterion to prove $f \notin \mathcal{R}[a,b]$ (Bartle Example 7.2.2(b), Exercises 7.2.1–7.2.2; PYQ MAT-22 Q4, MAT-22 MCQ 17, MAT-21 MCQ 12, 16).



What to notice: On the exact same partition of [0,1], choosing rational tags puts every rectangle at height 1 (S=1), while choosing irrational tags puts every rectangle at height 0 (S=0). **What it does not prove:** One partition is not enough; because rationals and irrationals are dense in every subinterval no matter how small $\eta > 0$ is, the gap $|1 - 0| = 1 \geq 1/2$ never shrinks.

NEGATION OF CAUCHY CRITERION (BARTLE EX 7.2.1) & EXAMPLE 7.2.2(B) C7

Negation of 7.2.1: $f \notin \mathcal{R}[a,b]$ if and only if there exists $\varepsilon_0 > 0$ such that for every $\eta > 0$ (or for every $n \in \mathbb{N}$ with $\eta = 1/n$) there exist tagged partitions $\dot{P}$ and $\dot{Q}$ with $\|P\| < \eta$ and $\|Q\| < \eta$ such that $|S(f;\dot{P}) - S(f;\dot{Q})| \geq \varepsilon_0$.

Example 7.2.2(b) (Dirichlet's function): Let f(x) := 1 if $x \in [0,1]$ is rational and f(x) := 0 if $x \in [0,1]$ is irrational. Then $f \notin \mathcal{R}[0,1]$.

In plain words: to disprove integrability, pick one fixed positive gap ε_0 (like 1/2) and show that no matter how fine the mesh η is, you can still find two taggings $\dot{P}, \dot{Q}$ whose sums differ by at least ε_0.

SKELETON · PROVE $f \notin \mathcal{R}[a,b]$ VIA CAUCHY CRITERION

Take $\varepsilon_0 =$? > 0. Given any $\eta > 0$, choose a partition P of [a,b] with $\|P\| < \eta$. Form $\dot{P}$ by choosing rational tags $t_i \in I_i$, giving $S(f;\dot{P}) =$?, and form $\dot{Q}$ by choosing irrational tags $s_i \in I_i$, giving $S(f;\dot{Q}) =$?. Then $|S(f;\dot{P}) - S(f;\dot{Q})| \geq \varepsilon_0$. By the Cauchy Criterion $(7.2.1)$, $f \notin \mathcal{R}[a,b]$. ∎

PROOF THAT DIRICHLET'S FUNCTION IS NOT IN $\mathcal{R}[0,1]$ (BARTLE P. 209)

1. Take $\varepsilon_0 := \frac{1}{2} > 0$, and let $\eta > 0$ be arbitrary. Choose any partition $P = \{[x_{i-1}, x_i]\}_{i=1}^n$ of [0,1] with $\|P\|$ < η(for instance, an equal partition with $n > 1/\eta$).
   Set the fixed gap $\varepsilon_0 = 1/2$ and let the opponent hand us any $\eta > 0$.

2. By the density of $\mathbb{Q}$ in $\mathbb{R}$, each subinterval $[x_{i-1}, x_i]$ contains a rational tag t_i. Let $\dot{P}$ be P with these rational tags; then $f(t_i) = 1$ for all i, so $S(f; \dot{P}) = \sum_{i=1}^n 1 \cdot (x_i - x_{i-1}) = 1 - 0 = 1$.
   Density of rationals (P14) and telescoping sum (M01).

3. By the density of $\mathbb{R} \setminus \mathbb{Q}$ in $\mathbb{R}$, each subinterval $[x_{i-1}, x_i]$ contains an irrational tag s_i. Let $\dot{Q}$ be P with these irrational tags; then $f(s_i) = 0$ for all i, so $S(f; \dot{Q}) = 0$.
   Density of irrationals (P14).

4. Both $\|P\|$ < η and $\|Q\|$ < η, yet $|S(f; \dot{P}) - S(f; \dot{Q})| = |1 - 0| = 1 \geq \frac{1}{2} = \varepsilon_0$. Thus the Cauchy Criterion 7.2.1 fails, and $f \notin \mathcal{R}[0,1]$. ∎
   Negation of Theorem 7.2.1.

WORKED EXAMPLE · H(X) = X + 1 ON RATIONALS, 0 ON IRRATIONALS (BARTLE EX 7.2.2; PYQ MAT-21 MCQ 16)

Let h(x) := x + 1 for $x \in [0,1] \cap \mathbb{Q}$ and h(x) := 0 for $x \in [0,1] \setminus \mathbb{Q}$. Take $\varepsilon_0 := 1$. For any $\eta > 0$, choose a partition P of [0,1] with $\|P\|$ < η. If $\dot{P}$ has all rational tags $t_i \in [x_{i-1}, x_i]$, then $h(t_i) = t_i + 1 \geq 1$ on [0,1], so $S(h; \dot{P}) = \sum_{i=1}^n (t_i + 1)(x_i - x_{i-1}) \geq \sum_{i=1}^n 1 \cdot (x_i x_{i-1}) = 1$. If $\dot{Q}$ has all irrational tags, then $S(h; \dot{Q}) = 0$. Hence $|S(h; \dot{P}) S(h; \dot{Q})| \geq 1 = \varepsilon_0$, so $h \notin \mathcal{R}[0,1]$.

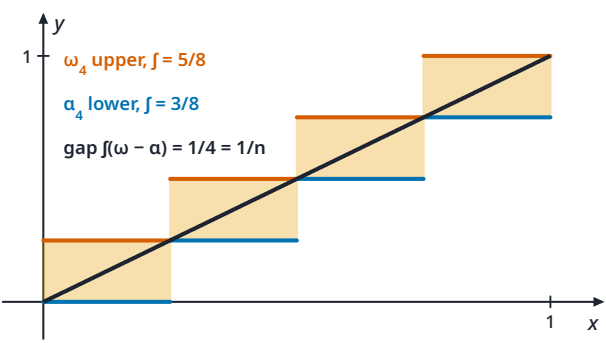
YOUR INTUITION WILL LIE TO YOU HERE

"There are vastly more irrationals than rationals, so Dirichlet's function is 'almost always 0' and its integral should be 0." That is true for the **Lebesgue** integral (taught in postgraduate Measure Theory), which gives 0. It is **false** for the Riemann integral, because Riemann sums let an opponent pick rational tags in every subinterval!

In the model paper: **Q24** (PYQ MAT-22 Q4, MAT-22 MCQ 17, MAT-21 MCQ 12, 16).

T12 The Squeeze Theorem for integrals

Proving $f \in \mathcal{R}[a,b]$ by trapping f between two simpler integrable functions $\alpha_\varepsilon \le f \le \omega_\varepsilon$ (Bartle Theorem 7.2.3, Example 7.2.6(b); PYQ MTS-26 Q18, MTS-22 Q6, MAT-21 Q7).



What to notice: On [0,1] with n=4, the line h(x)=x is trapped between the lower step function α_4 (integral 3/8) and upper step function ω_4 (integral 5/8); the shaded gap consists of 4 squares of area $(1/4)^2 = 1/16$, giving $\int_0^1 (\omega_4 - \alpha_4) = 1/4 = 1/n$. **What it does not prove:** One picture for n=4 does not prove $h \in \mathcal{R}[0,1]$; the Squeeze Theorem requires $\int_0^1 (\omega_n - \alpha_n) = 1/n < \varepsilon$ for every $\varepsilon > 0$ by choosing $n > 1/\varepsilon$.

THEOREM 7.2.3 · SQUEEZE THEOREM (BARTLE P. 209) · SAY IT WORD FOR WORD

Let f : $[a,b] \to \mathbb{R}$. Then $f \in \mathcal{R}[a,b]$ if and only if for every $\varepsilon > 0$ there exist functions α_ε and ω_ε in $\mathcal{R}[a,b]$ with $\alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x) \quad$ for all $x \in [a,b]$ and such that $\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon$.

In plain words: instead of wrestling with all possible tags of f, trap f between a floor function $\alpha_\varepsilon \in \mathcal{R}[a,b]$ and a ceiling function $\omega_\varepsilon \in \mathcal{R}[a,b]$ whose integrals are within ε of each other.

HYPOTHESES · THEOREM 7.2.3

☐ $\alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b]$ with $\alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x)$ for all $x \in [a,b]$
if $\alpha_\varepsilon, \omega_\varepsilon$ are not themselves integrable, trapping f between them tells us nothing

☐ $\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon$ holds for **every** $\varepsilon > 0$ ($so \alpha_\varepsilon, \omega_\varepsilon$ must be adjusted as ε shrinks)
Bartle Ex 7.2.4 trap: one fixed pair like $\alpha(x) =$ -x and $\omega(x) = x$ on [0,1] satisfies -x $\le f(x) \le x$, but $\int_0^1 (\omega - \alpha) = \int_0^1 2x\,dx = 1$, which is NOT < ε for $\varepsilon \le 1$. Indeed, f(x) = x on $\mathbb{Q}$ and -x on irrationals satisfies -x $\le f(x) \le x$ yet $f \notin \mathcal{R}[0,1]$!

SKELETON · PROVE $f \in \mathcal{R}[a,b]$ USING THE SQUEEZE THEOREM

Let $\varepsilon > 0$ be given. Define $\alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b]$ by ? . Then $\alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x)$ on [a,b], and $\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) =$? $< \varepsilon$. By the Squeeze Theorem (7.2.3), $f \in \mathcal{R}[a,b]$. ∎

Know ($\Leftarrow$) For every $\varepsilon > 0, \alpha_\varepsilon \le f \le \omega_\varepsilon$ with $\alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b]$ and $\int_a^b (\omega_\varepsilon \alpha_\varepsilon) < \varepsilon$.

Want Show $|S(f;\dot{P})S(f;\dot{Q})| < 3\varepsilon$ when $\|P\|, \|Q\| < \delta_\varepsilon$, and invoke the Cauchy Criterion 7.2.1.

PROOF OF THEOREM 7.2.3 (BARTLE PP. 209–210)

($\Rightarrow$)

1. If $f \in \mathcal{R}[a,b]$, simply take $\alpha_\varepsilon := f$ and $\omega_\varepsilon := f$ for all $\varepsilon > 0$; then $\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) = \int_a^b 0 = 0 < \varepsilon$.
   Trivial choice when f is already known to be integrable.

($\Leftarrow$)

1. Let $\varepsilon > 0$. Since $\alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b]$, there exists $\delta_\varepsilon > 0$ such that if $\dot{P}$ is any tagged partition with $\|P\| < \delta_\varepsilon$, then $|S(\alpha_\varepsilon; \dot{P}) - \int_a^b \alpha_\varepsilon| < \varepsilon \quad \text{and} \quad |S(\omega_\varepsilon; \dot{P}) - \int_a^b \omega_\varepsilon| < \varepsilon$.
   Definition 7.1.1 applied to α_ε and ω_ε, taking the minimum of the two deltas (M08).

2. Unpacking these two inequalities gives $\int_a^b \alpha_\varepsilon - \varepsilon < S(\alpha_\varepsilon; \dot{P})$ and $S(\omega_\varepsilon; \dot{P}) < \int_a^b \omega_\varepsilon + \varepsilon$. Since $\alpha_\varepsilon(x) \leq f(x) \leq \omega_\varepsilon(x)$ on [a,b], we have $S(\alpha_\varepsilon; \dot{P}) \leq S(f; \dot{P}) \leq S(\omega_\varepsilon; \dot{P})$, and therefore $\int_a^b \alpha_\varepsilon - \varepsilon < S(f; \dot{P}) < \int_a^b \omega_\varepsilon + \varepsilon$.
   Pointwise inequality $\alpha_\varepsilon \leq f \leq \omega_\varepsilon$ preserves Riemann sums (P02).

3. If $\dot{Q}$ is any other tagged partition with $\|Q\| < \delta_\varepsilon$, then the same bounds hold for $S(f; \dot{Q})$: $\int_a^b \alpha_\varepsilon - \varepsilon < S(f; \dot{Q}) < \int_a^b \omega_\varepsilon + \varepsilon$.
   Step 2 applied to $\dot{Q}$.

4. Subtracting the two inequalities and using linearity (Theorem 7.1.5) together with $\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon$ gives $|S(f; \dot{P}) - S(f; \dot{Q})| < \int_a^b \omega_\varepsilon - \int_a^b \alpha_\varepsilon + 2\varepsilon = \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) + 2\varepsilon < \varepsilon + 2\varepsilon = 3\varepsilon$.
   Two numbers in $(A - \varepsilon, B + \varepsilon)$ are within $(B - A) + 2\varepsilon$ of each other; Theorem 7.1.5(a),(b).

5. Since $\varepsilon > 0$ is arbitrary, the Cauchy Criterion (Theorem 7.2.1) implies that $f \in \mathcal{R}[a,b]$. ∎
   Theorem 7.2.1 (replacing ε by $\varepsilon/3$ if desired).

WORKED EXAMPLE · SQUEEZE APPLIED TO H(X)=X ON [0,1] (BARTLE EXAMPLE 7.2.6(B))

Partition [0,1] into n equal pieces using $P_n = (0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}, 1)$. On $[\frac{k-1}{n}, \frac{k}{n})$ for $k = 1, \ldots, n-1 (and$ on $[\frac{n-1}{n}, 1]$ for k=n), define step functions $\alpha_n(x) := \frac{k-1}{n} (the$ minimum of x on the piece) and $\omega_n(x) := \frac{k}{n} (the$ maximum of x on the piece). By Theorem 7.2.5 and P03,

$$\int_0^1 \alpha_n = \frac{1}{n}(0 + \frac{1}{n} + \cdots + \frac{n-1}{n}) = \frac{1}{n^2} \cdot \frac{(n-1)n}{2} = \frac{1}{2}(1 - \frac{1}{n}),$$

$$\int_0^1 \omega_n = \frac{1}{n}(\frac{1}{n} + \frac{2}{n} + \cdots + \frac{n}{n}) = \frac{1}{n^2} \cdot \frac{n(n+1)}{2} = \frac{1}{2}(1 + \frac{1}{n}).$$

Since $\alpha_n \le h \le \omega_n$ and $\int_0^1 (\omega_n - \alpha_n) = \frac{1}{n} < \varepsilon$ for any $n > 1/\varepsilon$, Theorem 7.2.3 gives $h \in \mathcal{R}[0,1]$. Moreover, $\frac{1}{2}(1 - \frac{1}{n}) \le \int_0^1 h \le \frac{1}{2}(1 + \frac{1}{n})$ for all $n \in \mathbb{N}$ forces $\int_0^1 x\,dx = \frac{1}{2}$.

In the model paper: **Q25** (PYQ MTS-26 Q18, MTS-22 Q6, MAT-21 Q7).

T13 Elementary step functions φ_J

A single flat block of height 1 on a subinterval J with endpoints c < d has integral d - c (Bartle Lemma 7.2.4, Ex 7.1.13; PYQ MAT-21 MCQ 10).

LEMMA 7.2.4 · ELEMENTARY STEP FUNCTIONS (BARTLE P. 210) · SAY IT WORD FOR WORD

If J is a subinterval of [a,b] having endpoints c < d, and if $\varphi_J(x) := 1$ for $x \in J$ and $\varphi_J(x) := 0$ elsewhere on [a,b], then $\varphi_J \in \mathcal{R}[a,b]$ and $\int_a^b \varphi_J = d - c$.

In plain words: whether the subinterval J is closed [c,d], open (c,d), or half-open [c,d) or (c,d], the indicator function φ_J is integrable and its integral is simply the length d - c of J.

SKELETON · PROVE LEMMA 7.2.4

First let J = [c,d]. For any $\dot{P}$ with $\|P\| < \delta$, the union of subintervals with tag in [c,d] contains $[c+\delta, d-\delta]$ and lies inside $[c-\delta, d+\delta]$, so $(d-c) - 2\delta \le S(\varphi_{[c,d]}; \dot{P}) \le (d-c) + 2\delta$. Choose $\delta_\varepsilon = \varepsilon/4$ to get $|S - (d-c)| < \varepsilon$. The other three intervals (c,d), [c,d), (c,d] differ from [c,d] at at most 2 endpoints, so Theorem 7.1.3 gives the same integral d - c. ∎

PROOF OF LEMMA 7.2.4 & EXERCISE 7.1.13 (BARTLE P. 210)

1. First suppose J = [c,d] with $a \le c < d \le b$. Let $\dot{P}$ be any tagged partition of [a,b] with $\|P\| < \delta$, and let $\dot{P}_1$ be the subset of subintervals whose tags belong to [c,d].
   Outside $\dot{P}_1$, $\varphi_{[c,d]}(t_i) = 0$, so $S(\varphi_{[c,d]}; \dot{P}) = S(\varphi_{[c,d]}; \dot{P}_1)$.

2. By M10 (Exercise 7.1.5), the union U_1 of the subintervals in $\dot{P}_1$ satisfies $[c+\delta, d-\delta] \subseteq U_1 \subseteq [c-\delta, d+\delta]$. Since $\varphi_{[c,d]}(t_i) = 1$ on $\dot{P}_1$, we have $(d-c) - 2\delta \le S(\varphi_{[c,d]}; \dot{P}) \le (d-c) + 2\delta$, so $|S(\varphi_{[c,d]}; \dot{P}) - (d-c)| \le 2\delta$.
   Length of $[c+\delta, d-\delta]$ is $(d-c) - 2\delta$ and length of $[c-\delta, d+\delta]$ is $(d-c) + 2\delta$.

3. Given $\varepsilon > 0$, choose $\delta_\varepsilon := \varepsilon/4 > 0$; then $\|P\| < \delta_\varepsilon \Rightarrow |S(\varphi_{[c,d]}; \dot{P}) - (d-c)| \le 2(\varepsilon/4) = \varepsilon/2 < \varepsilon$. Hence $\varphi_{[c,d]} \in \mathcal{R}[a,b]$ and $\int_a^b \varphi_{[c,d]} = d - c$.
   Definition 7.1.1 with L = d - c.

4. If J is (c,d), [c,d), or (c,d], then φ_J differs from $\varphi_{[c,d]}$ only at one or both of the endpoints c, d (at most two points). By Theorem 7.1.3, $\varphi_J \in \mathcal{R}[a,b]$ and $\int_a^b \varphi_J = \int_a^b \varphi_{[c,d]} = d - c$. ∎
   Changing a function at finitely many points (Theorem 7.1.3, T06).

CHECK YOURSELF · PYQ MAT-21 MCQ 10

Let g : $[0,4] \to \mathbb{R}$ be g(x) = 3 for $0 \le x \le 2$ and g(x) = 2 for $2 < x \le 4$. Express g using elementary step functions and find $\int_0^4 g$.

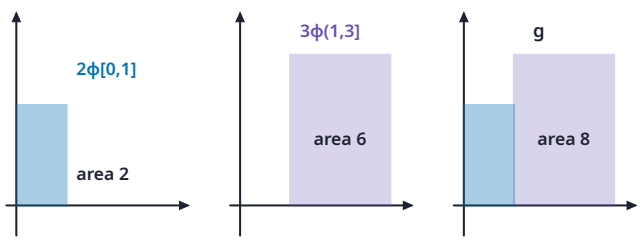
$g = 3\varphi_{[0,2]} + 2\varphi_{(2,4]}$. By Lemma 7.2.4 and linearity (7.1.5), $\int_0^4 g = 3(2-0) + 2(4-2) = 6 + 4 = 10$.

In the model paper: **Q8** (PYQ MAT-21 MCQ 10).

T14 Step functions are integrable

index ↑

Every step function is a linear combination of elementary step functions φ_{J_j} (Bartle Theorem 7.2.5, Example 7.2.6(a); PYQ MTS-26 Q5, MAT-22 Q16).



What to notice: The step function g of Example 7.1.4(b) splits into $2\varphi_{[0,1]}$ (area $2 \times 1 = 2$) plus $3\varphi_{(1,3]}$ (area $3 \times 2 = 6$), so $\int_0^3 g = 2 + 6 = 8$ without touching ε or δ. **What it does not prove:** This formula only works for a finite sum of interval indicators; a function that jumps infinitely often is not a step function.

THEOREM 7.2.5 · INTEGRABILITY OF STEP FUNCTIONS (BARTLE P. 210, EQS. (4)–(5))

If $\varphi : [a,b] \to \mathbb{R}$ is a step function, then $\varphi \in \mathcal{R}[a,b]$. Moreover, if $\varphi = \sum_{j=1}^m k_j \varphi_{J_j}$ where J_j has endpoints $c_j < d_j$, then $\int_a^b \varphi = \sum_{j=1}^m k_j (d_j - c_j)$.

In plain words: every step function is integrable, and its integral is the sum of (constant height k_j)$\times$ (interval length $d_j - c_j$).

HYPOTHESES · THEOREM 7.2.5

☐ φ is a **step function** on [a,b]: [a,b] is the union of **finitely many** non-overlapping intervals $J_1, \ldots, J_m$ on each of which φ is constant (Definition 5.4.9)

Bartle Ex 7.2.6 non-example: taking only finitely many values is NOT enough to be a step function! Dirichlet's function C7 takes only two values (0 and 1), yet the sets $\mathbb{Q} \cap [0,1]$ and $[0,1] \setminus \mathbb{Q}$ are not finite unions of intervals, and $f \notin \mathcal{R}[0,1]$.

PROOF OF THEOREM 7.2.5 (BARTLE PP. 210–211)

1. By Definition 5.4.9 (and Exercise 7.2.5), [a,b] is the union of finitely many non-overlapping subintervals $J_1, \ldots, J_m$ with endpoints $c_j \le d_j$ such that $\varphi(x) = k_j$ on J_j. Disjointness of the pieces (up to single endpoints, which can be absorbed into one piece or treated as single-point intervals) allows us to write $\varphi = \sum_{j=1}^{m} k_j \varphi_{J_j}$, where each φ_{J_j} is an elementary step function on [a,b].
   Decomposition into indicator functions (M06).

2. By Lemma 7.2.4, each φ_{J_j} belongs to $\mathcal{R}[a,b]$ and satisfies $\int_a^b \varphi_{J_j} = d_j - c_j$.
   Lemma 7.2.4 (T13).

3. By Theorem 7.1.5(a) and (b) (applied inductively to the finite sum of m terms, Exercise 7.1.7), the linear combination $\varphi = \sum_{j=1}^{m} k_j \varphi_{J_j}$ belongs to $\mathcal{R}[a,b]$ and $\int_a^b \varphi = \sum_{j=1}^{m} k_j \int_a^b \varphi_{J_j} = \sum_{j=1}^{m} k_j (d_j - c_j)$. ■
   Linearity of the integral (Theorem 7.1.5, T07).

WORKED EXAMPLES · BARTLE EXAMPLE 7.2.6(A) & EXERCISE 7.2.7

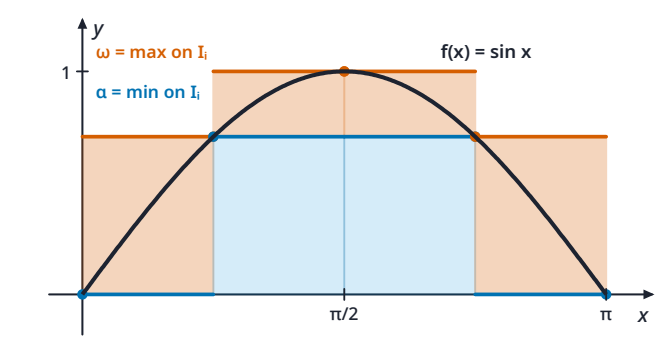
1. Example 7.2.6(a): The function g(x) = 2 on [0,1] and g(x) = 3 on (1,3] of Example 7.1.4(b) is the step function $2\varphi_{[0,1]} + 3\varphi_{(1,3]}$, so Theorem 7.2.5 immediately gives $\int_0^3 g = 2(1-0) + 3(3-1) = 8$ in one line!

2. Every Riemann sum is the integral of a step function (Bartle Ex 7.2.7): Given any f : $[a,b] \to \mathbb{R}$ and any tagged partition $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$, define the step function $\varphi(x) := f(t_i)$ on $[x_{i-1}, x_i)$ for $i = 1, \ldots, n-1$ and $\varphi(x) := f(t_n)$ on $[x_{n-1}, x_n]$. By Theorem 7.2.5, $\int_a^b \varphi = \sum_{i=1}^n f(t_i)(x_i - x_{i-1}) = S(f; \dot{P})$.

In the model paper: **Q26** (PYQ MTS-26 Q5, MAT-22 Q16).

T15 Continuous functions on [a,b] are integrable

Uniform Continuity + Max–Min + Step functions + Squeeze = Theorem 7.2.7 (PYQ MTS-25 Q18, MAT-21 Q30, MTS-22 Q19, MAT-22 MCQ 21, 23, MAT-21 MCQ 15).



What to notice: On each subinterval I_i, the continuous function $f(x) = \sin x$ attains its minimum at u_i (floor step α_ε) and maximum at v_i (ceiling step ω_ε); uniform continuity makes every vertical gap $f(v_i) - f(u_i) < \varepsilon/(b-a)$ once $\|P\| < \delta_\varepsilon$. **What it does not prove:** Continuity on an open interval (a,b) is not enough (1/x on (0,1] is not integrable).

THEOREM 7.2.7 · INTEGRABILITY OF CONTINUOUS FUNCTIONS (BARTLE P. 212) · SAY IT WORD FOR WORD

If f : $[a,b] \to \mathbb{R}$ is continuous on [a,b], then $f \in \mathcal{R}[a,b]$.

In plain words: every continuous function on a closed bounded interval is Riemann integrable.

HYPOTHESES & CONVERSE · THEOREM 7.2.7

☐ f is **continuous** on [a,b]

Dirichlet **C7** is defined on [0,1] and bounded, but discontinuous everywhere and $\notin \mathcal{R}[0,1]$

☐ the interval [a,b] is **closed and bounded**

f(x) = 1/x on (0,1] with f(0) = 0 **C16** is continuous on the half-open interval (0,1], yet unbounded and $\notin \mathcal{R}[0,1]$

☐ **Converse is FALSE (PYQ MAT-22 MCQ 21):** $f \in \mathcal{R}[a,b]$ does **not** imply f is continuous or uniformly continuous on [a,b]

the step function g of $7.1.4(b), \operatorname{sgn}$ **C9**, and Thomae's function **C8** are all in $\mathcal{R}[a,b]$ but discontinuous (and hence not uniformly continuous)

Know f is continuous on the closed bounded interval $[a,b]; \varepsilon > 0$ is given.

Want Construct step functions $\alpha_\varepsilon \le f \le \omega_\varepsilon$ on [a,b] with $\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon$, then apply the Squeeze Theorem 7.2.3.

SCRATCH · WHY WE NEED UNIFORM CONTINUITY (5.4.3) AND MAX–MIN (5.3.4)

On $I_i = [x_{i-1}, x_i]$, we want step heights $\alpha_\varepsilon(x) = \min_{I_i} f = f(u_i)$ and $\omega_\varepsilon(x) = \max_{I_i} f = f(v_i)$. How do we know u_i, v_i exist? Because f is continuous on the closed bounded interval I_i(**Max–Min Theorem 5.3.4**). Then $\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) = \sum_{i=1}^n (f(v_i) - f(u_i))(x_i x_{i-1})$. To make this sum < ε, we need $f(v_i) - f(u_i) < \varepsilon/(b-a)$ on **every** subinterval at once. Since $|v_i - u_i| \le x_i - x_{i-1} \le \|P\|$, the **Uniform Continuity Theorem 5.4.3** gives one $\delta_\varepsilon > 0$ that works across all $i = 1, \ldots, n$ simultaneously!

PROOF OF THEOREM 7.2.7 (BARTLE P. 212)

1. By the Uniform Continuity Theorem 5.4.3 (P08), since f is continuous on the closed bounded interval [a,b], f is uniformly continuous on [a,b]. Therefore, given any $\varepsilon > 0$, there exists $\delta_\varepsilon > 0$ such that if $u, v \in [a, b]$ and $|u - v| < \delta_\varepsilon$, then $|f(u) - f(v)| < \dfrac{\varepsilon}{b-a}$.
   Uniform Continuity Theorem 5.4.3 applied with tolerance $\varepsilon/(b-a) > 0$.

2. Let $P = \{I_i\}_{i=1}^n$ be a partition of [a,b] with $\|P\| < \delta_\varepsilon$, where $I_i = [x_{i-1}, x_i]$.
   Choose an equal partition with $n > (b-a)/\delta_\varepsilon$ so every subinterval has length $< \delta_\varepsilon$.

3. Since f is continuous on each closed bounded subinterval I_i, the Maximum–Minimum Theorem 5.3.4 (P07) implies that there exist points $u_i, v_i \in I_i$ where f attains its minimum and maximum values on I_i, respectively: $f(u_i) \le f(x) \le f(v_i)$ for all $x \in I_i$.
   Maximum–Minimum Theorem 5.3.4 on each $I_i = [x_{i-1}, x_i]$.

4. Define the step function $\alpha_\varepsilon : [a, b] \to \mathbb{R}$ by $\alpha_\varepsilon(x) := f(u_i)$ for $x \in [x_{i-1}, x_i) (i = 1, \ldots, n-1)$ and $\alpha_\varepsilon(x) := f(u_n)$ for $x \in [x_{n-1}, x_n]$. Similarly, define $\omega_\varepsilon(x) := f(v_i)$ for $x \in [x_{i-1}, x_i) (i = 1, \ldots, n-1)$ and $\omega_\varepsilon(x) := f(v_n)$ for $x \in [x_{n-1}, x_n]$. Then $\alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a, b]$ by Theorem 7.2.5, and $\alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x) \quad \text{for all } x \in [a, b]$.
   Construction of disjoint half-open intervals and Theorem 7.2.5 (T14).

5. Since $u_i, v_i \in I_i$ and $\|P\| < \delta_\varepsilon$, we have $|u_i - v_i| \le x_i - x_{i-1} < \delta_\varepsilon$, so $f(v_i) f(u_i) < \varepsilon/(b-a)$ by Step 1. Using formula (5) of Theorem 7.2.5, $0 \le \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) = \sum_{i=1}^n (f(v_i) - f(u_i))(x_i - x_{i-1}) < \sum_{i=1}^n \dfrac{\varepsilon}{b-a}(x_i - x_{i-1}) = \dfrac{\varepsilon}{b-a}(b-a) = \varepsilon$.
   Step-function integral formula (T14), Step 1 bound, and telescoping $\sum (x_i - x_{i-1}) =$ b-a (M01).

6 Since $\varepsilon > 0$ is arbitrary, the Squeeze Theorem 7.2.3 implies that $f \in \mathcal{R}[a,b]$. ∎

Squeeze Theorem 7.2.3 (T12).

WHERE EACH HYPOTHESIS WAS USED & TWO CLASSIC EXAM APPLICATIONS

Where hypotheses were used: Continuity of f and closedness/boundedness of [a,b] were used in Step 1 (Uniform Continuity Theorem 5.4.3) and in Step 3 (Maximum–Minimum Theorem 5.3.4 on each I_i).

Application $1 \cdot f \geq 0$ continuous and $\int_a^b f = 0 \Rightarrow f \equiv 0$ (Bartle Ex 7.2.8–9; PYQ MAT-22 MCQ 23, Q18): Suppose for contradiction that f(c) > 0 for some $c \in [a,b]$. By continuity at c with $\varepsilon = \frac{1}{2}f(c) > 0$, there is a subinterval $J = [c_1, d_1] \subseteq [a,b]$ of length $d_1 - c_1 > 0$ containing c on which $f(x) > \frac{1}{2}f(c)$. Then the elementary step function satisfies $0 \leq \frac{1}{2}f(c)\varphi_J(x) \leq f(x)$ on [a,b], so by Theorem 7.1.5(c) and Lemma 7.2.4, $\int_a^b f \geq \frac{1}{2}f(c)(d_1 - c_1) > 0$, contradicting $\int_a^b f = 0$! Hence f(x) = 0 for **all** $x \in [a,b]$. (Continuity cannot be dropped: f(0)=1 and f(x)=0 on (0,1] has $f \geq 0$ and $\int_0^1 f = 0$, yet $f(0) \neq 0$.)

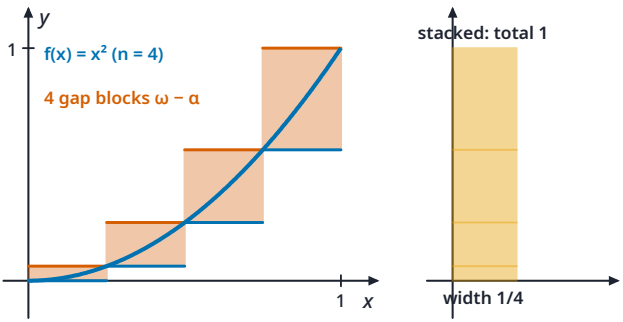
Application 2 · Mean Value Theorem for Integrals (Bartle Ex 7.2.16, Q20): If f is continuous on [a,b] (a < b), then by Max–Min (5.3.4) f attains its minimum m and maximum M on [a,b]. Since $m \leq f(x) \leq M$, Theorem 7.1.5(c) gives $m(b-a) \leq \int_a^b f \leq M(b-a)$, so $k := \frac{1}{b-a}\int_a^b f \in [m, M]$. By Bolzano's Intermediate Value Theorem 5.3.7, there exists $c \in [a,b]$ with f(c) = k, i.e. $\int_a^b f = f(c)(b-a)$.

In the model paper: **Q18**, **Q20**, **Q37** (PYQ MTS-25 Q18, MAT-21 Q30, MTS-22 Q19, MAT-22 MCQ 21, 23, MAT-21 MCQ 15).

T16 Monotone functions on [a,b] are integrable

Even with countably many jumps, an increasing or decreasing function on [a,b] is integrable because the staircase gap

telescopes (Bartle Theorem 7.2.8; PYQ MTS-26 Q19, MAT-22 Q30, MAT-22 MCQ 22).



What to notice: For the increasing function $f(x) = x^2$ on [0,1] with n=4 equal pieces, sliding all four error rectangles $(\omega - \alpha)$ horizontally stacks them into a single column of width (b-a)/n = 1/4 and height f(1)-f(0) = 1, so $\int_0^1 (\omega - \alpha) = 1/4$. **What it does not prove:** The picture shows n=4; the proof works by choosing $n > (b-a)(f(b) - f(a))/\varepsilon$ for any $\varepsilon > 0$.

THEOREM 7.2.8 · INTEGRABILITY OF MONOTONE FUNCTIONS (BARTLE P. 212) · SAY IT WORD FOR WORD

If f : $[a, b] \to \mathbb{R}$ is monotone on [a,b], then $f \in \mathcal{R}[a, b]$.

In plain words: every increasing or decreasing function on a closed bounded interval [a,b] is Riemann integrable.

HYPOTHESES & CONVERSE · THEOREM 7.2.8

☐ f is **monotone** (increasing or decreasing) on the closed bounded interval [a,b]

if f is only monotone on the open interval (0,1], e.g. f(x) = 1/x on (0,1] with f(0)=0 C16, f is unbounded and $\notin \mathcal{R}[0, 1]$. On a closed interval [a,b], a monotone function is automatically bounded between f(a) and f(b)!

☐ **Converse is FALSE (PYQ MAT-22 MCQ 22):** $f \in \mathcal{R}[a, b]$ does **not** imply f is monotone on [a,b]

$f(x) = \sin x$ on $[0, \pi]$ and $f(x) = x^2$ on [-1,1] are continuous (hence in $\mathcal{R}$) but not monotone; Thomae's function C8 is not monotone on any subinterval

Know f is increasing on [a,b] (for decreasing f, apply to -f and use Theorem $7.1.5(a)$); $\varepsilon > 0$ is given.

Want On n equal subintervals of length (b-a)/n, define step functions $\alpha \leq f \leq \omega$ with $\int_a^b (\omega - \alpha) = \dfrac{b-a}{n}(f(b) - f(a)) <$ ε.

PROOF OF THEOREM 7.2.8 (BARTLE P. 212)

1. Assume f is increasing on I = [a,b] (if f is decreasing, then -f is increasing, so -f $\in \mathcal{R}[a,b]$ and hence $f = -(-f) \in \mathcal{R}[a,b]$ by Theorem 7.1.5(a)). For $n \in \mathbb{N}$, partition I = [a,b] into n equal subintervals $I_k = [x_{k-1}, x_k]$ of length $x_k - x_{k-1} = (b-a)/n$ for $k = 1, \ldots, n$.
   Equal widths (b-a)/n allow us to factor out the common width and telescope the heights.

2. Since f is increasing on I_k, its minimum on I_k is attained at the left endpoint x_{k-1} and its maximum at the right endpoint x_k : $f(x_{k-1}) \leq f(x) \leq f(x_k)$ for all $x \in I_k$.
   Definition of increasing function (P10)—no continuity needed!

3. Define step functions $\alpha, \omega : [a,b] \to \mathbb{R}$ by $\alpha(x) := f(x_{k-1})$ and $\omega(x) := f(x_k)$ for $x \in [x_{k-1}, x_k) (k = 1, \ldots, n-1)$, and $\alpha(x) := f(x_{n-1}), \omega(x) := f(x_n)$ for $x \in [x_{n-1}, x_n]$. Then $\alpha, \omega \in \mathcal{R}[a,b]$ by Theorem 7.2.5 and $\alpha(x) \leq f(x) \leq \omega(x)$ on [a,b].
   Step-function integrability (Theorem 7.2.5, T14).

4. By Theorem 7.2.5, $\int_a^b \alpha = \dfrac{b-a}{n}(f(x_0) + f(x_1) + \cdots + f(x_{n-1}))$, $\int_a^b \omega = \dfrac{b-a}{n}(f(x_1) + \cdots + f(x_{n-1}) + f(x_n))$. Subtracting and cancelling the interior terms $f(x_1), \ldots, f(x_{n-1})$ gives $\int_a^b (\omega - \alpha) = \dfrac{b-a}{n}(f(x_n) - f(x_0)) = \dfrac{b-a}{n}(f(b) - f(a))$.
   Telescoping sum (M01, M05) with $x_0 = a$ and $x_n = b$.

5. Given $\varepsilon > 0$, choose $n \in \mathbb{N}$ such that $n > (b-a)(f(b) - f(a))/\varepsilon$. Then $\int_a^b (\omega - \alpha) < \varepsilon$. By the Squeeze Theorem 7.2.3, $f \in \mathcal{R}[a,b]$. ∎
   Archimedean property (P13) and Squeeze Theorem 7.2.3 (T12).

WORKED EXAMPLE · STAIRCASE FOR $f(x) = x^2$ ON [0,1] WITH N=4

With $P_4 = (0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1)$, the function $f(x) = x^2$ is increasing on [0,1]. The lower step heights are $0, \frac{1}{16}, \frac{1}{4}, \frac{9}{16}$ and upper step heights are $\frac{1}{16}, \frac{1}{4}, \frac{9}{16}, 1$, so:

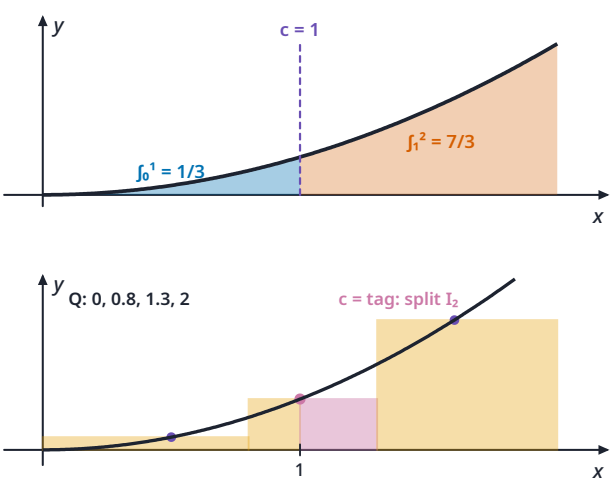
$$\int_0^1 \alpha = \frac{1}{4}(0 + \frac{1}{16} + \frac{4}{16} + \frac{9}{16}) = \frac{14}{64} = 0.21875, \qquad \int_0^1 \omega = \frac{1}{4}(\frac{1}{16} + \frac{4}{16} + \frac{9}{16} + 1) = \frac{30}{64} = 0.46875.$$

Their difference is $0.46875 - 0.21875 = 0.25 = \frac{1-0}{4}(f(1) - f(0))$, trapping the exact value $\int_0^1 x^2\,dx = \frac{1}{3} \approx 0.3333$.

In the model paper: **Q38** (PYQ MTS-26 Q19, MAT-22 Q30, MAT-22 MCQ 22).

T17 The Additivity Theorem

Splitting an integral at an interior point $c \in (a, b)$ (Bartle Theorem 7.2.9; PYQ MTS-26 Q25, MAT-22 Q37a, MTS-22 Q25).



What to notice: When c=1 falls inside the k-th subinterval $[x_{k-1}, x_k] = [0.8, 1.3]$ of $\dot{Q}$, we split that subinterval at c into ([0.8,1],1) and ([1,1.3],1) with tag c; only that single rectangle changes, by at most $2M(x_k - x_{k-1}) < \varepsilon/3$.
What it does not prove: You must prove both directions ($\Leftarrow$) and ($\Rightarrow$), and in ($\Leftarrow$) you must handle Case (ii) where c is not a partition point.

THEOREM 7.2.9 · ADDITIVITY THEOREM (BARTLE P. 213, EQ. (6)) · SAY IT WORD FOR WORD

Let f : $[a,b] \to \mathbb{R}$ and let $c \in (a,b)$. Then $f \in \mathcal{R}[a,b]$ if and only if its restrictions to [a,c] and to [c,b] are both Riemann integrable. In this case $\int_a^b f = \int_a^c f + \int_c^b f$.

In plain words: f is integrable on the whole interval [a,b] if and only if it is integrable on the left piece [a,c] and on the right piece [c,b], and the whole integral is the sum of the two piece integrals.

HYPOTHESES · THEOREM 7.2.9

☐ for (⇐): **both** restrictions $f|_{[a,c]} \in \mathcal{R}[a,c]$ and $f|_{[c,b]} \in \mathcal{R}[c,b]$

if f=0 on [0,1] and $f =$ Dirichlet C7 on [1,2], then $f|_{[0,1]} \in \mathcal{R}[0,1]$ but $f|_{[1,2]} \notin \mathcal{R}[1,2]$, so $f \notin \mathcal{R}[0,2]$

☐ boundedness of f on [a,b] in (⇐)

since $f|_{[a,c]} \in \mathcal{R}[a,c]$ and $f|_{[c,b]} \in \mathcal{R}[c,b]$, Theorem 7.1.6 implies each piece is bounded, so f is bounded on [a,b] by some M > 0 (state this in the exam—it carries a mark!)

Know (⇐)$f_1 = f|_{[a,c]} \in \mathcal{R}[a,c]$ with integral L_1, $f_2 = f|_{[c,b]} \in \mathcal{R}[c,b]$ with integral L_2.(⇒)$f \in \mathcal{R}[a,b]$.

Want (⇐) Show $|S(f;\dot{Q})(L_1+L_2)| < \varepsilon$ for $\|Q\| < \delta_\varepsilon$.(⇒) Use Cauchy 7.2.1 by adjoining the same tagged partition of [c,b] to $\dot{P}_1, \dot{Q}_1$.

SCRATCH · THE THREE $\varepsilon/3$ ERRORS IN (⇐) CASE (II)

When $c \in (x_{k-1}, x_k)$ is not a partition point of $\dot{Q}$, split I_k at c with tag c on both halves to get $\dot{Q}_1$ on [a,c] and $\dot{Q}_2$ on [c,b]. All other intervals match! So $S(f;\dot{Q}) - S(f_1;\dot{Q}_1) - S(f_2;\dot{Q}_2) = f(t_k)(x_k - x_{k-1}) - f(c)(c - x_{k-1}) - f(c)(x_k - c) = (f(t_k) - f(c))(x_k - x_{k-1})$. Bound this by $2M\|Q\| < \varepsilon/3$ by choosing $\delta_\varepsilon = \min\{\delta', \delta'', \varepsilon/(6M)\}$.

PROOF OF THEOREM 7.2.9 (BARTLE PP. 213–214)

($\Leftarrow$)

1. Suppose $f_1 := f|_{[a,c]} \in \mathcal{R}[a,c]$ with integral L_1 and $f_2 := f|_{[c,b]} \in \mathcal{R}[c,b]$ with integral L_2. Given $\varepsilon > 0$, choose $\delta' > 0$ such that $\|P_1\| < \delta' \Rightarrow |S(f_1; \dot{P}_1) - L_1| < \varepsilon/3$, and choose $\delta'' > 0$ such that $\|P_2\| < \delta'' \Rightarrow |S(f_2; \dot{P}_2) - L_2| < \varepsilon/3$.
   Definition 7.1.1 applied to f_1 and f_2 with tolerance $\varepsilon/3$.

2. By Theorem 7.1.6, f_1 is bounded on [a,c] and f_2 is bounded on [c,b], so f is bounded on [a,b]. Choose M > 0 such that $|f(x)| \leq M$ for all $x \in [a,b] (iff \equiv 0$, take M = 1), and define $\delta_\varepsilon := \min\{\delta',\, \delta'',\, \dfrac{\varepsilon}{6M}\} > 0$. Let $\dot{Q} = \{(I_i, t_i)\}_{i=1}^m$ be any tagged partition of [a,b] with $\|Q\| < \delta_\varepsilon$.
   Boundedness Theorem 7.1.6 on each piece; three-way minimum (M08).

3. **Case (i): c is a partition point of $\dot{Q}$.** Split $\dot{Q}$ at c into a tagged partition $\dot{Q}_1$ of [a,c] and a tagged partition $\dot{Q}_2$ of [c,b]. Since $S(f; \dot{Q}) = S(f_1; \dot{Q}_1) + S(f_2; \dot{Q}_2)$ and $\|Q_1\| < \delta'$, $\|Q_2\| < \delta''$, the Triangle Inequality and Step 1 give $|S(f; \dot{Q}) - (L_1 + L_2)| \leq |S(f_1; \dot{Q}_1) - L_1| + |S(f_2; \dot{Q}_2) L_2| < \dfrac{\varepsilon}{3} + \dfrac{\varepsilon}{3} < \varepsilon$.
   When c is already a cut point, no rectangle straddles c.

4. **Case (ii): c is not a partition point of $\dot{Q}$.** Then $c \in (x_{k-1}, x_k)$ for some $k \leq m$. Define tagged partitions $\dot{Q}_1$ of [a,c] and $\dot{Q}_2$ of [c,b] by $\dot{Q}_1 := \{(I_1, t_1), \ldots, (I_{k-1}, t_{k-1}), ([x_{k-1}, c], c)\}$, $\quad \dot{Q}_2 := \{([c, x_k], c), (I_{k+1}, t_{k+1}), \ldots, (I_m, t_m)\}$.
   Cut the straddling interval $[x_{k-1}, x_k]$ at c, using c as the tag for both pieces.

5. Since $(c - x_{k-1}) + (x_k - c) = x_k - x_{k-1}$, all terms of $S(f;\dot{Q})$ and $S(f_1;\dot{Q}_1) + S(f_2;\dot{Q}_2)$ cancel except the k-th term, giving $S(f;\dot{Q}) - S(f_1;\dot{Q}_1) - S(f_2;\dot{Q}_2) = (f(t_k) - f(c))(x_k - x_{k-1})$, and therefore $|S(f;\dot{Q}) - S(f_1;\dot{Q}_1) - S(f_2;\dot{Q}_2)| \leq (|f(t_k)| + |f(c)|)(x_k - x_{k-1}) \leq 2M\|Q\| < 2M \cdot \dfrac{\varepsilon}{6M} = \dfrac{\varepsilon}{3}$.
Triangle Inequality, $|f| \leq M$, and $\|Q\| < \varepsilon/(6M)$.

6. Since $\|Q_1\| \leq \|Q\| < \delta'$ and $\|Q_2\| \leq \|Q\| < \delta''$, Step 1 gives $|S(f_1;\dot{Q}_1)L_1| < \varepsilon/3$ and $|S(f_2;\dot{Q}_2) - L_2| < \varepsilon/3$. By the Triangle Inequality, $|S(f;\dot{Q}) - (L_1 + L_2)| \leq |S(f;\dot{Q}) - S(f_1;\dot{Q}_1) - S(f_2;\dot{Q}_2)| + |S(f_1;\dot{Q}_1) - L_1| + |S(f_2;\dot{Q}_2) - L_2| < \dfrac{\varepsilon}{3} + \dfrac{\varepsilon}{3} + \dfrac{\varepsilon}{3} = \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $f \in \mathcal{R}[a,b]$ and $\int_a^b f = L_1 + L_2 = \int_a^c f + \int_c^b f$.
Three-part Triangle Inequality and Definition 7.1.1 (uniqueness 7.1.2 identifies the integral as $L_1 + L_2$).

($\Rightarrow$)

1. Suppose $f \in \mathcal{R}[a,b]$ and let $\varepsilon > 0$ be given. By the Cauchy Criterion 7.2.1, there exists $\eta_\varepsilon > 0$ such that tagged partitions $\dot{P}, \dot{Q}$ of [a,b] with norms $< \eta_\varepsilon$ satisfy $|S(f;\dot{P}) - S(f;\dot{Q})| < \varepsilon$.
Cauchy Criterion 7.2.1($\Rightarrow$) applied to f on [a,b].

2. Let $\dot{P}_1, \dot{Q}_1$ be any tagged partitions of [a,c] with $\|P_1\| < \eta_\varepsilon$ and $\|Q_1\| < \eta_\varepsilon$. Choose a single tagged partition $\dot{R}$ of [c,b] with $\|R\| < \eta_\varepsilon$, and adjoin $\dot{R}$ to $\dot{P}_1$ and $\dot{Q}_1$ to obtain tagged partitions $\dot{P} = \dot{P}_1 \cup \dot{R}$ and $\dot{Q} = \dot{Q}_1 \cup \dot{R}$ of [a,b] with $\|P\| < \eta_\varepsilon$ and $\|Q\| < \eta_\varepsilon$.
Using the exact same $\dot{R}$ on [c,b] makes the [c,b] sums cancel when we subtract!

3. Then $S(f_1;\dot{P}_1) - S(f_1;\dot{Q}_1) = (S(f_1;\dot{P}_1) + S(f;\dot{R})) - (S(f_1;\dot{Q}_1) + S(f;\dot{R})) = S(f;\dot{P})S(f;\dot{Q})$, so $|S(f_1;\dot{P}_1) - S(f_1;\dot{Q}_1)| = |S(f;\dot{P}) - S(f;\dot{Q})| < \varepsilon$. By the Cauchy Criterion 7.2.1, $f_1 \in \mathcal{R}[a,c]$. By the same argument, $f_2 \in \mathcal{R}[c,b]$. Finally, the formula $\int_a^b f = \int_a^c f + \int_c^b f$ follows from the ($\Leftarrow$) part already proved. ∎
Cauchy Criterion 7.2.1($\Leftarrow$) on [a,c] and [c,b], then apply ($\Leftarrow$) for the equality.

In the model paper: **Q39** (PYQ MTS-26 Q25, MAT-22 Q37a, MTS-22 Q25).

T18 Restriction and subdivision corollaries

Restricting $f \in \mathcal{R}[a,b]$ to $[c,d] \subseteq [a,b]$ and splitting [a,b] at m points (Bartle Corollaries 7.2.10–7.2.11, Exercises 7.2.11–7.2.13; PYQ MAT-22 Q37b, MAT-22 MCQ 24, 25, MAT-21 MCQ 18).

COROLLARY 7.2.10 & COROLLARY 7.2.11 (BARTLE P. 214) · SAY THEM WORD FOR WORD

Corollary 7.2.10: If $f \in \mathcal{R}[a,b]$ and if $[c,d] \subseteq [a,b]$, then the restriction of f to [c,d] is in $\mathcal{R}[c,d]$.

Corollary 7.2.11: If $f \in \mathcal{R}[a,b]$ and if $a = c_0 < c_1 < \cdots < c_m = b$, then the restrictions of f to each subinterval $[c_{i-1}, c_i]$ are Riemann integrable and $\int_a^b f = \sum_{i=1}^{m} \int_{c_{i-1}}^{c_i} f$.

In plain words: integrability is inherited by every closed subinterval $[c,d] \subseteq [a,b]$, and an integral splits across any finite chain of cut points $c_0 < c_1 < \cdots < c_m$.

PROOFS OF COROLLARIES 7.2.10 AND 7.2.11 (BARTLE P. 214)

1. **Proof of 7.2.10 (PYQ MAT-22 Q37b):** Let $f \in \mathcal{R}[a,b]$ and $a \le c < d \le b$. If c > a, then $c \in (a,b)$, so Theorem 7.2.9($\Rightarrow$) implies that the restriction $f|_{[c,b]}$ belongs to $\mathcal{R}[c,b] (if c = a$, this holds by hypothesis). Next, if d < b, then $d \in (c,b)$, so applying Theorem 7.2.9($\Rightarrow$) to $f|_{[c,b]}$ at the point d shows that the restriction $f|_{[c,d]}$ belongs to $\mathcal{R}[c,d] (if d = b$, we already have $f|_{[c,b]} \in \mathcal{R}[c,b])$. ∎

   Two applications of the ($\Rightarrow$) direction of the Additivity Theorem 7.2.9 (T17).

2. **Proof of 7.2.11 (by induction on m; omitted in Bartle):** For $m = 1, [c_0, c_1] = [a,b]$, which is trivial; for m = 2, this is Theorem 7.2.9. Assume the result holds for m - 1 subintervals. Given $a = c_0 < c_1 < \cdots < c_{m-1} < c_m = b$, apply Theorem 7.2.9 at $c = c_{m-1} \in (a,b) : f$ is integrable on $[a, c_{m-1}]$ and on $[c_{m-1}, c_m]$, with $\int_a^b f = \int_a^{c_{m-1}} f + \int_{c_{m-1}}^{c_m} f$. Applying the induction hypothesis to $f|_{[a,c_{m-1}]}$ on $a = c_0 < \cdots < c_{m-1}$ gives $\int_a^b f = \sum_{i=1}^{m} \int_{c_{i-1}}^{c_i} f$.

   Mathematical induction on m using Theorem 7.2.9. ∎

IMPORTANT EXAM RESULT · INTEGRABLE ON EVERY [C,B] (BARTLE EX 7.2.11–7.2.13; PYQ MAT-22 MCQ 24, 25, Q28)

Statement (Bartle Ex 7.2.11): If f is **bounded** by M on [a,b] and $f|_{[c,b]} \in \mathcal{R}[c,b]$ for every $c \in (a,b)$, then $f \in \mathcal{R}[a,b]$ and $\lim_{c\to a^+} \int_c^b f = \int_a^b f$.

Proof: Given $\varepsilon > 0$, choose $c \in (a,b)$ with c - a < $\varepsilon/(4M)$. Define $\alpha_c(x) :=$ -M and $\omega_c(x) := M$ on [a,c), and $\alpha_c(x) := \omega_c(x) := f(x)$ on [c,b]. By Theorem 7.2.9($\Leftarrow$), $\alpha_c, \omega_c \in \mathcal{R}[a,b]$, $\alpha_c \le f \le \omega_c$, and $\int_a^b(\omega_c - \alpha_c) = \int_a^c 2M + \int_c^b 0 = 2M(c-a) < \varepsilon/2 < \varepsilon$. By the Squeeze Theorem 7.2.3, $f \in \mathcal{R}[a,b]$. Moreover, $|\int_a^b f - \int_c^b f| = |\int_a^c f| \le M(c-a) \to 0$ as $c \to a^+$.

Application to $\sin(1/x)$ (Bartle Ex 7.2.12; PYQ MAT-22 MCQ 25): Let $g(x) := \sin(1/x)$ for $x \in (0,1]$ and g(0) := 0. Then $|g(x)| \le 1$ on [0,1] (bounded!), and for every $c \in (0,1)$, g is continuous on [c,1] and hence in $\mathcal{R}[c,1]$ by Theorem 7.2.7. By Exercise 7.2.11, $g \in \mathcal{R}[0,1]$.

Why boundedness cannot be dropped (Bartle Ex 7.2.13): f(x) = 1/x on (0,1], f(0)=0 C16 is continuous (hence in $\mathcal{R}[c,1]$) for every $c \in (0,1)$, but f is unbounded on [0,1], so $f \notin \mathcal{R}[0,1]$.

In the model paper: **Q27** (PYQ MAT-22 Q37b), **Q28** (PYQ MAT-22 MCQ 24, 25, MAT-21 MCQ 18).

T19 Oriented integrals and additivity for any three points

> Defining $\int_\beta^\alpha f := -\int_\alpha^\beta f$ and $\int_\alpha^\alpha f := 0$ so additivity holds in any order (Bartle Definition 7.2.12, Theorem 7.2.13).

DEFINITION 7.2.12 & THEOREM 7.2.13 (BARTLE PP. 214–215, EQ. (8)) · SAY THEM WORD FOR WORD

Definition 7.2.12: If $f \in \mathcal{R}[a,b]$ and if $\alpha, \beta \in [a,b]$ with $\alpha < \beta$, we define $\int_\beta^\alpha f := -\int_\alpha^\beta f \qquad \text{and} \qquad \int_\alpha^\alpha f := 0.$

Theorem 7.2.13: If $f \in \mathcal{R}[a,b]$ and if α, β, γ are any numbers in [a,b], then $\int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f$, in the sense that the existence of any two of these integrals implies the existence of the third and the equality.

In plain words: integrating backwards from β to α flips the sign, integrating from α to α gives 0, and with those two conventions $\int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f$ holds no matter which of α, β, γ is smallest, middle, or largest.

PROOF OF THEOREM 7.2.13 (BARTLE PP. 214–215)

1. If any two of the numbers α, β, γ are equal, the formula $\int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f$ holds immediately by Definition 7.2.12 (for instance, if $\gamma = \beta$, it reads $\int_\alpha^\beta f = \int_\alpha^\beta f + 0$; if $\alpha = \beta$, it reads $0 = \int_\alpha^\gamma f - \int_\alpha^\gamma f$).
   Definition 7.2.12 handles all cases with repeated points.

2. Assume α, β, γ are all distinct. For any triple, define the cyclic sum $L(\alpha, \beta, \gamma) := \int_\alpha^\beta f + \int_\beta^\gamma f + \int_\gamma^\alpha f$. Since $\int_\gamma^\alpha f = -\int_\alpha^\gamma f$, the desired equation $\int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f$ holds if and only if $L(\alpha, \beta, \gamma) = 0$.
   Move $\int_\alpha^\gamma f + \int_\gamma^\beta f$ to the left-hand side using $\int_\alpha^\gamma f = -\int_\gamma^\alpha f$.

3. When $\alpha < \gamma < \beta$, the Additivity Theorem 7.2.9 gives $\int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f$, so $L(\alpha, \beta, \gamma) = 0$.
   Theorem 7.2.9 (T17) for the standard ordering $\alpha < \gamma < \beta$.

4. Cycling the three letters leaves the sum unchanged: $L(\beta, \gamma, \alpha) = L(\gamma, \alpha, \beta) = L(\alpha, \beta, \gamma)$. Swapping two adjacent letters reverses the orientation of every integral and negates the sum: $L(\beta, \alpha, \gamma) = L(\alpha, \gamma, \beta) = L(\gamma, \beta, \alpha) = -L(\alpha, \beta, \gamma)$.
   Commutativity of addition and $\int_u^v f = -\int_v^u f$.

5. Every one of the 6 possible orderings of three distinct numbers is obtained from $\alpha < \gamma < \beta$ by a cyclic shift or a swap, so $L(\alpha, \beta, \gamma) = 0$ for all configurations of α, β, γ. ∎
   0 and -0 are both 0.

WORKED EXAMPLE · NUMERICAL CHECK FOR $\alpha = 3, \beta = 1, \gamma = 0$ (MODEL PAPER Q9)

Let f(x) = x on [0,3], and take $\alpha = 3, \beta = 1, \gamma = 0 (so \gamma < \beta < \alpha)$. Using $\int_a^b x\,dx = \frac{1}{2}(b^2 - a^2)$ and Definition 7.2.12:

- Left-hand side: $\int_3^1 x\,dx = -\int_1^3 x\,dx = -\frac{1}{2}(3^2 - 1^2) =$ -4.
- Right-hand side: $\int_3^0 x\,dx + \int_0^1 x\,dx = -\int_0^3 x\,dx + \int_0^1 x\,dx = -\frac{9}{2} + \frac{1}{2} =$ -4. Both sides equal -4.

In the model paper: **Q9** .

The Fundamental Theorem

T20 Fundamental Theorem of Calculus (First Form)

index ↑

Evaluating $\int_a^b f = F(b)F(a)$ using the Mean Value Theorem (Bartle Theorem 7.3.1; PYQ MTS-26 Q8, MAT-22 MCQ 27, MAT-21 MCQ 19).

THEOREM 7.3.1 · FUNDAMENTAL THEOREM OF CALCULUS, FIRST FORM (BARTLE P. 216, EQ. (1)) · SAY IT WORD FOR WORD

Suppose there is a finite set E in [a,b] and functions f, F : $[a,b] \to \mathbb{R}$ such that:

(a) F is continuous on [a,b];

(b) F'(x) = f(x) for all $x \in [a,b] \setminus E$;

(c) f belongs to $\mathcal{R}[a,b]$.

Then we have $\int_a^b f = F(b) - F(a)$.

In plain words: if F is continuous on the whole closed interval [a,b], its derivative F' equals f except possibly at finitely many corner/endpoint exceptions E, and f is Riemann integrable, then the integral of f is simply F(b) - F(a).

HYPOTHESES · THEOREM 7.3.1 (EACH ONE IS ESSENTIAL!)

- ☐ (a) F is **continuous on all of [a,b]** (even at points of E)
  let $F(x) = \operatorname{sgn}(x)$ on [-1,1] and f(x) = 0; then F'(x) = 0 = f(x) on $[-1,1] \setminus \{0\} (E = \{0\})$ and $f \in \mathcal{R}[-1,1]$, but F jumps at 0 and $\int_{-1}^{1} 0 = 0 \neq 2 = F(1) - F(-1)$
- ☐ (b) F'(x) = f(x) for all $x \in [a,b] \setminus E$ where E is **finite**
  if E is uncountable (e.g. the Cantor function F on [0,1], which is continuous with F'=0 off the Cantor set), $\int_0^1 0 = 0 \neq 1 = F(1) - F(0)$
- ☐ (c) f belongs to $\mathcal{R}[a,b]$(**not automatic** even when $E = \varnothing$!)
  $K(x) = x^2 \cos(1/x^2), K(0) = 0$ C19 is differentiable at every point of $[0,1] (E = \varnothing)$, yet f = K' is unbounded near 0, so $K' \notin \mathcal{R}[0,1]$. Likewise $H(x) = 2\sqrt{x}$ has $H'(x) = 1/\sqrt{x} \notin \mathcal{R}[0,b]$

SKELETON · APPLY FTC 1 TO EVALUATE $\int_a^b f$

Define F(x) := ? on [a,b] and let E := ? (finite). Check: (a) F is continuous on [a,b]; (b) F'(x) = f(x) for all $x \in [a,b] \setminus E$; (c) $f \in \mathcal{R}[a,b]$ because ?. By FTC 1 (Theorem 7.3.1), $\int_a^b f = F(b) - F(a) =$?. ∎

Know F continuous on [a,b], F'(x) = f(x) on (a,b) (case $E = \{a, b\}$), and $f \in \mathcal{R}[a,b]$.

Want Show $|F(b) - F(a) - \int_a^b f| < \varepsilon$ for every $\varepsilon > 0$ by rewriting F(b) - F(a) as a Riemann sum $S(f; \dot{P}_u)$ via the Mean Value Theorem.

PROOF OF THEOREM 7.3.1 (BARTLE PP. 216–217)

1. We first prove the case where $E = \{a, b\}$. Let $\varepsilon > 0$ be given. Since $f \in \mathcal{R}[a, b]$ by hypothesis (c), there exists $\delta_\varepsilon > 0$ such that if $\dot{P}$ is any tagged partition of [a,b] with $\|P\| < \delta_\varepsilon$, then $|S(f; \dot{P}) - \int_a^b f| < \varepsilon$.
   Definition 7.1.1 (T03) applied to $f \in \mathcal{R}[a, b]$.

2. Let $P = \{[x_{i-1}, x_i]\}_{i=1}^n$ be a partition of [a,b] with $\|P\| < \delta_\varepsilon$. Since F is continuous on $[x_{i-1}, x_i]$ by (a) and differentiable on $(x_{i-1}, x_i) \subseteq [a, b] \setminus E$ by (b), the Mean Value Theorem 6.2.4 (P12) applied to F on $[x_{i-1}, x_i]$ provides a point $u_i \in (x_{i-1}, x_i)$ such that $F(x_i) - F(x_{i-1}) = F'(u_i)(x_i - x_{i-1}) = f(u_i)(x_i - x_{i-1}) \quad \text{for } i = 1, \ldots, n.$
   Mean Value Theorem 6.2.4 on each $[x_{i-1}, x_i]$; since $u_i \in (x_{i-1}, x_i) \subseteq (a, b), u_i \notin E$, so $F'(u_i) = f(u_i)$.

3. Adding these n equations and telescoping the left-hand side gives $F(b) - F(a) = \sum_{i=1}^n (F(x_i) - F(x_{i-1})) = \sum_{i=1}^n f(u_i)(x_i - x_{i-1}) = S(f; \dot{P}_u)$, where $\dot{P}_u := \{([x_{i-1}, x_i], u_i)\}_{i=1}^n$ is a tagged partition of [a,b] with norm $\|P\| < \delta_\varepsilon$.
   Telescoping sum (M01, M05) and definition of Riemann sum (T02).

4. Substituting $S(f; \dot{P}_u) = F(b) - F(a)$ into the inequality of Step 1 yields $|F(b) - F(a) - \int_a^b f| < \varepsilon$. Since $\varepsilon > 0$ is arbitrary, we conclude that $\int_a^b f = F(b) - F(a)$.
   Arbitrary ε rule (P05).

5. **General finite set E (Exercise 7.3.1):** Let $a = c_0 < c_1 < \cdots < c_m = b$ be the points of $E \cup \{a, b\}$ in increasing order. On each subinterval $[c_{k-1}, c_k]$, F is continuous, F'(x) = f(x) on (c_{k-1}, c_k), and $f|_{[c_{k-1}, c_k]} \in \mathcal{R}[c_{k-1}, c_k]$ by Corollary 7.2.10. Applying the case just proved on each $[c_{k-1}, c_k]$ and summing via Corollary 7.2.11 gives $\int_a^b f = \sum_{k=1}^m (F(c_k) - F(c_{k-1})) = F(b) - F(a)$. ∎
   Corollaries 7.2.10 and 7.2.11 (T18) plus telescoping over the m pieces.

EXAM TRAP · PYQ MAT-22 MCQ 27, MAT-21 MCQ 19

In the MCQ asking for $\int_a^b f(x)\,dx$ when f is continuous and F is an antiderivative of f, the answer is simply F(b) - F(a)—do **not** multiply by (b-a)!

In the model paper: **Q14** , **Q29** (PYQ MTS-26 Q8, MAT-22 MCQ 27, MAT-21 MCQ 19).

T21 Examples and limits of FTC 1: when it works and when it fails

> Checking all three hypotheses (a), (b), (c) in practice (Bartle Examples 7.3.2(a)–(e), Exercises 7.3.2–7.3.4, 7.3.8; PYQ MAT-22 Q35, MAT-22 MCQ 28, 29, MAT-21 MCQ 20).

EXAMPLES 7.3.2(A)–(E) (BARTLE PP. 217–218)

(a) $\int_a^b x\,dx = \frac{1}{2}(b^2 - a^2)$**:** $F(x) := \frac{1}{2}x^2$ has F'(x) = x for all $x \in [a,b] (E = \varnothing)$, and f(x) = x is continuous so $f \in \mathcal{R}[a,b] (7.2.7)$. More generally $(Ex 7.3.2), H_n(x) = \dfrac{x^{n+1}}{n+1}$ gives $\int_a^b x^n\,dx = \dfrac{b^{n+1} - a^{n+1}}{n+1}$ with $E = \varnothing$.

(b) $\int_a^b \dfrac{1}{x^2+1}\,dx = \operatorname{Arctan} b - \operatorname{Arctan} a$**:** $G(x) := \operatorname{Arctan} x$ has $G'(x) = \dfrac{1}{x^2+1}$ on $[a,b] (E = \varnothing)$, which is continuous and hence in $\mathcal{R}[a,b]$.

(c) $\int_{-10}^{10} \operatorname{sgn}(x)\,dx = 0$**:** $A(x) := |x|$ is continuous on $[-10,10]$, $A'(x) = \operatorname{sgn}(x)$ on $[-10,10] \setminus \{0\} (E = \{0\})$, and sgn is a step function so $\operatorname{sgn} \in \mathcal{R}[-10,10] (7.2.5)$. FTC 1 gives A(10) - A(-10) = 10 10 = 0.

(d) Failure for $H(x) := 2\sqrt{x}$ **on [0,b]:** H is continuous on [0,b] and $H'(x) = 1/\sqrt{x}$ on $(0,b) (E = \{0\})$, but h = H' is **unbounded** on (0,b), so $h \notin \mathcal{R}[0,b]$ (Theorem 7.1.6) no matter how h(0) is defined. Hypothesis (c) fails; FTC 1 does not apply.

(e) Failure for $K(x) := x^2 \cos(1/x^2) (x \in (0,1]), K(0) := 0$ C19**:** By the Product and Chain Rules, $K'(x) = 2x\cos(1/x^2) + (2/x)\sin(1/x^2)$ for x > 0, and at $0, \lim_{x\to 0} \dfrac{K(x) - K(0)}{x} = \lim_{x\to 0} x\cos(1/x^2) = 0$, so K'(0) = 0. Thus K is continuous and differentiable at **every** point of $[0,1] (E = \varnothing)$! However, K' is unbounded on $[0,1] (at x_m = (\dfrac{\pi}{2} + 2m\pi)^{-1/2} \to 0$, the second term is $2/x_m \to \infty)$, so K' $\notin \mathcal{R}[0,1]$ by Theorem 7.1.6 and FTC 1 does not apply.

In plain words: having an antiderivative F is never enough by itself—you must always check that F is continuous on [a,b] and that f = F' is bounded and integrable!

WORKED EXAMPLES · THREE PAST-PAPER FTC 1 PROBLEMS (Q14, Q15)

1. Bartle Ex 7.3.3 (PYQ MAT-22 MCQ 28, MAT-21 MCQ 20): Let g(x) = x for $|x| \geq 1$ and g(x) = -x for $|x|$ < 1, and let $G(x) = \frac{1}{2}|x^2 - 1|$ on [-2,3]. Then G is continuous on [-2,3]; for $|x| > 1, G(x) = \frac{1}{2}(x^2 - 1)$ has G'(x) = x = g(x), and for $|x|$ < 1, $G(x) = \frac{1}{2}(1 - x^2)$ has G'(x) = -x = g(x), so G'(x) = g(x) off the finite set $E = \{-1, 1\}$. Also g is continuous on [-2,-1], [-1,1], and [1,3] (up to the jump at -1), so $g \in \mathcal{R}[-2, 3]$. By FTC 1,

$\int_{-2}^{3} g(x)\,dx = G(3) - G(-2) = \frac{1}{2}|9 - 1| - \frac{1}{2}|4 - 1| = 4 - \frac{3}{2} = \frac{5}{2}$.

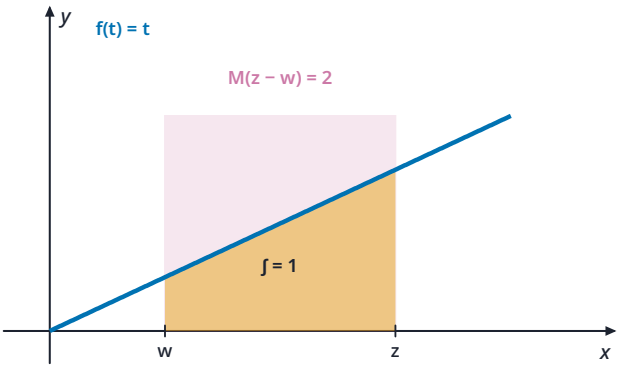
2. Bartle Ex 7.3.4 (PYQ MAT-22 MCQ 29): Let $B(x) = -\frac{1}{2}x^2$ for x < 0 and $B(x) = \frac{1}{2}x^2$ for $x \geq 0$. For x < 0, B'(x) = -x = $|x|$; for $x > 0, B'(x) = x = |x|$; and at x = 0, both one-sided derivatives equal $0 = |0|$. Thus $B'(x) = |x|$ for all $x \in [a, b] (E = \varnothing)$, and $|x|$ is continuous so $|x| \in \mathcal{R}[a, b]$. By FTC $1, \int_a^b |x|\,dx = B(b) - B(a)$.

3. PYQ MAT-22 Q35: Evaluate $\int_0^1 x^2(1 - \sqrt{x})\,dx$. Write $f(x) = x^2 - x^{5/2}$, which is continuous on [0,1] and hence in $\mathcal{R}[0, 1]$ (Theorem 7.2.7). Define $F(x) := \frac{1}{3}x^3 - \frac{2}{7}x^{7/2}$ on [0,1]; then F is continuous on [0,1] and $F'(x) = x^2 - x^{5/2} = f(x)$ on $[0, 1] (E = \varnothing)$. By FTC 1, $\int_0^1 x^2(1 - \sqrt{x})\,dx = F(1) - F(0) = (\frac{1}{3} - \frac{2}{7}) - 0 = \frac{1}{21}$.

In the model paper: **Q10** , **Q14** , **Q15** (PYQ MAT-22 Q35, MAT-22 MCQ 28, 29, MAT-21 MCQ 20).

T22 The indefinite integral is Lipschitz, hence continuous

> Accumulating area from a to z smooths out any integrable function $f \in \mathcal{R}[a, b]$ into a Lipschitz function $F(z) = \int_a^z f$ (Bartle Definition 7.3.3, Theorem 7.3.4; PYQ MAT-21 Q6).



What to notice: For f(t)=t on [0,2] (M=2) between w=0.5 and z=1.5, the shaded area $F(z) - F(w) = \int_w^z f = 1$ sits inside the bounding box of height M=2 and width z-w=1 (area 2). **What it does not prove:** Being Lipschitz makes F continuous everywhere, but it does not guarantee F is differentiable where f jumps.

DEFINITION 7.3.3 & THEOREM 7.3.4 (BARTLE P. 218, EQ. (3)) · SAY THEM WORD FOR WORD

Definition 7.3.3: If $f \in \mathcal{R}[a,b]$, the function defined by $F(z) := \int_a^z f \quad \text{for } z \in [a,b]$ is called the **indefinite integral** of f with **basepoint** a.

Theorem 7.3.4: The indefinite integral F defined by $F(z) := \int_a^z f$ is continuous on [a,b]. In fact, if $|f(x)| \leq M$ for all $x \in [a,b]$, then $|F(z) - F(w)| \leq M|z - w| \quad \text{for all } z, w \in [a,b]$.

In plain words: because $f \in \mathcal{R}[a,b]$ is bounded by M, the area added between w and z is at most height M times width $|z-w|$, so F is Lipschitz (and therefore uniformly continuous) on [a,b]—even if f itself is discontinuous!

HYPOTHESES · THEOREM 7.3.4

☐ $f \in \mathcal{R}[a,b]$ (which by Theorem 7.1.6 automatically supplies M > 0 with $|f(x)| \leq M$)

no continuity of f is required here: even for Thomae C8 or sgn C9, the indefinite integral F is continuous on [a,b]

PROOF OF THEOREM 7.3.4 (BARTLE P. 218)

1. By Corollary 7.2.10, for each $z \in [a,b]$ the restriction of f to [a,z] is in $\mathcal{R}[a,z]$, so $F(z) = \int_a^z f$ is well-defined (with $F(a) = \int_a^a f = 0$). By Theorem 7.1.6, f is bounded on [a,b], say $|f(x)| \le M$ for all $x \in [a,b]$.
   Corollary 7.2.10 (T18) and Boundedness Theorem 7.1.6 (T08).

2. Assume $w, z \in [a,b]$ with $w \le z$. By the Additivity Theorem 7.2.9, $F(z) = \int_a^z f = \int_a^w f + \int_w^z f = F(w) + \int_w^z f$, so $F(z) - F(w) = \int_w^z f$.
   Additivity Theorem 7.2.9 (T17) applied on [a,z] split at w.

3. Since -M $\le f(x) \le M$ for all $x \in [w,z]$, Theorem 7.1.5(c) and Example 7.1.4(a) give $-M(z-w) \le \int_w^z f \le M(z-w)$, whence $|F(z) - F(w)| = |\int_w^z f| \le M|z-w|$(*and* by symmetry the same holds if z < w).
   Monotonicity of the integral (7.1.5c) and constant integral (7.1.4a).

4. Thus F satisfies a Lipschitz condition with constant M on [a,b], and hence F is continuous on [a,b] (given $\varepsilon > 0$, take $\delta := \varepsilon/(M+1)$). ∎
   Every Lipschitz function is continuous (P09).

WORKED EXAMPLE · CHANGING THE BASEPOINT (BARTLE EXERCISES 7.3.6, 7.3.9)

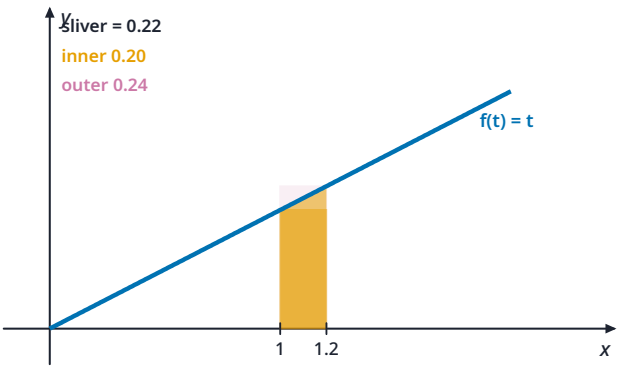
If $c \in [a,b]$ and $F_c(z) := \int_c^z f$ is the indefinite integral with basepoint c, Theorem 7.2.13 gives $F_a(z) = \int_a^z f = \int_a^c f + \int_c^z f = F_a(c) + F_c(z)$, so $F_c(z) = F_a(z) - F_a(c)$: changing the basepoint simply shifts F by the constant $-F_a(c)$. Similarly, $\int_x^b f = F_a(b) - F_a(x)$ and $\int_x^{\sin x} f = F_a(\sin x) - F_a(x)$.

In the model paper: **Q11** (PYQ MAT-21 Q6).

T23 Fundamental Theorem of Calculus (Second Form)

Differentiating the area accumulator $F(z) = \int_a^z f$ at a point c where f is continuous gives F'(c) = f(c) (Bartle Theorems 7.3.5–

7.3.6; PYQ MTS-25 Q25, MAT-21 Q37a, MAT-22 Q33, MAT-22 MCQ 30, MTS-22 Q14).



What to notice: Over the thin slice [c,c+h] (c=1, h=0.2), continuity of f traps f(x) between $f(c)-\varepsilon$ and $f(c)+\varepsilon$, so the sliver area $F(c+h)-F(c)=\int_c^{c+h} f$ is trapped between $(f(c)-\varepsilon)h$ and $(f(c)+\varepsilon)h$; dividing by h traps the difference quotient within ε of f(c). **What it does not prove:** If f jumps at c, the left and right slivers have different heights and F'(c) fails to exist.

THEOREM 7.3.5 & THEOREM 7.3.6 · FTC SECOND FORM (BARTLE P. 219) · SAY THEM WORD FOR WORD

Theorem 7.3.5 (Fundamental Theorem of Calculus, Second Form): Let $f \in \mathcal{R}[a,b]$ and let f be continuous at a point $c \in [a,b]$. Then the indefinite integral $F(z) := \int_a^z f$ is differentiable at c and F'(c) = f(c).

Theorem 7.3.6: If f is continuous on [a,b], then the indefinite integral $F(z) := \int_a^z f$ is differentiable on [a,b] and F'(x) = f(x) for all $x \in [a,b]$.

In plain words: at every point c where an integrable function f is continuous, the rate of change of the accumulated area $F(z)=\int_a^z f$ equals the height f(c) of the curve at c. If f is continuous on the whole interval [a,b], then F is an antiderivative of f on [a,b].

HYPOTHESES · THEOREM 7.3.5

- ☐ $f \in \mathcal{R}[a,b]$
  needed so that $F(z) = \int_a^z f$ is defined on [a,b]
- ☐ f is **continuous at the point c**
  for $f = \operatorname{sgn}$ on [-1,1] **C20**, f is discontinuous at c=0 and $F(x) = |x| - 1$ is not differentiable at 0; for Thomae h **C8**, $H(x) = \int_0^x h \equiv 0$ is differentiable everywhere, but at every rational $c \in \mathbb{Q} \cap [0,1], H'(c) = 0 \neq h(c)$!

Know $f \in \mathcal{R}[a,b]$, f continuous at $c \in [a,b]$, and $F(z) = \int_a^z f$.

Want Given $\varepsilon > 0$, find $\eta_\varepsilon > 0$ so that $0 < |h| < \eta_\varepsilon \Rightarrow$
$\left|\frac{F(c+h) - F(c)}{h} - f(c)\right| \leq \varepsilon$.

PROOF OF THEOREM 7.3.5 (BARTLE PP. 219–220)

1. Suppose first that $c \in [a, b)$ and consider the right-hand derivative of F at c. Since f is continuous at c, given $\varepsilon > 0$ there exists $\eta_\varepsilon > 0$ such that if $c \leq x < c + \eta_\varepsilon (x \in [a, b])$, then $f(c) - \varepsilon <$ f(x) $< f(c) + \varepsilon$.
   Definition of continuity of f at c (Module 1, 5.1.1).

2. Let h be any number with 0 < h < $\eta_\varepsilon (and c + h \leq b)$. By the Additivity Theorem 7.2.9 (and Corollary 7.2.10), f is integrable on [a,c], [a,c+h] and [c,c+h], and $F(c + h) - F(c) = \int_a^{c+h} f - \int_a^c f = \int_c^{c+h} f$.
   Additivity Theorem 7.2.9 (T17).

3. On the interval [c,c+h], the inequality of Step 1 holds, so Theorem 7.1.5(c) and Example 7.1.4(a) yield $(f(c) - \varepsilon) \cdot h \leq F(c + h) - F(c) = \int_c^{c+h} f \leq (f(c) + \varepsilon) \cdot h$.
   Monotonicity of the integral (7.1.5c) with constant bounds $f(c) \pm \varepsilon$ over an interval of length h.

4. Dividing through by h > 0 and subtracting f(c) gives $\left|\frac{F(c+h) - F(c)}{h} - f(c)\right| \leq \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $\lim_{h \to 0^+} \frac{F(c+h) - F(c)}{h} = f(c)$.
   h > 0 preserves inequalities; definition of right-hand limit.

5. If $c \in (a, b]$, the left-hand limit $\lim_{h \to 0^-} \frac{F(c+h) - F(c)}{h} = f(c)$ is proved in the exact same way ($for - \eta_\varepsilon <$ h < 0, on [c+h,c] we have $(f(c) - \varepsilon)(-h) \leq \int_{c+h}^c f = -(F(c + h) - F(c)) \leq (f(c) + \varepsilon)(-h)$, and dividing by -h > 0 gives the same bound). Thus F is differentiable at c and F'(c) = f(c). ∎
   Agreement of left-hand and right-hand derivatives at interior points (and one-sided derivative at an endpoint a or b). Theorem 7.3.6 follows immediately by applying Theorem 7.3.5 at every $x \in [a, b]$.

VARIABLE-LIMIT INTEGRALS (LEIBNIZ RULE, BARTLE EX 7.3.10–11, 7.3.15–16; PYQ MTS-22 Q14, MAT-22 MCQ 30)

1. Differentiating $\int_a^{v(x)} f(t)\,dt$ (Bartle Ex 7.3.10; PYQ MTS-22 Q14 "Leibniz rule"): If f is continuous on [a,b] and v : $[c,d] \to [a,b]$ is differentiable, define $F(u) := \int_a^u f(t)\,dt$. By Theorem 7.3.6, F'(u) = f(u). Since $G(x) := \int_a^{v(x)} f(t)\,dt = (F \circ v)(x)$, the Chain Rule 6.1.6 gives $G'(x) = F'(v(x))\,v'(x) = f(v(x))\,v'(x)$. More generally, for two differentiable limits u(x), v(x),

$$\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)\,dt = \frac{d}{dx}(F(v(x)) - F(u(x))) = f(v(x))\,v'(x) f(u(x))\,u'(x).$$

2. Bartle Ex 7.3.11 (Model Paper Q19):

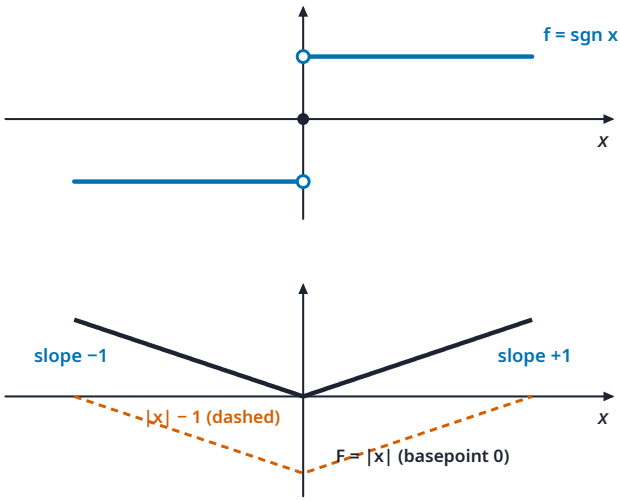
- $(a) F(x) = \int_0^{x^2} (1+t^3)^{-1}\,dt$: here $f(t) = (1+t^3)^{-1}$ is continuous on [0,1] and $v(x) = x^2$ has v'(x) = 2x, so $F'(x) = f(x^2) \cdot 2x = \dfrac{2x}{1+x^6}$.
- $(b) F(x) = \int_{x^2}^{x} \sqrt{1+t^2}\,dt$: here $f(t) = \sqrt{1+t^2}$ is continuous, so $F'(x) = f(x) \cdot 1 - f(x^2) \cdot 2x = \sqrt{1+x^2} - 2x\sqrt{1+x^4}$.

3. Bartle Ex 7.3.16 (PYQ MAT-22 MCQ 30, Model Paper Q17): Suppose f : $[0,1] \to \mathbb{R}$ is continuous and $\int_0^x f = \int_x^1 f$ for all $x \in [0,1]$. Since f is continuous, $f \in \mathcal{R}[0,1]$. Let $F(x) := \int_0^x f$; by Additivity, $\int_x^1 f = F(1) - F(x)$, so the hypothesis reads F(x) = F(1) - F(x), or 2F(x) = F(1) (a constant) for all $x \in [0,1]$. Differentiating via Theorem 7.3.6 gives 2f(x) = 2F'(x) = 0, so f(x) = 0 for all $x \in [0,1]$.

In the model paper: **Q17**, **Q19**, **Q40** (PYQ MTS-25 Q25, MAT-21 Q37a, MAT-22 Q33, MAT-22 MCQ 30, MTS-22 Q14).

T24 The indefinite integral need not be an antiderivative

> What happens to $F(z) = \int_a^z f$ when f is not continuous: corners and ignored spikes (Bartle Examples 7.3.7(a),(b), Exercises 7.3.12–7.3.13).



What to notice: The step function $f(x) = \operatorname{sgn} x$ jumps from -1 to +1 at x=0; its indefinite integral $F(x) = |x|$ (basepoint 0) or $|x| - 1$ (basepoint -1) is continuous everywhere (Lipschitz with M=1) but has a sharp corner at x=0 where slope -1 meets slope +1. **What it does not prove:** Having F'(x) exist everywhere still does not make F an antiderivative of f—for Thomae's function, H'(x)=0 exists everywhere but disagrees with h(x) at every rational!

EXAMPLES 7.3.7(A), (B) · WHEN F IS NOT AN ANTIDERIVATIVE (BARTLE P. 219)

(a) F'(c) fails to exist at a jump C20 **:** Let $f(x) := \operatorname{sgn} x$ on [-1,1]. Then $f \in \mathcal{R}[-1,1]$ and its indefinite integral with basepoint -1 is $F(x) = \int_{-1}^{x} \operatorname{sgn} = |x| - 1$ for $x \in [-1,1] (for x < 0, \int_{-1}^{x}(-1) = -(x+1) = |x| - 1$; for $x \geq 0, \int_{-1}^{0}(-1) + \int_{0}^{x} 1 =$ -1 $+ x = |x| - 1$). However, since F'(0) does not exist, F is **not** an antiderivative of f on [-1,1].

(b) F'(c) exists everywhere but does not equal f(c) C8 **:** Let h be Thomae's function (Example 7.1.7) and let $H(x) := \int_{0}^{x} h$. Since $\int_{0}^{1} h = 0$ and $h \geq 0$, we have H(x) = 0 for all $x \in [0,1]$. Thus H'(x) = 0 exists at every point of [0,1], but $H'(x) = 0 \neq h(x)$ at every rational $x \in \mathbb{Q} \cap [0,1]$, so H is **not** an antiderivative of h on [0,1].

In plain words: an antiderivative of f on [a,b] must satisfy F'(x) = f(x) at every point of [a,b]. If f jumps at c, F has a corner at c; if f has isolated spikes, F ignores them so $F'(c) \neq f(c)$ at the spikes.

WORKED EXAMPLE · BARTLE EXERCISE 7.3.12

Let f : $[0,3] \to \mathbb{R}$ be f(x) = x on [0,1), f(x) = 1 on [1,2), and f(x) = x on [2,3]. Find $F(x) = \int_0^x f$ and determine where F is differentiable:

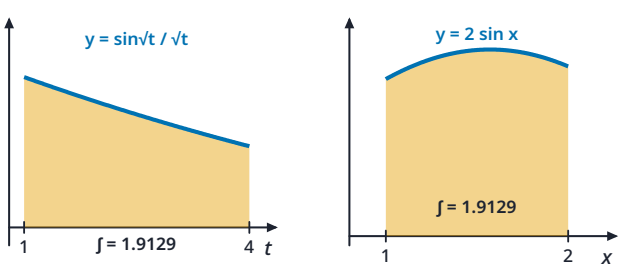
- For $0 \le x \le 1$: $F(x) = \int_0^x t\,dt = \frac{1}{2}x^2$.
- For $1 \le x \le 2$: $F(x) = \int_0^1 t\,dt + \int_1^x 1\,dt = \frac{1}{2} + (x-1) = x - \frac{1}{2}$.
- For $2 \le x \le 3$: $F(x) = F(2) + \int_2^x t\,dt = \frac{3}{2} + \frac{1}{2}(x^2 - 4) = \frac{1}{2}(x^2 - 1)$.

Notice that f is continuous at $x = 1 (\lim_{x \to 1^-} x = 1 = f(1))$, so by Theorem 7.3.5, F is differentiable at x = 1 with F'(1) = 1. At x = 2, f jumps from 1 to 2, so $F'_-(2) = 1 \ne 2 = F'_+(2)$: F is differentiable on $[0,3] \setminus \{2\}$ with F'(x) = f(x) there, and not differentiable at x = 2.

In the model paper: **Q12** , **Q40** .

T25 The Substitution Theorem

Reversing the Chain Rule to change variables in a definite integral (Bartle Theorem 7.3.8, Examples 7.3.9(a),(b), Exercises 7.3.17–7.3.19; PYQ MTS-25 Q6, MAT-22 Q18, MAT-21 Q19, MAT-22 MCQ 19, 26).



What to notice: Setting $x = \varphi(t) = \sqrt{t}$ transforms $\int_1^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt$ (left panel) into $\int_1^2 2\sin x\,dx$ (right panel) with the exact same area $2(\cos 1 - \cos 2) \approx 1.9129$. **What it does not prove:** Theorem 7.3.8 cannot be used on [0,4] with $\varphi(t) = \sqrt{t}$ because $\varphi'(t) = 1/(2\sqrt{t})$ blows up at t=0; on [0,4] we must use FTC 1 with $E = \{0\}$ instead.

THEOREM 7.3.8 · SUBSTITUTION THEOREM (BARTLE P. 220, EQ. (5)) · SAY IT WORD FOR WORD

Let $J := [\alpha, \beta]$ and let $\varphi : J \to \mathbb{R}$ have a continuous derivative on J. If f : $I \to \mathbb{R}$ is continuous on an interval I containing $\varphi(J)$, then $\int_{\alpha}^{\beta} f(\varphi(t))\,\varphi'(t)\,dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx$.

In plain words: if $x = \varphi(t)$ has a continuous derivative $\varphi'(t)$ on the entire closed interval $[\alpha, \beta]$ and f is continuous on the range of φ, you may replace $f(\varphi(t))\varphi'(t)\,dt$ by $f(x)\,dx$ and change the limits from $t = \alpha, \beta$ to $x = \varphi(\alpha), \varphi(\beta)$.

HYPOTHESES · THEOREM 7.3.8

☐ φ is differentiable on $J = [\alpha, \beta]$ and φ' **is continuous on the closed interval** $[\alpha, \beta]$

for $\int_0^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt$ with $\varphi(t) = \sqrt{t}$ (Example 7.3.9(b) & Ex 7.3.19$(a), (b)$), $\varphi'(t) = 1/(2\sqrt{t})$ does not exist at t = 0; for $\varphi(t) = |t|$ on $[-1, 1] (Ex 7.3.19(c))$, $\varphi'(0)$ does not exist; for $\varphi(t) = \operatorname{Arcsin} t$ on $[0, 1] (Ex 7.3.19(d))$, $\varphi'(t) = 1/\sqrt{1 - t^2}$ blows up at t = 1

☐ f is **continuous** on an interval $I \supseteq \varphi(J)$

used in Step 2 of the proof so that $F(u) = \int_{\varphi(\alpha)}^{u} f(x)\,dx$ satisfies F'(u) = f(u) everywhere on I by Theorem 7.3.6

PROOF OF THEOREM 7.3.8 (BARTLE EXERCISE 7.3.17, P. 224)

1. Since f is continuous on the interval $I \supseteq \varphi(J)$, define the indefinite integral $F(u) := \int_{\varphi(\alpha)}^{u} f(x)\,dx$ for $u \in I$ (using Definition 7.2.12 if u < $\varphi(\alpha)$).
   Theorem 7.2.7 (T15) and Definition 7.3.3 (T22).

2. By the Second Form of the Fundamental Theorem (Theorem 7.3.6), F is differentiable on I and F'(u) = f(u) for all $u \in I$.
   Theorem 7.3.6 (T23) since f is continuous on I.

3. Define the composite function $H(t) := F(\varphi(t))$ for $t \in J = [\alpha, \beta]$. Since φ is differentiable on J and F is differentiable at $\varphi(t) \in I$, the Chain Rule 6.1.6 (P11) gives $H'(t) = F'(\varphi(t))\,\varphi'(t) = f(\varphi(t))\,\varphi'(t) \quad \text{for all } t \in J$.
   Chain Rule 6.1.6 (P11) and Step 2.

4. Since f, φ, and φ' are continuous, the function $t \mapsto f(\varphi(t))\varphi'(t)$ is continuous on $J = [\alpha, \beta]$ and hence belongs to $\mathcal{R}[\alpha, \beta]$ by Theorem 7.2.7.
   Composition and product of continuous functions are continuous; Theorem 7.2.7 (T15).

5. Applying the First Form of the Fundamental Theorem (Theorem 7.3.1 with $E = \varnothing$) to H on $[\alpha, \beta]$ and noting that $H(\alpha) = F(\varphi(\alpha)) = \int_{\varphi(\alpha)}^{\varphi(\alpha)} f = 0$ gives $\int_{\alpha}^{\beta} f(\varphi(t))\,\varphi'(t)\,dt = H(\beta) - H(\alpha) = F(\varphi(\beta)) - 0 = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx$. ■
   Theorem 7.3.1 (T20) and Definition 7.2.12.

WORKED EXAMPLES · BARTLE EXAMPLES 7.3.9(A),(B) & EXERCISE 7.3.18 (PYQ MTS-25 Q6, MAT-22 Q18, MAT-21 Q19)

1. Example $7.3.9(a)(\int_1^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt$**, PYQ MTS-25 Q6, MAT-21 Q19):** On J = [1,4], set $\varphi(t) := \sqrt{t}$; then $\varphi'(t) = \frac{1}{2\sqrt{t}}$ is continuous on [1,4]. With $f(x) := 2\sin x$ (continuous on $\mathbb{R}$), we have $f(\varphi(t))\varphi'(t) = (2\sin\sqrt{t})\frac{1}{2\sqrt{t}} = \frac{\sin\sqrt{t}}{\sqrt{t}}, \varphi(1) = 1,$ and $\varphi(4) = 2$. By Theorem 7.3.8 and FTC 1,

$\int_1^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt = \int_1^2 2\sin x\,dx = [-2\cos x]_1^2 = 2(\cos 1 - \cos 2) \approx 1.912898.$

2. Example $7.3.9(b)(\int_0^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt$**, PYQ MAT-22 Q18):** On $[0,4], \varphi(t) = \sqrt{t}$ does **not** have a continuous derivative at t = 0, so Theorem 7.3.8 cannot be applied! Instead: (i) let $g(t) := \frac{\sin\sqrt{t}}{\sqrt{t}}$ for $t \in (0,4]$ and g(0) := 1; since $|\sin u| \le |u|, |g(t)| \le 1$ is bounded on [0,4] and continuous on every [c,4] (c>0), so $g \in \mathcal{R}[0,4]$ by Exercise 7.2.11 (T18). (ii) Define $F(t) := -2\cos\sqrt{t}$ on [0,4]; then F is continuous on [0,4] and $F'(t) = (2\sin\sqrt{t})\frac{1}{2\sqrt{t}} = g(t)$ for all $t \in (0,4](E = \{0\})$. By FTC 1 (Theorem 7.3.1),

$\int_0^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt = F(4) - F(0) = -2\cos 2 - (-2\cos 0) = 2(1 - \cos 2) \approx 2.832294.$

3. Four drills from Bartle Ex 7.3.18:

- $(a)\ \int_0^1 t\sqrt{1+t^2}\,dt : \varphi(t) = 1+t^2, f(x) = \frac{1}{2}\sqrt{x} \Rightarrow \int_1^2 \frac{1}{2}x^{1/2}\,dx = [\frac{1}{3}x^{3/2}]_1^2 = \frac{2\sqrt{2}-1}{3} \approx 0.609476.$
- $(b)\ \int_0^2 t^2(1+t^3)^{-1/2}\,dt : \varphi(t) = 1+t^3, f(x) = \frac{1}{3}x^{-1/2} \Rightarrow \int_1^9 \frac{1}{3}x^{-1/2}\,dx = [\frac{2}{3}\sqrt{x}]_1^9 = \frac{4}{3}.$
- $(c)\ \int_1^4 \frac{\sqrt{1+\sqrt{t}}}{\sqrt{t}}\,dt : \varphi(t) = 1+\sqrt{t}$ on $[1,4], f(x) = 2\sqrt{x} \Rightarrow \int_2^3 2x^{1/2}\,dx = [\frac{4}{3}x^{3/2}]_2^3 = \frac{4}{3}(3\sqrt{3} - 2\sqrt{2}) \approx 3.156967.$

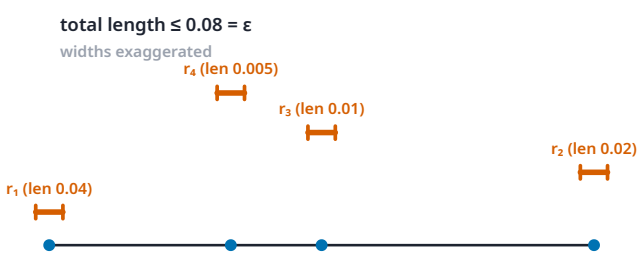
- $(d)\ \int_1^4 \frac{\cos\sqrt{t}}{\sqrt{t}}\,dt : \varphi(t) = \sqrt{t}$ on $[1,4], f(x) = 2\cos x \Rightarrow \int_1^2 2\cos x\,dx = 2(\sin 2 - \sin 1) \approx 0.135653.$

4. Symmetry and Translation (PYQ MAT-22 MCQ 19, 26; Q16): With $\varphi(x) = \frac{\pi}{2} - x (\varphi' = -1), \int_0^{\pi/2} f(\cos x)\,dx = -\int_{\pi/2}^0 f(\sin u)\,du = \int_0^{\pi/2} f(\sin u)\,du$. Splitting $\int_0^{\pi} f(\sin x)\,dx = \int_0^{\pi/2} + \int_{\pi/2}^{\pi}$ and substituting $u = \pi - x$ on $[\pi/2, \pi]$ shows $\int_{\pi/2}^{\pi} f(\sin x)\,dx = \int_0^{\pi/2} f(\sin u)\,du$, so $\int_0^{\pi/2} f(\cos x)\,dx = \frac{1}{2}\int_0^{\pi} f(\sin x)\,dx (and$ is NOT equal to $\int_0^{\pi} f(\sin x)\,dx)$. For the c-translate g(y) := f(y-c) on [a+c,b+c] (Bartle Ex 7.1.15), shifting any tagged partition of [a+c,b+c] by -c gives $\int_{a+c}^{b+c} g = \int_a^b f$.

In the model paper: **Q16** , **Q30** (PYQ MTS-25 Q6, MAT-22 Q18, MAT-21 Q19, MAT-22 MCQ 19, 26).

T26 Null sets and "almost everywhere"

> When is a set of bad points small enough for the Riemann integral to ignore? (Bartle Definition 7.3.10, Example 7.3.11; PYQ MTS-25 Q7, MTS-26 Q7, MTS-22 MCQ 10).



What to notice: Enumerating $\mathbb{Q} \cap [0,1] = \{r_1, r_2, r_3, \dots\}$ and covering r_k by an open interval J_k of width $\varepsilon/2^k$ makes the total width $\sum_{k=1}^{\infty} \varepsilon/2^k = \varepsilon$, even though $\mathbb{Q} \cap [0,1]$ is dense in [0,1]! **What it does not prove:** Being a null set does not mean the set is empty ($\varnothing$) or even countable—the Cantor set is uncountable and null.

DEFINITION 7.3.10 · NULL SET AND ALMOST EVERYWHERE (BARTLE P. 220, EQ. (6)) · SAY IT WORD FOR WORD

(a) A set $Z \subseteq \mathbb{R}$ is said to be a **null set** if for every $\varepsilon > 0$ there exists a countable collection $\{(a_k, b_k)\}_{k=1}^{\infty}$ of open intervals such that $Z \subseteq \bigcup_{k=1}^{\infty}(a_k, b_k) \qquad \text{and} \qquad \sum_{k=1}^{\infty}(b_k a_k) \leq \varepsilon.$

(b) If Q(x) is a statement about the point $x \in I$, we say that Q(x) holds **almost everywhere** on I (or for almost every $x \in I$), written Q(x) for a.e. $x \in I$, if there exists a null set $Z \subseteq I$ such that Q(x) holds for all $x \in I \setminus Z$.

In plain words: a set Z is a null set (has "length zero") if, whatever tiny total length $\varepsilon > 0$ you are handed, you can cover all of Z with a sequence of open intervals whose lengths add up to at most ε.

PROOF OF EXAMPLE 7.3.11 · $\mathbb{Q} \cap [0,1]$ IS A NULL SET (BARTLE P. 221; PYQ MTS-25 Q7)

1. The set $\mathbb{Q}_1 := \mathbb{Q} \cap [0,1]$ is countable (P13), so its elements can be enumerated as a sequence $r_1, r_2, \ldots, r_k, \ldots$.
   Countability of the rationals (P13).

2. Let $\varepsilon > 0$ be given. For each $k \in \mathbb{N}$, let J_k be the open interval $J_k := (r_k - \frac{\varepsilon}{2^{k+1}},\ r_k + \frac{\varepsilon}{2^{k+1}})$. Then J_k contains r_k and has length $b_k - a_k = 2 \cdot \frac{\varepsilon}{2^{k+1}} = \frac{\varepsilon}{2^k}$.
   The $\varepsilon/2^k$ geometric covering trick (M07).

3. The union $\bigcup_{k=1}^{\infty} J_k$ of this countable collection of open intervals contains every point of $\mathbb{Q}_1$, and the sum of the lengths is $\sum_{k=1}^{\infty}(b_k - a_k) = \sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon \sum_{k=1}^{\infty} \frac{1}{2^k} = \varepsilon \cdot 1 = \varepsilon$. Since $\varepsilon > 0$ is arbitrary, $\mathbb{Q}_1 = \mathbb{Q} \cap [0,1]$ is a null set. ∎
   Geometric series $\sum_{k=1}^{\infty} 1/2^k = 1 (P03)$ and Definition 7.3.10(a).

RESULT · FOUR KEY PROPERTIES OF NULL SETS (BARTLE P. 221; PYQ MTS-22 MCQ 10)

- **1. Subsets:** Every subset $A \subseteq Z$ of a null set Z is a null set (any open cover of Z also covers A).
- **2. Unions:** The union $Z_1 \cup Z_2$ of two null sets (or countably many null sets) is a null set (cover Z_1 with total length $\leq \varepsilon/2$ and Z_2 with total length $\leq \varepsilon/2$).
- **3. Every countable set is null:** The exact same $\varepsilon/2^k$ argument of Example 7.3.11 shows that **every finite or countably infinite set** in $\mathbb{R}$ is a null set.
- **4. Not every null set is countable (MTS-22 MCQ 10 trap!):** The **Cantor set** (Bartle Definition 11.1.10) is an **uncountable** null set in [0,1]. On the other hand, a non-degenerate interval [a,b] (a < b) is **never** a null set (its length is b - a > 0).

In the model paper: **Q13** (PYQ MTS-25 Q7, MTS-26 Q7, MTS-22 MCQ 10).

T27 Lebesgue's Integrability Criterion

The complete "if and only if" verdict on which bounded functions are Riemann integrable (Bartle Theorem 7.3.12, Examples 7.3.13(a)–(e); PYQ MAT-22 Q6).

THEOREM 7.3.12 · LEBESGUE'S INTEGRABILITY CRITERION (BARTLE P. 221) · SAY IT WORD FOR WORD

A bounded function f : $[a, b] \to \mathbb{R}$ is Riemann integrable if and only if it is continuous almost everywhere on [a,b].

In plain words: $f \in \mathcal{R}[a, b]$ if and only if (1) f is bounded on [a,b], and (2) the set $D(f) = \{x \in [a, b] : f \text{ is discontinuous at } x\}$ is a null set. (The proof of 7.3.12 is in Bartle Appendix C and is not in the syllabus; we quote 7.3.12 to test functions and prove Theorems 7.3.14–7.3.16.)

HYPOTHESES · THEOREM 7.3.12

☐ f is **bounded** on [a,b]

f(x) = 1/x on (0,1], f(0) = 0 **C16** has discontinuity set $D(f) = \{0\}$(a 1-point set, hence null!), yet $f \notin \mathcal{R}[0,1]$ because f is unbounded! Never forget to check boundedness first.

☐ the discontinuity set D(f) is a **null set**

Dirichlet's function **C7** is bounded ($0 \leq f \leq 1$), but D(f) = [0,1], which is not a null set, so $f \notin \mathcal{R}[0,1]$

SKELETON · TEST INTEGRABILITY USING LEBESGUE'S CRITERION

Step 1 (Boundedness): Check that $|f(x)| \leq M$ on [a,b] (if unbounded, $f \notin \mathcal{R}[a,b]$ by Theorem 7.1.6). **Step 2 (Discontinuity set):** Find the set D of points of [a,b] where f is discontinuous: D = ? . Since D is finite / countable / null / not null, by Lebesgue's Integrability Criterion (7.3.12), f is in / not in $R[a, b]$. ∎

WORKED EXAMPLES · BARTLE EXAMPLES 7.3.13(A)–(E) (MODEL PAPER Q31)

- **(a) Step functions:** The step function *g* of Example 7.1.4(b) is bounded ($|g| \leq 3$) and discontinuous only at x = 1. In general, any step function is bounded and has at most finitely many discontinuities (a null set), so every step function is in $\mathcal{R}[a,b]$.
- **(b) Monotone functions:** A monotone function on [a,b] is bounded between f(a) and f(b), and by Theorem 5.6.4 (P10) its set of discontinuities is at most countable, hence a null set. By 7.3.12, every monotone function on [a,b] is in $\mathcal{R}[a,b]$.
- $(c) G(1/n) = 1/n (n \in \mathbb{N}), G(x) = 0$ **elsewhere on [0,1] (Example 7.1.4(d)):** G is bounded ($0 \leq G \leq 1$), continuous at 0 (since $\lim_{x\to 0} G(x) = 0 = G(0)$), and discontinuous precisely on the countable set $D = \{1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{n}, \ldots\}$. Since every countable set is a null set, $G \in \mathcal{R}[0,1]$.
- **(d) Dirichlet's function** C7 **:** Discontinuous at **every** point of [0,1] (D = [0,1], not a null set), so $f \notin \mathcal{R}[0,1]$.
- **(e) Thomae's function h** C8 **:** Bounded ($0 \leq h \leq 1$) and discontinuous precisely on $D = \mathbb{Q} \cap [0,1]$, which is a null set by Example 7.3.11. By 7.3.12, $h \in \mathcal{R}[0,1]$.

In the model paper: **Q31** (PYQ MAT-22 Q6).

T28 The Composition Theorem

> Composing $f \in \mathcal{R}[a,b]$ with an outer continuous function φ preserves integrability (Bartle Theorem 7.3.14; PYQ MAT-21 Q4).

THEOREM 7.3.14 · COMPOSITION THEOREM (BARTLE P. 222) · SAY IT WORD FOR WORD

Let $f \in \mathcal{R}[a,b]$ with $f([a,b]) \subseteq [c,d]$, and let $\varphi : [c,d] \to \mathbb{R}$ be continuous. Then the composition $\varphi \circ f$ belongs to $\mathcal{R}[a,b]$.

In plain words: if f is integrable and φ is continuous on the range of f, then $\varphi(f(x))$ is also integrable—because $\varphi \circ f$ can only be discontinuous where f was already discontinuous!

HYPOTHESES · THEOREM 7.3.14

☐ inner function $f \in \mathcal{R}[a,b]$ and outer function $\varphi : [c,d] \to \mathbb{R}$ is **continuous**

Why continuity of φ cannot be dropped (Bartle Remark p. 222 / Ex 7.3.22): Let $h \in \mathcal{R}[0,1]$ be Thomae's function C8, and let $\varphi : [0,1] \to \mathbb{R}$ be $\varphi(0) := 0$ and $\varphi(y) := 1$ for $y \in (0,1]$. Notice that $\varphi \in \mathcal{R}[0,1]$ (discontinuous only at y=0, a single point!). Yet for $x \in [0,1]$, $(\varphi \circ h)(x) = 1$ if $x \in \mathbb{Q} \cap [0,1]$ (since h(x) = 1/n > 0) and $(\varphi \circ h)(x) = \varphi(0) = 0$ if x is irrational: $\varphi \circ h$ is **Dirichlet's function** C7, which is NOT in $\mathcal{R}[0,1]$! Thus the composition of two integrable functions need not be integrable unless the outer one is continuous.

Know $f \in \mathcal{R}[a,b] (sof$ is bounded and D(f) is null by 7.3.12), and $\varphi : [c,d] \to \mathbb{R}$ is continuous.

Want Show $\varphi \circ f$ is bounded and $D(\varphi \circ f) \subseteq D(f)$ is null, then invoke Lebesgue's Criterion 7.3.12.

PROOF OF THEOREM 7.3.14 (BARTLE P. 222)

1. If f is continuous at a point $u \in [a,b]$, then since φ is continuous at $f(u) \in [c,d]$, the composition $\varphi \circ f$ is also continuous at u.
   Composition of continuous functions is continuous (Module 1, Theorem 5.2.6 / P07).

2. Taking the contrapositive: if $\varphi \circ f$ is discontinuous at u, then f must be discontinuous at u. Thus the set D of points of discontinuity of $\varphi \circ f$ on [a,b] is a subset of the set D(f) of points of discontinuity of f.
   $u \notin D(f) \Rightarrow u \notin D$, so $D \subseteq D(f)$.

3. Since $f \in \mathcal{R}[a,b]$, Lebesgue's Integrability Criterion 7.3.12 implies that D(f) is a null set. Since any subset of a null set is a null set (T26), D is also a null set.
   Theorem 7.3.12($\Rightarrow$) and subset property of null sets (T26).

4. Furthermore, since φ is continuous on the closed bounded interval $[c,d]$, φ is bounded on [c,d] by the Boundedness Theorem 5.3.2 (P07), so $\varphi \circ f$ is bounded on [a,b]. By Lebesgue's Integrability Criterion 7.3.12, $\varphi \circ f \in \mathcal{R}[a,b]$. ∎
   Boundedness Theorem 5.3.2 (P07) and Theorem 7.3.12($\Leftarrow$).

In the model paper: **Q32** (PYQ MAT-21 Q4).

T29 Absolute value and the triangle inequality for integrals

If $f \in \mathcal{R}[a,b]$, then $|f| \in \mathcal{R}[a,b]$ and $|\int_a^b f| \leq \int_a^b |f| \leq M(b-a)$ (Bartle Corollary 7.3.15; PYQ MTS-25 Q8, MTS-22 Q7, MAT-21 Q20).

COROLLARY 7.3.15 (BARTLE P. 222) · SAY IT WORD FOR WORD

Suppose $f \in \mathcal{R}[a,b]$. Then its absolute value $|f|$ is in $\mathcal{R}[a,b]$, and $|\int_a^b f| \le \int_a^b |f| \le M(b-a)$, where $|f(x)| \le M$ for all $x \in [a,b]$.

In plain words: taking the absolute value of an integrable function keeps it integrable, and the absolute value of the integral never exceeds the integral of the absolute value (because positive and negative areas can cancel in $\int_a^b f$, but never cancel in $\int_a^b |f|$).

DIRECTION & CONVERSE · COROLLARY 7.3.15

☐ $f \in \mathcal{R}[a,b] \Rightarrow |f| \in \mathcal{R}[a,b]$

Converse is FALSE: $|f| \in \mathcal{R}[a,b]$ does **not** imply $f \in \mathcal{R}[a,b]$! Define f(x) := 1 for rational $x \in [0,1]$ and f(x) := -1 for irrational $x \in [0,1]$. Then $|f(x)| = 1$ for all $x \in [0,1]$, so $|f| \in \mathcal{R}[0,1]$ with $\int_0^1 |f| = 1$, yet $f \notin \mathcal{R}[0,1]$ (rational tags give S = 1, irrational tags give S = -1, gap = 2).

PROOF OF COROLLARY 7.3.15 (BARTLE P. 222)

1. Since $f \in \mathcal{R}[a,b]$, Theorem 7.1.6 implies that f is bounded on [a,b], say $|f(x)| \le M$ for all $x \in [a,b]$, so $f([a,b]) \subseteq [-M, M]$.
   Boundedness Theorem 7.1.6 (T08).

2. The function $\varphi(t) := |t|$ is continuous on [-M,M]. By the Composition Theorem 7.3.14, $|f| = \varphi \circ f$ belongs to $\mathcal{R}[a,b]$.
   Continuity of $|t|$ (Module 1, T04) and Composition Theorem 7.3.14 (T28).

3. Since $-|f(x)| \le f(x) \le |f(x)|$ for all $x \in [a,b]$, Theorem 7.1.5(a) and (c) give -\int_a^b \lvert f\rvert \le \int_a^b f \le \int_a^b \lvert f\rvert, \quad \text{which\ means} \quad $|\int_a^b f| \le \int_a^b |f|$.
   Monotonicity and linearity of the integral (Theorem 7.1.5, T07) and P05.

4. Finally, since $|f(x)| \le M$ on [a,b], Theorem 7.1.5(c) and Example 7.1.4(a) yield $\int_a^b |f| \le \int_a^b M = M(b-a)$. ∎
   Constant integral $\int_a^b M = M(b-a)$ $(T05)$.

In the model paper: **Q33** (PYQ MTS-25 Q8, MTS-22 Q7, MAT-21 Q20).

T30 The Product Theorem

index ↑

The product fg of two integrable functions is integrable via the polarisation identity $\text{fg} = \frac{1}{2}[(f+g)^2 - f^2 - g^2]$ (Bartle Theorem 7.3.16; PYQ MTS-26 Q6, MAT-22 Q17, MAT-21 Q18).

THEOREM 7.3.16 · PRODUCT THEOREM (BARTLE P. 222) · SAY IT WORD FOR WORD

If f and g belong to $\mathcal{R}[a,b]$, then the product fg belongs to $\mathcal{R}[a,b]$.

In plain words: squaring an integrable function keeps it integrable (by the Composition Theorem with $\varphi(t) = t^2$), and any product fg can be written in terms of squares as $\frac{1}{2}[(f+g)^2 - f^2 g^2]$.

HYPOTHESES & TRAP · THEOREM 7.3.16

☐ $f, g \in \mathcal{R}[a,b] \Rightarrow \text{fg} \in \mathcal{R}[a,b]$

Converse is FALSE: $f^2 \in \mathcal{R}[a,b]$ does NOT imply $f \in \mathcal{R}[a,b]$ (take $f = \pm 1$ on $rationals/irrationals$: $f^2 \equiv 1 \in \mathcal{R}, f \notin \mathcal{R}$). **Warning:** $\int_a^b \text{fg} \neq (\int_a^b f)(\int_a^b g)$ in general! For f(x) = g(x) = x on $[0,1]$, $\int_0^1 x^2\,dx = \frac{1}{3} \neq \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

PROOF OF THEOREM 7.3.16 (BARTLE P. 222)

1. Since $f, g \in \mathcal{R}[a,b]$, we also have $f + g \in \mathcal{R}[a,b]$ by Theorem 7.1.5(b). By the Boundedness Theorem 7.1.6, f, g, and f+g are bounded on [a,b]; choose M > 0 so that f([a,b]), g([a,b]), and (f+g)([a,b]) all lie in [-M,M].
   Linearity 7.1.5(b) (T07) and Boundedness Theorem 7.1.6 (T08).

2. The squaring function $\varphi(t) := t^2$ is continuous on [-M,M]. Applying the Composition Theorem 7.3.14 to f, g, and f+g shows that $f^2 = \varphi \circ f, g^2 = \varphi \circ g$, and $(f+g)^2 = \varphi \circ (f+g)$ all belong to $\mathcal{R}[a,b]$.
   Composition Theorem 7.3.14 (T28) with $\varphi(t) = t^2$.

3. From the algebraic identity $\text{fg} = \frac{1}{2}[(f+g)^2 - f^2 - g^2]$, linearity of the integral (Theorem 7.1.5(a),(b)) implies that $\text{fg} \in \mathcal{R}[a,b]$. ∎
   Expand $(f+g)^2 = f^2 + 2\text{fg} + g^2$ and apply Theorem 7.1.5 (T07).

In the model paper: **Q34** (PYQ MTS-26 Q6, MAT-22 Q17, MAT-21 Q18).

T31 Integration by Parts

Integrating the Product Rule (FG)' = fG + Fg via the Product Theorem and FTC 1 (Bartle Theorem 7.3.17).

THEOREM 7.3.17 · INTEGRATION BY PARTS (BARTLE P. 222, EQ. (7)) · SAY IT WORD FOR WORD

Let F, G be differentiable on [a,b] and let f := F' and g := G' belong to $\mathcal{R}[a,b]$. Then $\int_a^b$ fG $= [FG]_a^b - \int_a^b$ Fg.

In plain words: if F and G are differentiable on [a,b] and their derivatives f = F' and g = G' are Riemann integrable on [a,b], then $\int_a^b F'(x)G(x)\,dx = F(b)G(b) - F(a)G(a)\,\int_a^b F(x)G'(x)\,dx$.

HYPOTHESES · THEOREM 7.3.17

☐ F, G are **differentiable on [a,b]** and f = F', g = G' $\in \mathcal{R}[a,b]$

remember from Example 7.3.2(e) C19 that F differentiable does not automatically make F' $\in \mathcal{R}[a,b]$, so $f, g \in \mathcal{R}[a,b]$ must be assumed (it is automatic when f, g are continuous)

PROOF OF THEOREM 7.3.17 (BARTLE P. 223)

1. By the Product Rule for derivatives (Theorem 6.1.3(c), P11), the product FG is differentiable on [a,b] and $(FG)'(x) = F'(x)G(x) + F(x)G'(x) = f(x)G(x) + F(x)g(x) \quad \text{for all } x \in [a,b]$.
   Product Rule 6.1.3(c) (P11).

2. Since F and G are differentiable on [a,b], they are continuous on [a,b] by Theorem 6.1.2 (P11) and hence belong to $\mathcal{R}[a,b]$ by Theorem 7.2.7. Since $f, g \in \mathcal{R}[a,b]$ by hypothesis, the Product Theorem 7.3.16 implies that fG and Fg both belong to $\mathcal{R}[a,b]$.
   Theorem 6.1.2 (differentiable ⇒ continuous), Theorem 7.2.7 (T15), and Product Theorem 7.3.16 (T30).

3. By Theorem 7.1.5(b), (FG)' = fG + Fg $\in \mathcal{R}[a,b]$. Applying the First Form of the Fundamental Theorem 7.3.1 (with $E = \varnothing$) to FG gives $[FG]_a^b = F(b)G(b) - F(a)G(a) = \int_a^b (FG)' = \int_a^b$ fG $+ \int_a^b$ Fg. Subtracting $\int_a^b$ Fg yields $\int_a^b$ fG $= [FG]_a^b - \int_a^b$ Fg. ∎
   Linearity 7.1.5(b) (T07) and FTC 1 (Theorem 7.3.1, T20).

WORKED EXAMPLE · EVALUATE $\int_0^1 x e^x \, dx$ (MODEL PAPER Q35)

Take $F(x) := e^x$ and G(x) := x on [0,1]. Both are differentiable on [0,1] with $f(x) = F'(x) = e^x$ and g(x) = G'(x) = 1, which are continuous and hence in $\mathcal{R}[0,1]$. By Theorem 7.3.17 and FTC 1,

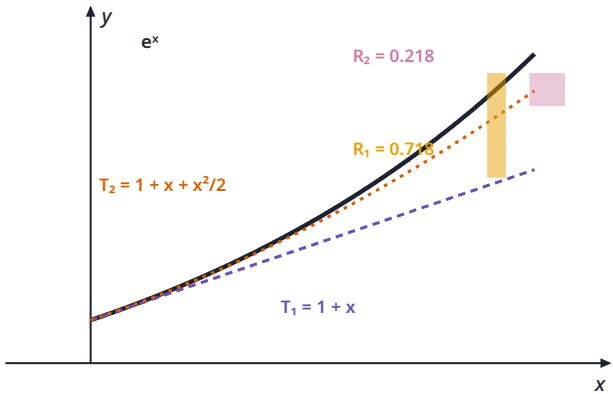
$\int_0^1 e^x \cdot x \, dx = [e^x \cdot x]_0^1 - \int_0^1 e^x \cdot 1 \, dx = (e \cdot 1 - 1 \cdot 0)[e^x]_0^1 = e - (e-1) = 1.$

In the model paper: **Q35** .

T32 Taylor's Theorem with the Integral Remainder

index ↑

Repeated Integration by Parts yields an exact integral formula for the Taylor remainder R_n (Bartle Theorem 7.3.18; PYQ MTS-25 Q19, MAT-22 Q32).



What to notice: For $f(x) = e^x$ on [0,1] (a=0, b=1), the linear Taylor polynomial $T_1(1) = 2$ leaves remainder $R_1 = \int_0^1 e^t(1-t)\,dt = e - 2 \approx 0.7183$, and the quadratic polynomial $T_2(1) = 2.5$ leaves remainder $R_2 = \frac{1}{2}\int_0^1 e^t(1-t)^2\,dt = e - 2.5 \approx 0.2183$. **What it does not prove:** Unlike the Lagrange remainder (Module 2 / §6.4, out of scope), this formula has no unknown point c; it requires $f^{(n+1)} \in \mathcal{R}[a,b]$.

THEOREM 7.3.18 · TAYLOR'S THEOREM WITH THE REMAINDER (BARTLE P. 223, EQS. (8)–(9)) · SAY IT WORD FOR WORD

Suppose that f', . . . , $f^{(n)}$, $f^{(n+1)}$ exist on [a,b] and that $f^{(n+1)} \in \mathcal{R}[a,b]$. Then we have $f(b) = f(a) + \frac{f'(a)}{1!}(b-a) + \cdots + \frac{f^{(n)}(a)}{n!}(b-a)^n + R_n$, where the remainder is given by $R_n = \frac{1}{n!}\int_a^b f^{(n+1)}(t)\,(b-t)^n\,dt$.

In plain words: if you integrate $R_n = \frac{1}{n!}\int_a^b f^{(n+1)}(t)(b-t)^n\,dt$ by parts n times, each step peels off one Taylor term $-\frac{f^{(k)}(a)}{k!}(b-a)^k$ until only $R_0 = \int_a^b f'(t)\,dt = f(b) - f(a)$ remains.

HYPOTHESES · THEOREM 7.3.18

☐ f', f'', . . . , $f^{(n+1)}$ exist at every point of [a,b] and $f^{(n+1)} \in \mathcal{R}[a,b]$

for $k \le n$, $f^{(k)}$ is differentiable on [a,b] (since $f^{(k+1)}$ exists), hence continuous on [a,b] (Theorem 6.1.2) and in $\mathcal{R}[a,b]$ (Theorem 7.2.7); only the top derivative $f^{(n+1)}$ needs the extra assumption $f^{(n+1)} \in \mathcal{R}[a,b]$

PROOF OF THEOREM 7.3.18 (BARTLE P. 223, WITH MISPRINT CORRECTED)

1. Since $f^{(n)}$ and $t \mapsto (b-t)^n/n!$ are differentiable on [a,b] with derivatives $f^{(n+1)} \in \mathcal{R}[a,b]$ and $-(b-t)^{n-1}/(n-1)! \in \mathcal{R}[a,b]$ (continuous), we apply Integration by Parts (Theorem 7.3.17) to R_n with $F(t) := f^{(n)}(t) (f(t) = F'(t) = f^{(n+1)}(t))$ and $G(t) := (b-t)^n/n! (g(t) = G'(t) = -(b-t)^{n-1}/(n-1)!)$.
   Check hypotheses of Theorem 7.3.17 (T31) for $F(t) = f^{(n)}(t)$ and $G(t) = (b-t)^n/n!$.

2. Since $G(b) = (b-b)^n/n! = 0$ and $G(a) = (b-a)^n/n!$, Theorem 7.3.17 gives $R_n = [\frac{f^{(n)}(t)(b-t)^n}{n!}]_{t=a}^{t=b} - \int_a^b f^{(n)}(t)(-\frac{(b-t)^{n-1}}{(n-1)!})dt = -\frac{f^{(n)}(a)}{n!}(b-a)^n + \frac{1}{(n-1)!}\int_a^b f^{(n)}(t)(b-t)^{n-1}\,dt$. That is, $R_n = -\frac{f^{(n)}(a)}{n!}(b-a)^n + R_{n-1}$, where $R_{n-1} = \frac{1}{(n-1)!}\int_a^b f^{(n)}(t)(b-t)^{n-1}\,dt$.
   Evaluate the boundary term at t=b and t=a. (**Bartle misprint note:** the first displayed integral on p. 223 prints $(b-a)^{n-1}$ by mistake; the variable of integration is t, so it must be $(b-t)^{n-1}$, as Bartle's next line confirms.)

3. For each k = n, n-1, . . . , 1, $f^{(k)}$ is continuous on [a,b] (since $f^{(k+1)}$ exists) and therefore belongs to $\mathcal{R}[a,b]$. Repeating this integration by parts n times yields $R_k = -\frac{f^{(k)}(a)}{k!}(b-a)^k + R_{k-1}$ for k = n, n-1, . . . , 1, down to $R_0 = \int_a^b f'(t)\,dt = f(b) - f(a)$ by the First Form of the Fundamental Theorem 7.3.1.
   Induction backwards from k=n to k=1, ending with FTC 1 (T20) for $R_0 = \int_a^b f'(t)\,dt$.

4. Rearranging $R_{k-1} = \frac{f^{(k)}(a)}{k!}(b-a)^k + R_k$ from k=1 to k=n gives $f(b) - f(a) = R_0 = \frac{f'(a)}{1!}(b-a) + R_1 = \cdots = \sum_{k=1}^{n} \frac{f^{(k)}(a)}{k!}(b-a)^k + R_n$, which is Taylor's formula (8). ∎
   Back-substitute $R_1, R_2, \ldots, R_n$ into $f(b) = f(a) + R_0$.

CHECK YOURSELF · VERIFY THEOREM 7.3.18 FOR $f(x) = e^x, a = 0, b = 1, n = 1 (Q36)$

Compute the Taylor polynomial f(0) + f'(0)(1-0) and the integral remainder $R_1 = \int_0^1 e^t(1-t)\,dt$, and check their sum.

f(0) + f'(0)(1) = 1 + 1 = 2. By parts with $F(t) = e^t, G(t) = 1 - t$: $R_1 = [e^t(1-t)]_0^1 - \int_0^1 e^t(-1)\,dt = (0-1) + (e-1) = e - 2$. Sum: 2 + (e - 2) = e = f(1).

In the model paper: **Q36** (PYQ MTS-25 Q19, MAT-22 Q32).

REFERENCE · COUNTEREXAMPLES

Module 3 Counterexample Library

> Know these six C-coded functions and four auxiliary non-examples cold: every "Is the converse true?" or "Can this hypothesis be dropped?" question uses one of them.

CODE & FUNCTION ON [A,B]	PROPERTIES	WHAT IT DISPROVES
C7 **Dirichlet** on $[0,1] : f(x) = 1(x \in \mathbb{Q}), 0(x \notin \mathbb{Q})$	Bounded ($0 \le f \le 1$); takes only 2 values; discontinuous everywhere (D=[0,1] not null); S=1 (rational tags) vs S=0 (irrational tags).	Bounded $\not\Rightarrow f \in \mathcal{R}[0,1] (T08)$; finitely many values $\not\Rightarrow$ step function (T14); Cauchy & Lebesgue fail (T11, T27).
C8 **Thomae (popcorn)** on [0,1]: h(m/n)=1/n (lowest terms), $h(0) = 1, h(x) = 0 (x \notin \mathbb{Q})$	$h \in \mathcal{R}[0,1]$ with $\int_0^1 h = 0$; continuous at irrationals, discontinuous on $D = \mathbb{Q} \cap [0,1]$ (null); $H(x) = \int_0^x h \equiv 0$.	$f \in \mathcal{R} \not\Rightarrow$ continuous, uniformly continuous, or monotone $(T15, T16)$; $H'(x) = 0 \neq h(x)$ on $\mathbb{Q}$, so H is not an antiderivative of h (T24).
C9 **Signum / Step** on $[-1,1] : f(x) = \operatorname{sgn} x (-1, 0, 1)$	Step function, monotone increasing, $f \in \mathcal{R}[-1,1], \int_{-1}^1 \operatorname{sgn} = 0$; discontinuous at 0.	$f \in \mathcal{R} \not\Rightarrow$ continuous (T15); MVT for integrals fails without continuity (on [0,2], g=1 on [0,1), 2 on [1,2] never takes average value 3/2).

C16 **Unbounded 1/x** on $[0,1]$: $f(x) = 1/x (x \in (0,1]), f(0) = 0$	Continuous on $(0,1]$; $D = \{0\}$ is null; $f\|_{[c,1]} \in \mathcal{R}[c,1]$ for every $c \in (0,1)$; **unbounded**.	$f \notin \mathcal{R}[0,1]$ by 7.1.6 (T08); continuous on $(a,b] \not\Rightarrow \mathcal{R}[a,b] (T15)$; $\mathcal{R}[c,b]\ \forall c > a \not\Rightarrow \mathcal{R}[a,b]$ without boundedness (T18); Lebesgue needs bounded (T27).
C19 **Unbounded derivative** on $[0,1]$: $K(x) = x^2 \cos(1/x^2), K(0) = 0$	K is continuous and differentiable at **every** $x \in [0,1] (E = \varnothing)$, but $K'(x) = 2x \cos(1/x^2) + (2/x) \sin(1/x^2)$ is unbounded.	$K' \notin \mathcal{R}[0,1]$: FTC 1 hypothesis $(c) f \in \mathcal{R}[a,b]$ is NOT automatic even when F'=f everywhere (T20, T21).
C20 **Indefinite integral of** sgn: $f =$ sgn on $[-1,1], F(x) = \|x\| - 1$	$F(x) = \int_{-1}^{x} \text{sgn} = \|x\| - 1$ is Lipschitz (M=1) and continuous, but F'(0) DNE (slopes ∓ 1).	FTC 2 needs continuity of f at c: when f jumps at 0, F has a corner at 0 and is not an antiderivative on [-1,1] (T23, T24).
Aux 1 · Countable spikes: G(1/n)=1/n vs H(1/k)=k (0 else on [0,1])	Both have $D = \{1/k : k \in \mathbb{N}\}$ (countable, hence null). G is bounded (≤ 1); H is unbounded.	$G \in \mathcal{R}[0,1]$ with $\int_0^1 G = 0 (7.1.4d$, T06, $T27)$; $H \notin \mathcal{R}[0,1] (7.1.6, T08)$. Changing g=0 on a countable set can destroy integrability!
Aux $2 \cdot \pm 1$ **Dirichlet**: $f(x) = 1 (x \in \mathbb{Q})$, -1 $(x \notin \mathbb{Q})$	$\|f(x)\| = 1$ and $f(x)^2 = 1$ for all $x \in [0,1]$, so $\|f\|, f^2 \in \mathcal{R}[0,1]$, but $f \notin \mathcal{R}[0,1]$.	Converse of Corollary $7.3.15 (\|f\| \in \mathcal{R} \not\Rightarrow f \in \mathcal{R}, T29)$ and converse of squaring $(f^2 \in \mathcal{R} \not\Rightarrow f \in \mathcal{R}, T30)$.
Aux 3 · Bad outer φ: h Thomae, $\varphi(0) = 0, \varphi(y) = 1 (y > 0)$	Both $h \in \mathcal{R}[0,1]$ and $\varphi \in \mathcal{R}[0,1]$, but φ is discontinuous at 0.	$\varphi \circ h =$ Dirichlet $\notin \mathcal{R}[0,1]$: continuity of outer φ in Composition Theorem 7.3.14 cannot be dropped (T28).
Aux 4 · Onepoint spike: f(0)=1, f(x)=0 on (0,1]	$f(x) \geq 0$ on [0,1] and $\int_0^1 f = 0 (by 7.1.3)$, yet $f(0) = 1 \neq 0$.	Shows "$f \geq 0$ and $\int_a^b f = 0 \Rightarrow f \equiv 0$" fails if continuity of f is dropped (Bartle Ex 7.2.9, T15).

INTERLEAVED DRILL · WHICH TOOL?

"Which tool?" mixed recognition drill

In the exam, questions are not labelled by section. Read each prompt, name the exact theorem/tool and the hypotheses to check, then uncover the answer.

DRILL 1 · INTEGRABILITY OF A MESSY BOUNDED FUNCTION

Show that $g(x) = \sin(1/x)(x \in (0,1]), g(0) = 0$ is in $\mathcal{R}[0,1]$. Which two tools work?

Tool A (Lebesgue's Criterion 7.3.12, T27): $|g(x)| \leq 1$ (bounded) and g is discontinuous only at $\{0\}(a$ 1-point null set). **Tool B (Bartle Ex 7.2.11, T18):** $|g| \leq 1$ on [0,1] and g is continuous (hence integrable by 7.2.7) on [c,1] for every $c \in (0,1)$.

DRILL 2 · DISPROVING INTEGRABILITY

Show that $H(1/k) = k(k \in \mathbb{N}), 0$ elsewhere on [0,1] is not in $\mathcal{R}[0,1]$; then show h(x) = x + 1 on $\mathbb{Q}, 0$ on irrationals is not in $\mathcal{R}[0,1]$.

For H: **Boundedness Theorem 7.1.6 (T08)**—H is unbounded $(H(1/k) = k \to \infty)$, so $H \notin \mathcal{R}[0,1]$ in one line! For h: h is bounded $(0 \leq h \leq 2)$, so 7.1.6 does not apply; use the **Cauchy Criterion negation (7.2.1, T11)** with $\varepsilon_0 = 1$ (rational tags give $S \geq 1$, irrational tags give S = 0), or **Lebesgue 7.3.12 (T27)** (D = [0,1], not null).

DRILL 3 · FTC 1 VS SUBSTITUTION AT A SINGULAR ENDPOINT

Compare $\int_1^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt$ and $\int_0^4 \frac{\sin\sqrt{t}}{\sqrt{t}}\,dt$. Why do they need different theorems?

On $[1,4], \varphi(t) = \sqrt{t}$ has $\varphi'(t) = \frac{1}{2\sqrt{t}}$ continuous on [1,4], so **Substitution (Theorem 7.3.8, T25)** works directly. On $[0,4], \varphi'(0)$ does not exist, so 7.3.8 fails; instead use **Ex 7.2.11 (T18)** for integrability $(|\sin\sqrt{t}/\sqrt{t}| \leq 1)$ and **FTC 1 (Theorem 7.3.1, T20)** with $F(t) = -2\cos\sqrt{t}$ and exceptional set $E = \{0\}$.

DRILL 4 · DERIVATIVE OF AN INTEGRAL WITH VARIABLE LIMITS

Find F'(x) for $F(x) = \int_{x^2}^{x} \sqrt{1+t^2}\,dt$. Which theorems do you cite?

Split via **Theorem 7.2.13 (T19)** as $F(x) = G(x)G(x^2)$ where $G(u) = \int_0^u \sqrt{1+t^2}\,dt$. Cite **FTC 2 (Theorem 7.3.6, T23)** (since $\sqrt{1+t^2}$ is continuous) to get $G'(u) = \sqrt{1+u^2}$, and the **Chain Rule 6.1.6 (P11)** on $G(x^2)$ to get $F'(x) = \sqrt{1+x^2} - 2x\sqrt{1+x^4}$.

DRILL 5 · INTEGRAL EQUALS ZERO ON ALL SUBINTERVALS

If f : $[0,1] \to \mathbb{R}$ is continuous and $\int_0^x f = \int_x^1 f$ for all $x \in [0,1]$, how do you prove $f \equiv 0$? What if instead $f \geq 0$ is continuous and $\int_a^b f = 0$?

First prompt: put $F(x) = \int_0^x f$; by **Additivity 7.2.9 (T17)**, $F(x) = F(1) - F(x) \Rightarrow F(x) = \frac{1}{2}F(1)$ constant, then differentiate via **FTC 2 (7.3.6, T23)** to get f(x) = F'(x) = 0. Second prompt: only one integral $\int_a^b f = 0$ is given, so use **continuity at c + Monotonicity 7.1.5(c) + Lemma 7.2.4 (T15, Ex 7.2.8)**: if f(c) > 0, then $f > \frac{1}{2}f(c)$ on some interval J of length $\ell > 0$, giving $\int_a^b f \geq \frac{1}{2}f(c)\ell > 0$.

REVISION · THEOREM SHEET

All Definitions & Theorems (Word for Word)

> Cover the right column and recite each statement and its exact hypotheses before the exam.

REF & TOPIC	EXACT STATEMENT & HYPOTHESES
Def 7.1.1 T03 Riemann Integrable	$f:[a,b]\to\mathbb{R}$ is in $\mathcal{R}[a,b]$ with $L=\int_a^b f$ iff $\exists L\in\mathbb{R}$ such that $\forall\varepsilon>0\ \exists\delta_\varepsilon>0$: for every tagged partition $\dot{P}$ of [a,b] with $\|P\|<\delta_\varepsilon, \lvert S(f;\dot{P})-L\rvert<\varepsilon$.
Thm 7.1.2 T04 Uniqueness	**Hyp:** $f\in\mathcal{R}[a,b]$. **Concl:** The value of $\int_a^b f$ is uniquely determined.
Thm 7.1.3 T06 Finite Change	**Hyp:** $g\in\mathcal{R}[a,b]$ and f(x)=g(x) except for a **finite** number of points in [a,b]. **Concl:** $f\in\mathcal{R}[a,b]$ and $\int_a^b f=\int_a^b g$.
Thm 7.1.5 T07 Linearity & Order	**Hyp:** $f,g\in\mathcal{R}[a,b], k\in\mathbb{R}$. **Concl:** $(a)kf\in\mathcal{R}[a,b], \int_a^b kf=k\int_a^b f; (b)f+g\in\mathcal{R}[a,b], \int_a^b(f+g)=\int_a^b f+\int_a^b g; (c)f(x)\le g(x)$ on $[a,b]\Rightarrow\int_a^b f\le\int_a^b g$.
Thm 7.1.6 T08 Boundedness	**Hyp:** $f\in\mathcal{R}[a,b]$. **Concl:** f is bounded on [a,b].
Thm 7.2.1 T10 Cauchy Criterion	$f\in\mathcal{R}[a,b]\iff\forall\varepsilon>0\ \exists\eta_\varepsilon>0$ such that for any tagged partitions $\dot{P},\dot{Q}$ of [a,b] with $\|P\|<\eta_\varepsilon$ and $\|Q\|<\eta_\varepsilon, \lvert S(f;\dot{P})-S(f;\dot{Q})\rvert<\varepsilon$.
Thm 7.2.3 T12 Squeeze Theorem	$f\in\mathcal{R}[a,b]\iff\forall\varepsilon>0\ \exists\alpha_\varepsilon,\omega_\varepsilon\in\mathcal{R}[a,b]$ with $\alpha_\varepsilon(x)\le f(x)\le\omega_\varepsilon(x)$ on [a,b] and $\int_a^b(\omega_\varepsilon-\alpha_\varepsilon)<\varepsilon$.
Lem 7.2.4 / Thm 7.2.5 T13 T14	If $J\subseteq[a,b]$ has endpoints c
Thm 7.2.7 / 7.2.8 T15 T16	**7.2.7:** f continuous on $[a,b]\Rightarrow f\in\mathcal{R}[a,b]$. **7.2.8:** f monotone on $[a,b]\Rightarrow f\in\mathcal{R}[a,b]$.
Thm 7.2.9 T17 Additivity	**Hyp:** $f:[a,b]\to\mathbb{R}, c\in(a,b)$. **Concl:** $f\in\mathcal{R}[a,b]\iff f\lvert_{[a,c]}\in\mathcal{R}[a,c]$ and $f\lvert_{[c,b]}\in\mathcal{R}[c,b]$, and then $\int_a^b f=\int_a^c f+\int_c^b f$.(7.2.10 : $[c,d]\subseteq[a,b]\Rightarrow f\lvert_{[c,d]}\in\mathcal{R}[c,d]$.)
Thm 7.3.1 T20 FTC First Form	**Hyp:** $E\subset[a,b]$ finite, (a) F continuous on [a,b], (b) F'(x)=f(x) for all $x\in[a,b]\setminus E, (c)f\in\mathcal{R}[a,b]$. **Concl:** $\int_a^b f=F(b)-F(a)$.

Thm 7.3.4 / 7.3.5 / 7.3.6 T22 T23 FTC Second Form	Let $f \in \mathcal{R}[a,b]$ and $F(z) := \int_a^z f$. **7.3.4:** $\lvert f\rvert \le M \Rightarrow \lvert F(z)-F(w)\rvert \le M\lvert z-w\rvert$ (F continuous). **7.3.5:** If f is continuous at $c \in [a,b]$, then F'(c)=f(c). **7.3.6:** If f is continuous on [a,b], then F'(x)=f(x) on [a,b].
Thm 7.3.8 T25 Substitution	**Hyp:** $\varphi : [\alpha,\beta] \to \mathbb{R}$ has continuous derivative φ' on $[\alpha,\beta]$; $f : I \to \mathbb{R}$ is continuous on an interval $I \supseteq \varphi([\alpha,\beta])$. **Concl:** $\int_\alpha^\beta f(\varphi(t))\varphi'(t)\,dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx$.
Def 7.3.10 / Thm 7.3.12 T26 T27 Lebesgue Criterion	$Z \subseteq \mathbb{R}$ is **null** if $\forall \varepsilon > 0\ \exists \{(a_k,b_k)\}_{k=1}^\infty$ with $Z \subseteq \bigcup (a_k,b_k)$ and $\sum (b_k-a_k) \le \varepsilon$. **7.3.12:** Bounded $f : [a,b] \to \mathbb{R}$ is in $\mathcal{R}[a,b] \iff f$ is continuous a.e. on [a,b] (D(f) is null).
Thm 7.3.14 / 7.3.15 / 7.3.16 T28 – T30	**7.3.14:** $f \in \mathcal{R}[a,b], f([a,b]) \subseteq [c,d], \varphi : [c,d] \to \mathbb{R}$ continuous $\Rightarrow \varphi \circ f \in \mathcal{R}[a,b]$. **7.3.15:** $f \in \mathcal{R}[a,b] \Rightarrow \lvert f\rvert \in \mathcal{R}[a,b]$ and $\lvert\int_a^b f\rvert \le \int_a^b \lvert f\rvert \le M(b-a)$. **7.3.16:** $f,g \in \mathcal{R}[a,b] \Rightarrow fg \in \mathcal{R}[a,b]$.
Thm 7.3.17 / 7.3.18 T31 T32 Parts & Taylor	**7.3.17:** F,G differentiable on $[a,b], f = F', g = G' \in \mathcal{R}[a,b] \Rightarrow \int_a^b$ fG $= [FG]_a^b - \int_a^b$ Fg. **7.3.18:** $f', \dots, f^{(n+1)}$ exist on $[a,b], f^{(n+1)} \in \mathcal{R}[a,b] \Rightarrow f(b) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!}(b-a)^k + R_n, R_n = \frac{1}{n!}\int_a^b f^{(n+1)}(t)(b-t)^n\,dt$.

SPACED RECALL PLAN · DAY 0 / 1 / 3 / 7 / 14

- **Day 0 (today):** Read through with a blank sheet; cover the proof of 7.2.7 (T15), 7.2.8 (T16), 7.2.9 (T17), and 7.3.5 (T23) and re-explain every "Why" line aloud.
- **Day 1:** Write from memory the exact definitions of $\mathcal{R}[a,b]$(7.1.1), Cauchy Criterion (7.2.1), Squeeze Theorem (7.2.3), Null Set (7.3.10), and Lebesgue's Criterion (7.3.12).
- **Day 3:** Write out the four 10-mark proofs (Q37–Q40: Continuous $\Rightarrow \mathcal{R}$, Monotone $\Rightarrow \mathcal{R}$, Additivity, FTC 2) from their skeletons without looking.
- **Day 7:** Do the "Which tool?" drill above and Section A (Q1–Q20) of the Model Paper.
- **Day 14 / Exam week:** Recite the Counterexample Library (C7 – C20) and sit Sections B and C of the Model Paper timed.

Past Questions (PYQ) → Topic & Model Paper Map

> All 74 in-scope Module 3 questions from MTS 6B 10 (2022, 2025, 2026) and MAT 6B 09 (2021, 2022).

PAST PAPER TAGS (74 TOTAL)	TOPIC	PAPER Q
MTS-26 Q4; MAT-22 Q3, MCQ 16; MAT-21 Q5	T01 Partition & norm	Q1
MTS-22 Q4	T02 Riemann sum	Q2
MAT-21 MCQ 9	T03 Def 7.1.1($\mathcal{R}[a,b]$)	Q3
MTS-25 Q4	T05 Constant function $\in \mathcal{R}$	Q5
MAT-22 MCQ 18; MAT-21 MCQ 13; MTS-22 MCQ 7	T06 Finite exception set (7.1.3)	Q6
MTS-25 Q5; MAT-21 Q29; MTS-22 Q5; MAT-22 MCQ 14	T07 Linearity & monotonicity (7.1.5)	Q21
MAT-22 MCQ 15; MAT-21 MCQ 11, 17; MTS-22 MCQ 8, 9	T08 Boundedness Theorem (7.1.6)	Q7
MAT-21 Q8	T10 Cauchy Criterion (7.2.1)	Q23
MAT-22 Q4, MCQ 17; MAT-21 MCQ 12, 16	T11 Dirichlet & x+1 not in $\mathcal{R}$	Q24
MTS-26 Q18; MTS-22 Q6; MAT-21 Q7	T12 Squeeze Theorem (7.2.3)	Q25
MAT-21 MCQ 10	T13 Elementary step φ_J(7.2.4)	Q8
MTS-26 Q5; MAT-22 Q16	T14 Step functions $\in \mathcal{R}$(7.2.5)	Q26
MTS-25 Q18; MAT-21 Q30; MTS-22 Q19; MAT-22 MCQ 21, 23; MAT-21 MCQ 15	T15 Continuous $\Rightarrow \mathcal{R}$(7.2.7) & $\int f = 0$	Q18 Q37
MTS-26 Q19; MAT-22 Q30, MCQ 22	T16 Monotone $\Rightarrow \mathcal{R}$(7.2.8)	Q38
MTS-26 Q25; MAT-22 Q37a; MTS-22 Q25	T17 Additivity Theorem (7.2.9)	Q39

MAT-22 Q37b; MAT-22 MCQ 24, 25; MAT-21 MCQ 18	T18 Restriction (7.2.10) & [c,b] (Ex 7.2.11)	Q27 Q28
MTS-26 Q8; MAT-22 Q35, MCQ 27, 28, 29; MAT-21 MCQ 19, 20	T20 T21 FTC First Form (7.3.1) & evaluations	Q14 Q15 Q29
MAT-21 Q6	T22 Indefinite integral (7.3.3, 7.3.4)	Q11
MTS-25 Q25; MAT-21 Q37a; MAT-22 Q33, MCQ 30; MTS-22 Q14	T23 FTC Second Form (7.3.5) & Leibniz	Q17 Q19 Q40
MTS-25 Q6; MAT-22 Q18; MAT-21 Q19; MAT-22 MCQ 19, 26	T25 Substitution Theorem (7.3.8)	Q16 Q30
MTS-25 Q7; MTS-26 Q7; MTS-22 MCQ 10	T26 Null sets & $\mathbb{Q} \cap [0,1] (7.3.10, 7.3.11)$	Q13
MAT-22 Q6	T27 Lebesgue Criterion (7.3.12)	Q31
MAT-21 Q4	T28 Composition Theorem (7.3.14)	Q32
MTS-25 Q8; MTS-22 Q7; MAT-21 Q20	T29 Absolute Value $\lvert f \rvert \in \mathcal{R} (7.3.15)$	Q33
MTS-26 Q6; MAT-22 Q17; MAT-21 Q18	T30 Product Theorem $fg \in \mathcal{R} (7.3.16)$	Q34
MTS-25 Q19; MAT-22 Q32	T32 Taylor's Theorem (7.3.18)	Q36